Ethiopia NHDW

High-Level Requirements

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Executive Summary

The Ethiopia National Health Data Warehouse (NHDW) High-Level Requirements have been developed by the Ministry of Health as the first phase of a project to develop a comprehensive health data warehouse for Ethiopia. The project's vision is to provide a comprehensive data platform for the health sector in Ethiopia, bringing together all relevant data sources from the health and other related sectors. The main objective of having a robust NHDW is to provide the health sector with a single point of truth and analytical platform for health-related activities that will contribute to strengthening patient care, policy-making, and providing advanced data analytic capability. The need for a NHDW has been identified and explained in several Ministry of Health strategy and policy documents, including the Information Revolution, Health Sector Development mid-term Investment Plan (HSDIP), National eHealth Architecture, and the Digital Health Blueprint. It is also envisaged that the NHDW will be consistent with other eGovernment initiatives in Ethiopia, including the eGovernment National Data Warehouse project.

The scope of the National Health Data Warehouse (NHDW) includes aggregate and individual-level routine data in addition to other health sector data sources, such as laboratory, pharmacy, and supply chain, being collected and used across the public and private health sectors. The warehouse will also include data from other areas related to health and collected by other government agencies, such as climate data and census or statistical data. The warehouse will also include relevant health research and survey data (non-routine), other data marts, and data repositories as data sources. The NHDW includes methods to structure, transform, analyse, and visualise the data in ways that serve the goals of the key stakeholders of the warehouse.

Development of the High Level Requirements was led by the Strategic Affairs Executive Office (SAEO) and the Digital Health Lead Executive Office (DHLEO) in the Ministry of Health, assisted by a Technical Working Group (TWG) comprising health data experts from the Ministry and partner organisations. The high-level requirements were developed over a period of months, including two workshops and on-the-ground assessments of relevant facilities, agencies, and organisations in Addis Ababa, and were based on a Terms of Reference originally developed by the MoH and TWG and having as reference a roadmap document also developed by the MoH and TWG with a vision for the overall development, deployment, and maintenance plan for the NHDW.

This high-level requirements document forms part of the first phase, being the planning and requirements gathering phase.

The document comprises twelve chapters devoted to specific aspects of the project, as follows:

Chapter One introduces the project and provides the background to the project as well as the goals, objectives, scope, and intended audience of the project.

Chapter Two provides an overall description of the project, the guiding principles, methodologies, and development approaches used in developing the high-level requirements, as well as the high-level business requirements with use case examples.

Chapter Three provides the data source requirements, including details of the identified data sources and the important metrics pertaining to the data resources (ownerships, volume, update frequency, etc.).

Chapters Four and Five provide a listing and description of the high-level functional and nonfunctional requirements, respectively, that were identified based on the discussions carried on for the project. Chapter Five also provides a description of the quality assurance requirements.

Chapter Six covers the technical considerations for the implementation of the NHDW, including the architectural options and technology choices, including ingestion, storage, processing, analytics, security, and governance of the data platform.

Chapter Seven provides details of the hosting structure requirements, including the data centre assessment, computer resource requirements, and building infrastructure requirements.

Chapter Eight introduces some of the governance aspects of the NHDW, including applicable legislation and proclamations and metrics used to evaluate the effectiveness of governance practices and data quality.

Chapter Nine introduces elements of risk management, the risk management process, and actions to mitigate risks.

Chapter Ten provides a high-level roadmap for implementation of the NHDW using a phase-based approach.

This High-Level Requirements document also includes selected references used to compile the document (Chapter Eleven) and various Annexures, including the results of the data source assessment and Industry Standards and Best Practices for Data Warehousing (Chapter Twelve).

In summary, these NHDW High-Level Requirements have focused on producing a snapshot of the current priorities in the MoH and TWG of the vision for the NHDW through a high-level requirements lens. Recommendations are documented at relevant places in the text and will serve as a guideline to prospective implementers of the NHDW. Implementation of the initial version of the NHDW is estimated to take between three and five years, depending on the extent of data analysis envisaged.

Chapter 1: Introduction

1.1 Background

The Ethiopian Ministry of Health (MoH) has been working to strengthen its health system for more than two decades. This has included investing significantly in digital health systems and subsystems to be able to capture and generate accurate and reliable data that can inform program design and decision-making to improve health outcomes, efficiency, and quality of healthcare service delivery across all health sectors. Through these efforts, numerous health management and service delivery information systems have been developed, deployed, and used. These information systems, in addition to facilitating service delivery, have produced a wide variety of data that can, potentially, provide additional insights into the health system. However, health data is often stored in disconnected silos in multiple different formats. It is also fragmented across multiple different sources and lacks integration and interoperability of systems. This hampers comprehensive analysis and the ability to unlock insights from multiple data sources and types.

1.2 Ethiopia's Digital Health Policy Landscape

Ethiopia has set an aggressive agenda for improving the health of the population in its five-year strategic plans from 2015/16-2019/20 (HSTP I), 2020/21-24/25 (HSTP II), and 2024-2026 (HSDIP) [1], [2]. In response to one of the four transformation agendas identified in the two HSTPs II, the MoH has developed a roadmap for implementation for the period 2016-2020 and the HIS Strategic Plan for the period from 2018 to 2025 [3] [4]. In 2021, it also developed a Digital Health Blueprint for Ethiopia [5] that envisages a Digital Ethiopia by 2025. An eHealth Architecture (eHA) document developed in 2019 provides technical architectures for the current ("as-is") and future ("to-be") eHealth architectures in Ethiopia [6]. The latter architecture can be used to guide the implementation and integration of digital health systems in Ethiopia to realise the goal of interoperable digital health systems. The eHA aims to facilitate data analytics, enable data-informed decision-making by health system managers and stakeholders, and promote research efforts. Previous work on a concept note for a data warehouse was conducted by the JSI/Data Use Partnership [7]. Ethiopia has also developed an enterprise architecture project to help with designing the national digital health ecosystem [8].

The Need for a National Health Data Warehouse

The concept of a Data Warehouse is identified in the strategic documents mentioned above. HSTP II identifies the need to establish a data warehouse as a major strategic initiative. The Digital Health Blueprint identifies a data warehouse in Pillar IV: Data Hubs. The future architecture described in the eHealth Architecture includes a Data Warehouse in a general Analytics and Business Intelligence cluster.

According to the Ethiopian Health Sector Development mid-term Investment Plan (HSDIP), one of the major strategic initiatives is to develop a data warehouse for the Ethiopia Ministry of Health that "serves as a centralised repository for all healthcare information that is retrieved from multiple sources with the purpose of strengthening evidence-based decision-making [9].

Based on the National Health Data Access and Sharing guideline, Strategy 2 aims to develop a national health data repository/warehouse, data integration, and data interoperability, improve data recording, management, access, safety, data disaster recovery, and data backups. In addition, the guideline recommends the provision of dedicated servers with designed rooms/space or cloud servers and backups, guaranteeing that the data collection, generation, storage, and transfer follow standards.

As part of this strategic initiative, the development and implementation of data warehousing and analytics platforms, integrated with other digital health solutions, have been identified as major activities, as prioritised in the HSDIP. A National Health Data Warehouse (NHDW) is a strategic part of this, acting as a centralised repository that consolidates and integrates data from various systems and sources.

The Ethiopian MoH aims to have a NHDW that provides a single source of truth by ingesting, validating, transforming, and integrating health data from the current and future operational and other systems across the health sector and external data sources [10]. The desire is to enable longitudinal analysis, monitoring trends, drill-down queries, and other analytical use cases that are difficult to achieve when data is scattered in silos, and preventing parallel reporting systems and data inconsistencies The warehouse schema and data model selected should satisfy these use cases. Relevant dimensions and attributes must be included that allow analysis and visualisation of data across multiple parameters, including time, geography, facilities, population demographics, etc. The technology platform must provide scalable storage and processing to handle large volumes of multivariate health data that will accumulate over time [10]. The Ministry of Health has previously developed a data warehouse requirements document [11].

1.3 Purpose

The main purpose for the development of Ethiopian NHDW is to have a central health data repository where data from diverse datasets (e.g., clinical, demographic, and socioeconomic data) are integrated and consolidated in one place, providing a more holistic view of health issues and producing multiple data outputs for health systems planning, program management, and program implementation. The NHDW will make secured, well-integrated individual and aggregate data from different sources with improved quality available; it will serve to produce reports, conduct various analytics and data visualisation, and produce insights, patterns, and models to improve existing diagnosis or clinical treatment algorithms and monitor the performance and outcomes of healthcare service delivery. Developing the NHDW serves several important purposes, such as providing a rich source of data for research for epidemiological studies, health outcomes research, policy evaluation leading to evidence-based decision-making, and making data available for health facility-based patient interventions. Policymakers can analyse data to allocate resources more effectively, identify health disparities, track health trends more efficiently, monitor outbreaks, and assess the overall health status of the population, aiding in timely public health responses. Overall, the development of the NHDW will be a valuable tool to drive the Ethiopian health system through evidence-based decisions to enhance public health infrastructure, efficiency of the health system, and improving the quality of health services and outcomes.

1.4 Aims and Objectives

The overall goal of the Ethiopian NHDW development is to have centralised and consolidated health data from various sources, providing a single access point for data analysts, researchers, policymakers, and healthcare providers.

The NHDW aims at achieving:

Data Integration and Interoperability

- Facilitate the integration of diverse datasets from the different data sources available in Ethiopia

- Promoting seamless data sharing and communication among different health information systems.

Enhanced patient care and outcomes

- Actionable insights for clinical decisions in providing preventive therapeutic and palliative cares
- Improved resource allocation, health quality and equitable service coverage and utilisation

Improved clinical and public health decision-making

- Access to comprehensive, integrated health data
- Improve the quality of health services and outcomes by identifying health disparities and target interventions effectively to enhance the overall health of the population.
- Improved data quality to enable healthcare organisations to benchmark performance and implement best practices based on data-driven insights.
- Improve monitoring of health trends, new patterns, outbreaks, and population health, enabling timely public health interventions.

Operational efficiency

- Streamlined data access and analysis processes
- Reduced duplication of efforts across agencies and streamline data management processes, which can also lead to better budget management and cost savings in healthcare administration.
- Support advanced analytics, including longitudinal analysis and trend monitoring;
- Promote transparency in health data reporting, fostering trust among the public and stakeholders.

Support for research and innovation

- A platform for advanced analytics and AI applications
- Facilitation of health system research and policy analysis
- Support research and analysis that informs health policies, resource allocation, and healthcare practices based on solid evidence
- Provide a richer resource for researchers conducting epidemiological studies, clinical trials, and health services research.
- Encourage the development of new health technologies and solutions through access to comprehensive health data.

The NHDW aims to enhance the effectiveness and quality of healthcare delivery, improve population health, and support informed decision-making at all levels of the healthcare system. To achieve the above-described goals, the NHDW should support advanced analytics such as enabling AI-driven insights, data science applications, and longitudinal trend analysis and monitoring.

The more specific objectives for developing an NHDW requirement are:

- Determine the NHDW business requirements for the health system.
- Identify aggregate and individual-level data from both internal and external data sources.
- Collect and integrate health data from identified data sources that have been previously assessed, including systems/applications data such as data from electronic health records (EHRs), lab systems, and supply chains available at hospitals, health facilities, clinics, health

agencies, and research institutions, including aggregate and individual-level data into a unified system.

- Standardise the datasets and create a data dictionary by establishing common data definitions, formats, and standards to ensure consistency and comparability across the different datasets.
- Implement processes for data validation and cleaning, ensuring high-quality, reliable data for analysis.
- Design how data will be stored, processed, and accessed, making sure that the architecture of the NHDW meets the specific needs of the Ministry of Health as well as users and administrators, facilitating efficient data management and analysis.
- Design the conceptual data model, develop Extraction, transformation and loading processes, and build physical data model and databases of the NHDW
- Build the NHDW with hybrid approach and test the data models and APIs
- Implement deployment strategies and establish operational procedures and maintenance protocols
- Provide accurate training and capacity building and provide resources such as user manuals and documentation for users to effectively utilise NHDW and the data produced to foster a data-driven culture in healthcare.
- Select appropriate BI, analytics and visualisation tools
- Deploy and run a secured and robust NHDW for data analytics and reporting needs as a single source of truth.

1.5 Project Scope

The Ethiopia MoH NHDW development and implementation roadmap [10] specifies five NHDW development phases, i.e., 1) planning and requirement gathering; 2) data warehouse design and development; 3) Data modelling and analysis; 4) deployment and maintenance; and 5) quality assurance and implementation. This document forms part of phase one of this development process for the planning and requirement gathering phase. Phase one includes the background and rationale for development of the NHDW, an overall description of the NHDW, guiding principles and a development framework for the NHDW, high-level business requirements, example use case descriptions, a data sources assessment, functional and non-functional requirements for the NHDW, a detailed initial NHDW architecture design, IT infrastructure requirements, governance and risk management principles, and an implementation roadmap. The project may also include strengthening the ICT infrastructure at the lower level of the data source, including at health facilities, as well as improving the data quality of the source data systems. Defining the scope of detailed use case definitions and development will occur in a phased and iterative manner.

1.6 Document Convention

This document describes a collection of high-level requirements for the NHDW obtained from the revision of the existing documentation from Ethiopia and other countries, requirements gathered from the workshops, and infrastructure and data source assessment findings. This high-level requirements document has been written based on the Institute of Electrical and Electronics Engineers (IEEE) Software Requirements Specifications template and adapted as per the needs of documenting the high-level requirements for a national data warehouse, including chapters such as risk management, implementation roadmap, and costing. This document uses Arial font size 11 throughout the text and

Arial font size 10 on the tables. The text is single-spaced and maintains 1-inch margins. All the terminology used is detailed in subchapter 1.8.

1.7 Intended Audience and Reading Suggestions

This document is intended for the following audience: Ministry of Health (MOH) leadership, managers and experts, data analysts, researchers, academicians, health information system partners, health sector stakeholders, and donors. This document is to create alignment across multiple stakeholders on the understanding of the high-level requirements for the establishment of a national data warehouse and to provide a foundation for the start of the project implementation. It contains high-level functional, non-functional, and technical approaches. It also contains the main findings of the data sources and infrastructure assessment under the data sources requirements and hosting infrastructure requirements, respectively, as well as the needs for the implementation of the project. It has been added at the end of the document aspects related to the governance needed for the implementation and success of this project, the proposed implementation roadmap, and the costs and estimations. For those interested in the technology, we suggest a detailed reading of chapter 3, where the data source requirements are well detailed, as well as chapters 6 and 7 that refer to the architectural details and the hosting infrastructure assessment findings and requirements needs.

1.8 Summary: Chapter 1

The Ethiopian Ministry of Health is working to strengthen its information systems, which has included investing significantly in digital health systems to be able to capture and generate accurate and reliable data that can inform program design and decision-making to improve health outcomes, efficiency, and quality of healthcare service delivery across all health sectors. Through these efforts, numerous health management and service delivery information systems have been developed, deployed, and used. These information systems, in addition to facilitating service delivery, have produced a wide variety of data that can, potentially, provide additional insights into the health system. However, health data is often stored in disconnected silos in multiple different formats. It is also fragmented across multiple different sources and lacks integration and interoperability of systems.

In response to this, the Ethiopian MoH aims to develop a National Health Data Warehouse that provides a single source of truth by ingesting, validating, transforming, and integrating health data from current and future operational and other systems across the health sector and external data sources. The desire is to enable longitudinal analysis that monitors trends, drill-down queries, and other analytical use cases that are difficult to achieve when data is scattered in silos and prevents parallel reporting systems and data inconsistencies. The technology platform must provide scalable storage and processing to handle large volumes of multivariate health data that will accumulate over time.

The NHDW aims to make secured, well-integrated individual and aggregate data from different sources with improved quality available; it will serve to produce reports, conduct various analytics and data visualisation, and produce insights, patterns, and models to improve existing diagnosis or clinical treatment algorithms and monitor the performance and outcomes of healthcare service delivery. Developing the NHDW serves several important purposes, such as providing a rich source of data for research for epidemiological studies, health outcomes research, policy evaluation leading to evidence-based decision-making, and making data available for health facility-based patient interventions. Policymakers can analyse data to allocate resources more effectively, identify health disparities, track health trends more efficiently, monitor outbreaks, and assess the overall health status of the population, aiding in timely public health responses.

The NHDW goals are to:

- Produce data integration and interoperability
- Enhance patient care and outcomes
- Improve clinical and public health decision-making
- Improve operational efficiency
- Provide support for research and innovation

This high-level requirements document includes the guiding principles and a development framework for the NHDW, high-level business requirements, a data sources assessment, functional and non-functional requirements for the NHDW, a detailed initial NHDW architecture design, IT infrastructure requirements, governance and risk management principles, and an implementation roadmap for the development of the NHDW.

Chapter 2: Overall Descriptions

This chapter provides an overall perspective of the development and implementation of an NHDW for Ethiopia. It includes a description of users and their characteristics, high-level guiding principles in the development of the NHDW, the documentation that will be delivered and available for the NHDW's users, assumptions and dependencies in the development of the NHDW, and high-level business requirements for the NHDW.

2.1 The NHDW Perspective

The NHDW is an initiative to consolidate existent disparate data sources for analysis and strategic decision-making. The data warehouse is envisioned in the Ethiopian National eHealth Architecture in the business intelligence area [6]. To enable broad cross-cutting data integration and analysis, the NHDW ingests as a data source every other system's data that exists in the national health sector, which spans over multiple agencies and institutions handling health data. Data sources external to the health sector are also to be considered, such as climate data, animal health data, and population statistics. The MOH has produced a NHDW Strategic Roadmap, which provides further important contextual information [10].

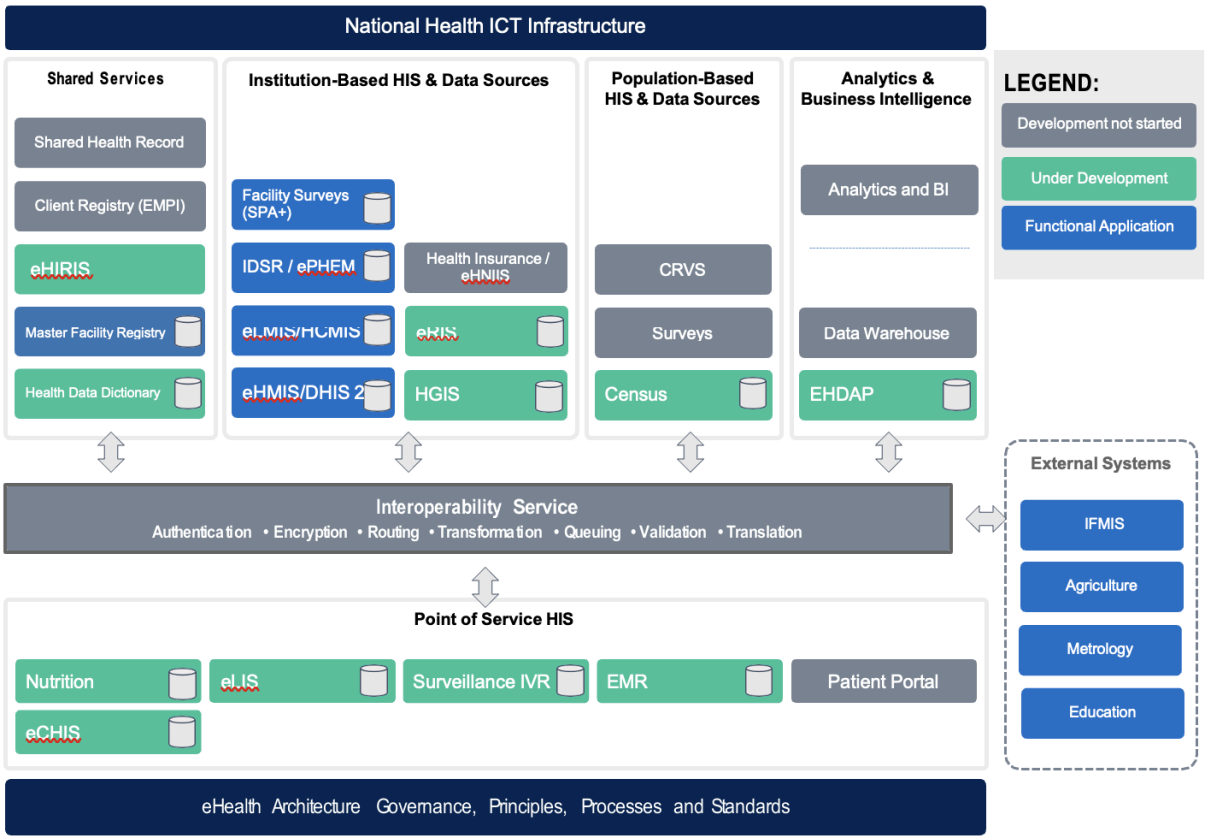


Figure 1: Ethiopian National eHealth Architecture

2.2 User Classes and Characteristics

The Ethiopia National Health Data Warehouse project in general will involve several user classes with their distinct characteristics and roles to ensure efficient data management and utilisation throughout its lifecycle, including stakeholders that play a crucial role in the success of its implementation. These user classes ensure that the NHDW serves its purpose effectively by providing the right level of access to different users according to the requirements described in chapters four and five. The user classes for this project and their roles and characteristics and access level to the NHDW are described below:

Stakeholders

Stakeholders will provide valuable knowledge and insights for better decisions and assess and share resource availability, such as equipment, space, or even staff.

Roles and characteristics: uphold ethical standards and corporate social responsibility; build strong relationships and facilitate the communications with relevant individuals and organisations for strategic partnerships and long-term success of the Ethiopian NHDW, providing continuity and sustainability; contribute to sustainable practices and long-term viability of the project; support the TWG and other partner to better understand the project and main pain points, helps to determine the Project's Scope and its implementation and get buy-In from other users and stakeholders, lead the testing and implementation process and provide support and maintenance to the product during its entire life cycle. In summary, as seen in Table 1 below, the stakeholders can play different roles throughout the project. Also see Annexure 3 for a more detailed stakeholder list.

Table 1: Organisational Type and Role

Organisation Type	Role in NDHW
MOH	Data Contributor
	Data User
	Research
	System Development
	Budget allocation
	Infrastructure
Agencies	Data Contributor
	Data User
	Research
Ethio telecom	Infrastructure
Universities and HDSS Sites	Research
	System Development
	Data User
Donor (CDC, USAID, GF, BMGF)	Budget allocation
	Data User
Implementing partners	System Development
	Data User

INSA	System Development
AI Institute	System Development
Regional Health Bureau	Data Contributor
	Data User
Health Institution	Data Contributor
	Data User
Central Statistics (CRVS, Census)	Data Contributor
National ID Program	Data Contributor
Metrology/Climate Institute	Data Contributor

Access level: according to the matrix above

Administrators:

Characteristics: All-level hospital and health clinics administrators and health system managers who oversee operations with the existing health information systems and resource allocation, including MOH, EPHI, National Health Insurance, EHIS, EFDA EPSS (Fanos, Vitas, mBrana), National Blood Bank Services and Central Statistics

Access Level: High, with permissions to access aggregated data for decision-making and reporting.

Data Providers:

Characteristics: All level hospitals, health clinics, laboratories, pharmacies, and including all agencies and Metrology/Climate Institute using health information systems, with the responsibility of submitting data to the national data warehouse. The users must be familiar with clinical data standards, indicators in use, and reporting. It will be required that these users are trained and regularly refreshed on the data submissions.

Access Level: High, with permissions to access aggregated data for decision-making and reporting.

Healthcare Providers

Characteristics: doctors/physicians, nurses, and other medical staff at the facility level who need access to real-time patient medical records, treatment histories, and diagnostic data for patient care improvement and clinical decision-making.

Access Level: High, with permissions to read and update patient data and outcomes

Data Analysts:

Characteristics: Professionals who analyse data for insights, such as agencies, epidemiologists, and university researchers who are proficient in statistical software and data visualisation tools, with strong understanding of data interpretation and reporting.

Access Level: Moderate to high, with access to build/generate complex queries, analytic functions, and data visualisation tools.

Researchers:

Characteristics: Medical researchers, academics, and public health analysts conducting research on health outcomes who require access to large datasets for studies and analysis, such as HDSS sites, the University of Gondar, Addis Ababa University, Hawassa University, Mekelle University, Jimma University, and Haramaya University, with proficiency in research methodologies and statistical analysis.

Access Level: Moderate to high, with access to comprehensive datasets for analysis.

Policy Makers:

Characteristics: Government officials and health policy makers who use data to inform public health strategies and policies.

Access Level: Moderate, with access to summary and trend data rather than individual patient records.

IT and Data Management Staff:

Characteristics: IT professionals and data managers responsible for technical aspects, maintaining the data warehouse infrastructure, and ensuring data integrity, with skills in database management, data security, and system integration, with the role of guaranteeing data quality, interoperability, data hosting, and system performance. Users with this characteristics included Ethio Telecom, MOH ICT,

Access Level: High, with full access to all data for maintenance and troubleshooting purposes and backend systems and administrative tools.

General Public and Patients:

Characteristics: Individuals accessing their own health records and treatment histories. Citizens seeking information about health trends and statistics throughout simplified data presentations and summaries. The users might need a basic understanding of health data interpretations.

Access Level: Low, with permissions to view their personal health information, such as public-facing dashboards and reports, and main results and outcomes of the country in situations such as disease outbreaks (COVID, MPox, etc)

2.3 User Documentation

The documentation that will be delivered and available for the NHDW's users will include the following:

- Data Dictionary: A comprehensive document that provides detailed information about the data elements stored within the warehouse. It serves as a reference guide for users, helping them understand the structure, content, and meaning of the data.
- API Documentation: Instructions for developers on how to interact with the data warehouse programmatically, including endpoints, authentication, and sample queries.
- Technical Specifications: Details on the architecture, technologies used, and system requirements for accessing the data warehouse.
- Use Cases: prioritised use cases that will be used for the NHDW and that illustrate what data to integrate, how to query and analyse data for various applications, such as health service

delivery or medical products, vaccines and technology innovations, data quality management, and data warehouse administration.

- User Manual: Comprehensive guide detailing how to access, navigate, and utilise the data warehouse, including step-by-step instructions on how to access and produce reports and data visualisations.
- Frequently Asked Questions: Address common questions users may have about the NHDW.
- Training Materials: Resources such as PPT presentations, tutorials, job aids, and videos that allow users to be trained and to build capacity for using, monitoring, and administering the data warehouse; standard procedures for the implementation of the infrastructure; and FAQs to help users understand how to effectively use the warehouse.
- Troubleshooting Guide: Common issues users may encounter, along with solutions for basic incidents and a list of contact information of the support team members.
- Access, privacy and Security Guidelines: Information on user roles, permissions, access, privacy and security protocols to protect sensitive health data, and standards of procedures for how to manage data produced by the warehouse, specifically for the ones with PII.
- Governance Framework: Documentation outlining strategy, policy, and regulation documents that guide the implementation and governance of the NHDW, including data management practices such as data quality standards, privacy regulations, and compliance requirements.
- Implementation Roadmap: Phased implementation plans and timelines, detailing incremental deployments and integration steps, and guidelines on how to manage the data sources during the implementation phases, including strategies for transitioning from legacy systems to the new data warehouse, including data migration plans and resource allocation.
- Change Management: Procedures for managing changes to the data warehouse, including structured approaches to transitioning individuals, teams, and organisations to a new system, impact assessments, and communication plans.
- Change Log: A record of updates, enhancements, and changes made to the data warehouse, including version history.
- Requirements documentation and traceability: detailed and structured documentation of the business, information, application, infrastructure, and technology requirements, traced throughout the development lifecycle.
- Risk Assessment and Mitigation Plan: list of the identified and expected risks associated with the data warehouse, including strategies for mitigating it, including data security and disaster recovery plans.
- Backup and Recovery guidelines: Detailed backup and recovery guidelines for the NHDW.

2.4 Assumptions and Dependencies

The following assumptions have been made regarding the analysis of the business processes and user requirements:

- An iterative approach of requirement specification and software specification will take place centred on specific business use cases as the implementation of the NHDW progresses in future phases.
- Business requirement specifications and use case definitions will be formalised in the early stages of a future phase that will be implementation-orientated.

- Detailed functional and non-functional requirements will be defined with more precision and mapped to specific use cases and business requirements after the business requirements specification process is conducted.

An important policy assumption is that the NHDW project will have the necessary approvals to store the data of all its intended data sources. Many data sources have as owners and originators different agencies and institutions. This data is currently hosted in different data centres. Hence, the official data owners will need to pass approval for the NHDW technical team to store a copy of their data in a central location. Similar data sharing agreements for secondary use of the data will need to be established. This will be especially applicable for data sources containing personally identifiable information (PIIs).

Technical assumptions:

- A data centre of sufficient capability is employed, fulfilling the infrastructure specifications described in Chapter 7 below.
- Stable internet with sufficient bandwidth is made available for robust data transmission from data sources to NHDW servers.
- The NHDW administrator will have staff capacity to support the data transmission and standardization process from the data source location.

2.5 Principles

2.5.1 Guiding principles of the project

Guiding frameworks adopted:

- The Open Group Architecture Framework (TOGAF) [12].
- Principles for Digital Development [13].
- Hybrid methodology: Combining top-down (Bill Inmon) [14] and bottom-up (Ralph Kimball) [15] approaches to development.
- FAIR principles: Data management principles [16] [17].
- Data Vault 2.0 Methodology: Guides data warehouse modelling and management [18] [19].
- Health Data Principles: The Health Data Governance Principles bring a human rights and equity lens to the use of data within and across health systems [20].

The following Table are high-level guiding principles in the development of the Ethiopian NHDW and throughout the project's lifetime and implementation. Many of these principles have been sourced from the Health Data Governance Principles [21], stewarded by Transform Health under the leadership of its Policy Circle with over 200 contributors & 130 organisations. You can read more about them here.

Table 2: Guiding Principles

Principle	Statement	Rationale	Implication
Data is an asset	Data is an asset that has value to the Health sector and is managed accordingly	Data is a valuable resource for the health sector; it has real, measurable value. The purpose of data is to inform decision-making. Accurate, complete, and timely data is critical to inform decisions. Most of the health sector's assets are carefully managed, and data is no exception. Data is the foundation of our decision-making, so we must also carefully manage data to ensure that we know where it is, can rely upon its accuracy, and can obtain it when and where we need it	All stakeholders of the NHDW should understand the relationship between value of data, sharing of data, and accessibility to data. Stewards of data must have the authority and means to manage the data for which they are accountable. The role of data stewards is critical because obsolete, incorrect, or inconsistent data may be passed to health program managers and adversely affect important regional and national health program decisions that may affect health program performance. Procedures must be developed and used to prevent and correct errors in the health data and improve those processes that produce flawed information. Data quality will need to be measured and steps taken to improve data quality, e.g. Ethiopian DHIS2 data should be audited on a regular basis and processes implemented to improve source data quality. The Ethiopia MoH should decide on process changes to enhance data quality and procedures.

Protect Individuals & communities	Establish governance early on. Health data governance must protect individuals & communities from harm & violations at every stage of the data lifecycle.	This balance requires rigorous evaluation and risk assessment of data practices to identify and mitigate potential harm, which should be built into every stage of the data lifecycle.	When data is not personally identifiable, health data governance should mitigate collective risks, including risks of discrimination to specific groups or communities and broader society. Protection of health data requires secure methods of collecting data and secure data storage. Consideration should be given to how long data is stored, with guidance on a reasonable timeframe after which data should be removed from the system. Because personal health data is "lifetime" data, the data retention policies should not create gaps in longitudinal health records. Comprehensive data security policies should also respond to data transfer approaches and the evolving health innovation ecosystem. For every stage of the data lifecycle, health data governance must indicate where deidentification and anonymisation are necessary. In addition, health data governance should outline protections for de-identified and anonymised data, as even this data may expose sensitive information. The possibility of re-identification resulting from rapid technological developments should also be considered (e.g., by data analysis algorithms or triangulation of data sources). Data privacy needs to comply with the Ethiopian Personal Data Protection Proclamation of 2024 [42].

Build trust in Data Systems	Well-developed health data systems should reinforce trust in data systems and practices.	Developing health data governance systems in a participatory and transparent manner, followed by a prioritised and iterative approach, can help build trust. Transparency in health data governance can create buy-in from stakeholders around data processes, thus enabling better re-use of data, greater collaboration on data analysis methodologies, and higher-quality insights from data. Trust requires safeguarding data, preserving privacy, and establishing transparent and inclusive data collection, processing, storage, analysis, use, sharing, and disposal processes. It also requires responsiveness to questions from data subjects in the health system and other stakeholders to address potential grievances.	Health data stewards should apply existing—and establish new—best practices to protect individual and collective data. This includes both technical approaches to data collection and storage (encryption, de-identification) and policies and processes related to how data is accessed and used (e.g., security policies). Health data governance should align with and learn from established policies and regulations for a NHDW. Trust in data systems also requires trust in the quality of data. Health data governance should advance overall improvement in data quality and make data more available and accessible, as appropriate. Global standards must be adopted in the development of the NHDW to promote high-quality, accurate, and reliable data collection processes. Establish transparent & accessible processes and systems. It is also important to monitor the usage of the data warehouse. This helps ensure the data is being used appropriately and that sensitive information is not being accessed without permission. Additionally, monitoring usage can help identify potential security risks or unauthorised access attempts. Institute feedback and accountability mechanisms through which individuals and communities can report data misuse, make enquiries into health data structures and processes, remove their data from a system, and provide general feedback.

Ensure Data Security	Data security is an essential component of health data governance, encapsulating technical and procedural requirements. Any technical processes employed to collect, process, store, use, or share data should employ robust security mechanisms. This may include password requirements, security keys, and data encryption.	Data security builds trust with stakeholders, including healthcare professionals, patients, and policymakers. This includes applying best practices for the collection, processing, storage, analysis, use, sharing, and disposal of health data. The NHDW must bring together multiple, autonomous data sources to allow cross-system sharing while appropriately adapting efficient data practices across different sectors. An approach that prioritises data security may maximise the value and use of data and create new opportunities to generate insights from multiple stakeholders across sectors.	Data storage and processing facilities must be secured as per global or national standards. Data security must be instituted right from the design of technologies and processes to foster trust. Data security, data rights, and data ownership may be enhanced through federated storage, processing, and use of data. Define roles and responsibilities for those who will be involved in the data warehouse. It is important to identify who will be responsible for managing the data, who will have access to it, and how it will be used. This includes ensuring that only authorised users have access to the data and that all data is encrypted, which helps ensure a clear understanding of who is responsible for what. Establish clear documentation, provide stakeholders with information on how data is used, and offer channels for addressing data concerns. Data security audits should be undertaken periodically. Use auditing tools. Auditing tools help track data access, when it was accessed, and what was accessed. This information can be invaluable in identifying any potential security risks or unauthorised access attempts. Consider how to minimise the impact of potential security breaches. This may include using unique identifiers in place of an individual's name; placing limits on how long data may be stored; adding enhanced security measures for sensitive data; safe storage guidelines; and efforts to protect the right to privacy as a system design principle. Security-related best practices must continually evolve as new technologies are introduced. Additionally, it is important to have a disaster recovery plan in place in case of any unforeseen events. When data breaches do occur, health data governance should require stakeholders to inform the individuals and communities affected and report the breach to the concerned regulators

Enhance Health Systems & Services	Health data governance should encourage a culture of exploring, developing, and using data insights at all levels to enable the meaningful use of data to enhance health system efficiency and resilience, improve health access, and advance health equity towards achieving universal health coverage.	Health data adds significant value to health systems and services, leading to personal and public health improvements. In addition to improved access to health services, health data offers opportunities for greater quality, efficiency, effectiveness, and sustainability of health systems.	Health data governance should include operational improvement when defining data needs. Appropriate health data governance is a prerequisite to a resilient and responsive health system and can improve the efficiency and effectiveness of health services. These benefits can extend to all operational components of a health system (e.g., supply chain, health workforce management). Health data governance should be designed in such a way to build confidence in data users and decision-makers. Data sharing is also needed to support public health informatics and data-led actions. Data sharing and access policies should be designed to enable such use of data. Data use and re use creates opportunities for innovation and advancements in the medical sciences. The value offered by these advancements should be considered when defining the potential use of health data. For example, data may be needed by research institutions and academia for research and development purposes. These stakeholders may legitimately require appropriate, secure access to data, but individuals and communities who contribute data must also fully understand how their data may contribute to research and development.

Promote Data Sharing & Interoperability	Data sharing is a prerequisite for creating value from health data.	At a national level, data sharing allows for deeper and more significant insights related to health needs and challenges. Systems designed for interoperability, around common protocols, structures, and definitions, enable continual sharing of data as well as ensuring data quality. This promotes greater standardisation and comparability of health data, which allows for greater system interoperability, data sharing, data quality, and data hygiene.	Health data governance should include data sharing rules and guidelines that address a range of data sharing scenarios that should minimise risk even as they enhance public health. Stewards should ensure there are limitations on data access, identifying which stakeholders will have access to various levels of data, to balance the need to minimise risk of exposure while maximising sharing that can lead to additional value. Leveraging data sharing to use previously collected data can even reduce the need and extent of new data collection. Interoperability will make sharing data between systems simpler and more secure while preventing potential errors during manual data transfers. This is accomplished through the application of recognised standards and system design. Common data structures will support data sharing by allowing predictability and easier consolidation of data from a variety of data systems. Such common structures enable opportunities for deeper and more complex insights that add value and efficiency to medical sciences and improve health outcomes. Health data governance should utilise existing terminology and definitions of key concepts like data types, system protocols, and stakeholder roles and responsibilities. Such definitions are provided by global normative bodies and are widely used in both the public and private sectors. Existing data standards and frameworks (including ISO/TS 22220:201118, HL7 FHIR19, OpenHIE20, GS121, and others) may be applied whenever possible.

Scalability & flexibility	Environments change, and new demands emerge. Services or processes require adjustments; assets last only a limited period and need maintenance.	A scalable design allows the data warehouse to adapt to future demands and prevent performance issues or severe cost overruns as needs grow.	The NHDW architecture needs to be eminently scaleable; the system is national in scope and needs to be able to cater for the massive digital and data revolution that is presently underway in all countries and ecosystems. The stewards of the system must operate under the assumption that storage, compute, and network resource requirements will increase over time. Design for a modular system and flexible storage solutions, enabling smooth scalability without costly overhauls as data volumes and users increase. The NHDW will be a long-term project and core component of the national health information system architecture and needs to be designed and developed to be sustainable
Focus on the person	Design the system around the patient and health professional, not the disease.	Widespread system adoption and effective data-driven insights will lead to effective decision-making and improved health outcomes.	The architecture needs to encapsulate the unique challenges and opportunities of national health information systems in the country. Keep in mind the health professional who provides the service and the patient who received the service. Use data-driven insights to guide program decisions and evaluate the impact on health outcomes. Engage users in design and testing processes to ensure the system supports straightforward access to relevant data for effective decision-making. Be collaborative by designing and implementing the solution in an iterative manner with all stakeholders involved.

2.5.2 Digital Development Principles

In addition to the guiding principles of the implementation of the NHDW, the actual data warehouse development would follow the digital development principles [19]. These include:

1. **Design with the User:** this principle is embodied in the "top-down" component of the hybrid approach to NHDW development
2. **Understand the Ecosystem:** the NHDW architecture needs to encapsulate the unique challenges and opportunities of national health information systems in low and middle-income countries.
3. **Design for Scale:** the NHDW architecture needs to be eminently scaleable; the system is national in scope and needs to be able to cater for the massive digital and data revolution that is presently underway in all countries and ecosystems.
4. **Build for Sustainability:** The NHDW will be a long-term project and core component of the national health information system architecture and needs to be designed and developed to be sustainable
5. **Be Data-Driven:** the NHDW is clearly data-centric and should be developed with all kinds of data in mind.
6. **Use Open Data, Open Standards, Open Source, and Open Innovation:** the NHDW needs to embrace the open approach to systems development and take advantage of open sharing of information, experiences, tools and technologies that accompany this approach.
7. **Reuse and Improve:** The NHDW should reuse existing tools that are openly available and have proven utility and also embrace the notion of giving back so that other countries can benefit from the experiences of this project
8. **Address Privacy and Security:** The NHDW needs to have stringent measures in place to protect, restrict and secure access to data, especially personal identifiable information and other sensitive information about health
9. **Be collaborative:** the NHDW developers should collaborate with all stakeholders involved.

The NHDW development guiding principles have a broad scope and describe overall principles regarding the value and use of health-related data. The Digital Development Principles have a narrower scope and focus on the development and implementation of a digital solution. Both principles emphasise the important needs of data security, and the protection of privacy of personal information. These two sets of principles complement each other and will enhance the development and implementation process of the NHDW.

2.6 The Open Group Architecture Framework (TOGAF)

The NHDW project is closely related to and informed by the national enterprise architecture project because it is part of the national enterprise and has many architectural design elements associated with it. As a result, many of the guiding principles of the NHDW are aligned with the core concepts included in the Open Group Architecture Framework (TOGAF) [12].

The TOGAF Standard is a framework for Enterprise Architecture (EA) and an approach to achieving it. In this context, an enterprise is defined as the health ecosystem within a single jurisdiction, e.g., a country. A separate EA may be defined across multiple cooperating countries, e.g., in a regional economic community or a continental framework. The EA helps to optimise fragmented legacy processes and systems across the enterprise and to create an integrated environment that is responsive to change and supports delivery of the health system strategy. It also supports effective data use and

digital transformation, which are key factors to health system success. It also helps with documenting processes around personal data as required by Ethiopian legislation. Key elements of TOGAF include the Enterprise Continuum and the Architecture Development Method. Central to TOGAF is the Architecture Repository, which facilitates knowledge sharing, reuse of architecture assets, and overall consistency.

The Enterprise Continuum (EC) sets the broader context for an architecture and how generic solutions can be leveraged and specialised in order to support the requirements of an individual organisation. It includes two complementary concepts: the Architecture Continuum and the Solutions Continuum. At an architectural level, it enables an enterprise (country) to use generic architectures, e.g., the Open Health Information Exchange (OpenHIE), to guide the design of its own EA and the data vault architecture to design its own NHDW architecture. At a solution level, the enterprise continuum assists with the adoption and localisation of reusable software building blocks and digital public goods.

Enterprise Architecture Capability and Governance. The TOGAF Standard—Enterprise Architecture Capability and Governance assists with establishing appropriate organisation structures, processes, roles, responsibilities, and governance in order to successfully operate an architecture function within an enterprise. These can be effectively realised using the Architecture Development Method (ADM), a core part of the TOGAF framework. This provides a tested and repeatable process for developing architectures. These include establishing an architecture framework, developing architecture content, transitioning, and governing the realisation of architectures. These activities are carried out within an iterative cycle of continuous architecture definition and realisation, allowing the organisation to transform its enterprise in a controlled manner in response to prioritised goals and opportunities.

The ADM comprises a number of phases shown in the diagram below.

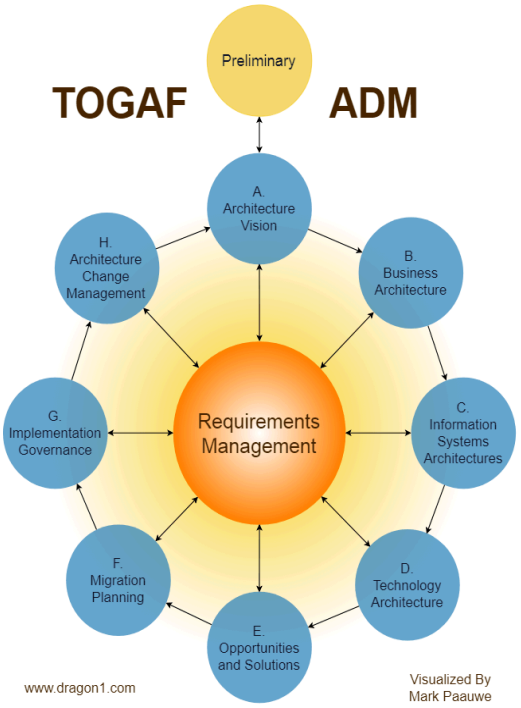


Figure 2. The TOGAF Architecture Development Method (ADM) [12].

Applying TOGAF to the NHDW

- **Preliminary Phase**:
  - **Define the scope and objectives**: Clearly define the scope of the data warehousing project, including its purpose, target users, and desired outcomes. This aligns with TOGAF's Architecture Vision phase, ensuring the project has a clear direction and business justification.
  - **Identify stakeholders and concerns**: Identify all stakeholders involved in the data warehousing project and understand their concerns and requirements. This helps in addressing potential conflicts and ensuring buy-in from key players.
  - **Establish an Architecture Principles**: Define a set of principles that will guide the design and implementation of the data warehouse. These principles should align with the overall enterprise architecture principles and address concerns like data security, data quality, and data governance.
- **Architecture Development Method (ADM)**:
  - **Business Architecture**: Model the business processes that the data warehouse will support. Identify the information flows and how the data warehouse will contribute to achieving business goals.
  - **Data Architecture**: This is a crucial phase to define the structure of the data warehouse, including:
    - Data entities and relationships
    - Data sources and integration methods

        - Data storage and access mechanisms
        - Data quality and governance processes
    - **Application Architecture**: Define the applications that will interact with the data warehouse, such as reporting tools, data mining applications, and data visualisation tools.
    - **Technology Architecture**: Specify the technology infrastructure required to support the data warehouse, including hardware, software, and network components.
- **Implementation and Migration**:
    - **Develop a detailed implementation plan**: Outline the steps involved in building and deploying the data warehouse, including data migration, testing, and user training.
    - **Establish a governance structure**: define roles and responsibilities for managing and maintaining the data warehouse, including data quality control, data security, and access management.
    - **Monitor and evaluate**: Continuously monitor the performance and effectiveness of the data warehouse and make adjustments as needed to ensure it meets business requirements.
- **Architecture Repository**:
    - **Document all architecture artefacts**: Maintain a centralised repository of all architecture documents, including data models, data flow diagrams, technology specifications, and governance policies.
    - **Facilitate knowledge sharing and reuse**: The repository promotes knowledge sharing and reuse of architecture assets, streamlining future data warehousing initiatives.

2.7 NHDW Development Approach

The Ethiopia MoH NHDW development and implementation roadmap [10] specifies five development phases in the overall development of the NHDW, i.e.

1. Planning and requirement gathering
2. Data warehouse design and development
3. Data modelling and analysis
4. Deployment and maintenance, and
5. Quality assurance and implementation

This section describes noteworthy frameworks to guide the development process of the NDHW data platform.

Globally, there are three known Data warehouse building approaches with their own pros and cons [15], [16]: Top-down, where first the data warehouse is designed using relational model and then developing the data marts for different subject areas; Bottom-up, the data marts are built using dimensional models and later the data warehouse will be built; the third one is the hybrid one. Depending on the country's context, the advantages and disadvantages of each approach are described below in detail. The justification for selected (hybrid approach) also clearly mentioned in this document.

2.7.1 The top-down approach

This approach, advocated by Bill Inmon [15] (often called the "father of data warehousing"), focuses on building a **centralised, enterprise-wide data warehouse** first. From this central repository, **data marts** are then created to serve specific business units or departments.

Key Concepts:

- **Enterprise Data Warehouse (EDW):** The first step is creating a large, centralised data warehouse that integrates data from all of the organisation's source systems.
- **Data Marts:** Once the EDW is in place, smaller, department-specific data marts are built for specialised use cases. These data marts draw data from the centralised warehouse.
- **Normalised Data Model:** The data in the enterprise data warehouse is stored in a **normalised** form (usually third normal form, 3NF) to reduce redundancy and maintain consistency.
- **Integration Focus:** Inmon's approach emphasises data integration, ensuring that the data warehouse is a unified, consistent source of truth across the organisation.

Steps in Top-Down Approach:

1. **Build the Enterprise Data Warehouse (EDW):** Create a comprehensive data warehouse to store all organisational data in a structured, integrated way.
2. **Data Modelling:** Use a normalised data model (typically 3NF) to structure the data, reducing redundancy.
3. **Create Data Marts:** Build smaller, departmental data marts that use data from the EDW to address specific reporting or analytical needs.
4. **Reporting and Analysis:** Business users can generate reports and analyses from data marts, which are fed by the enterprise data warehouse.

Pros of the Top-Down Approach:

- **Data Consistency:** A single version of the truth is ensured because all data flows through a centralised data warehouse.
- **Integration Across Departments:** Since the focus is on creating a unified data model, data from various departments is well integrated, reducing data silos.
- **Scalable and Flexible:** The EDW can be expanded over time to integrate more data sources and provide additional data marts as needed.

Cons of the Top-Down Approach:

- **Time-consuming:** Building a fully integrated enterprise data warehouse is a long process, often taking months or years before business users can access data marts.
- **Costly:** Due to the complexity and scale of the EDW, the initial implementation can be expensive.
- **Rigid Design:** The highly structured, normalised model may limit flexibility when it comes to quickly adapting to new business requirements.

Best Suited For:

- Large organisations that need a unified data warehouse for consistent reporting and analytics across all departments.
- Organisations that prioritise long-term, integrated data solutions over quick, short-term results.

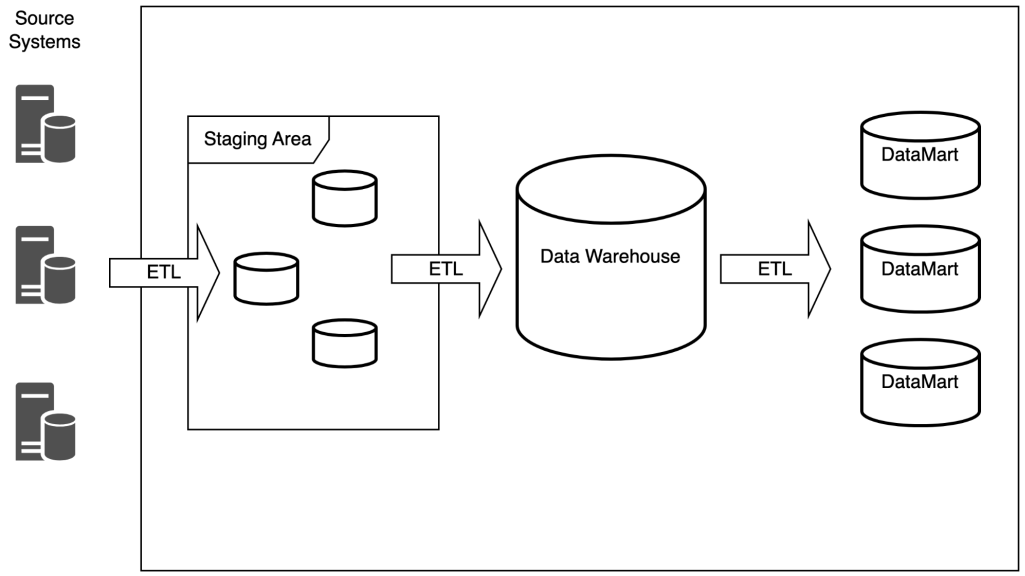


Figure 3: Top-Down Approach

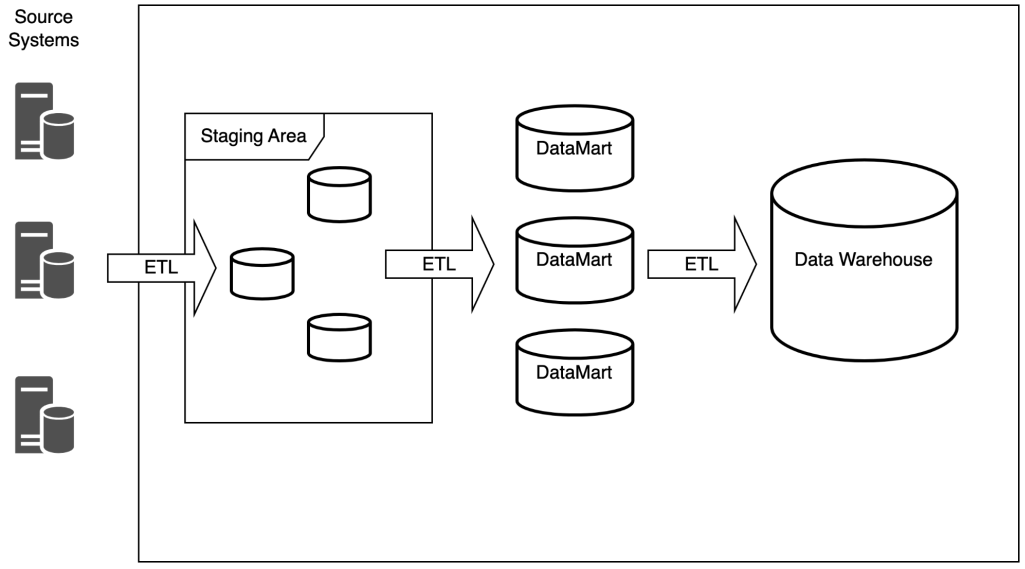


Figure 4: Bottom-Up Approach

2.7.2 The bottom-up approach

Popularised by Ralph Kimball [16], it focuses on building **data marts** first. These data marts are designed to meet the needs of specific business areas (e.g., finance, sales, marketing) and are later integrated into a **data warehouse**. In this approach, the data warehouse is essentially a collection of integrated data marts.

Key Concepts:

- **Data Marts First:** Instead of starting with a central data warehouse, the bottom-up approach begins by creating smaller, department-specific data marts that serve the immediate needs of users.
- **Dimensional Data Model:** Kimball advocates for the use of a **dimensional model**, often referred to as a **star schema**, where data is stored in facts and dimensions, making it simpler for users to query.
- **Data Warehouse as a Collection of Data Marts:** Once the data marts are built, they are integrated to form the overall data warehouse. In this sense, the data warehouse is a collection of interconnected data marts.

Steps in the Bottom-Up Approach:

1. **Build Data Marts:** Start by creating data marts that address specific business needs (e.g., sales reporting, inventory management).
2. **Use Dimensional Modelling:** Design each data mart using a star or snowflake schema with facts and dimensions, which makes querying and reporting more user-friendly.
3. **Integrate Data Marts:** As more data marts are developed, they can be integrated into a cohesive data warehouse with shared dimensions across data marts.
4. **Reporting and Analysis:** Each data mart can immediately serve the reporting and analysis needs of its business unit, offering quicker access to data.

Pros of the Bottom-Up Approach:

- **Faster Time to Value:** Data marts can be built relatively quickly, allowing business users to start generating reports and gaining insights sooner.
- **Lower Initial Cost:** The bottom-up approach can be more cost-effective initially because it allows for incremental development based on immediate needs.
- **User-Friendly Design:** The dimensional model (star schema) makes it easier for non-technical users to query and understand the data.
- **Agility:** This approach allows for greater flexibility, as data marts can be developed based on changing business requirements.

Cons of the Bottom-Up Approach:

- **Data Silos:** Without proper integration, data marts may become isolated, leading to data silos where each department operates independently without a unified data model.
- **Inconsistent Data:** If not well integrated, different data marts may use inconsistent definitions of key metrics or dimensions (e.g., sales figures in different departments might not match).

- **Limited Scope Initially:** Since the focus is on individual data marts, the early versions of the data warehouse might lack the full, integrated view of the entire organisation.

Best Suited For:

- Organisations that need quick access to specific data for immediate business needs.
- Companies that are willing to build the data warehouse incrementally and value short-term gains over long-term integration.
- Businesses that prioritise agility and the ability to quickly adapt to changing reporting and analytics requirements.

Table 3: Summary Comparison: Top-Down (Inmon) vs Bottom-Up (Kimball)

Aspect	Top-Down (Inmon)	Bottom-Up (Kimball)
Initial Focus	Enterprise Data Warehouse (EDW)	Departmental Data Marts
Data Model	Normalised (3NF)	Dimensional (Star Schema)
Data Marts	Created after EDW is built	Created first and integrated into a data warehouse
Time to Value	Longer to implement (months/years)	Faster to implement (weeks/months)
Integration	Centralised, unified data across departments	Data marts are integrated later
Data Consistency	High consistency due to centralization	Risk of inconsistency between data marts
User Accessibility	More technical to query initially	Easier for users to query due to dimensional model
Cost	High initial cost, scalable long-term	Lower initial cost, more flexible
Best For	Large enterprises with complex data needs	Smaller to mid-sized organisations needing fast results
Example	A global retailer building a single source of truth for all data across departments	A fast-growing startup building sales and marketing data marts first

2.7.3 Hybrid approach

A hybrid, top-down and bottom-up approach strategy will be used to define the detailed requirements of the NHDW and guide its development. This approach ensures that both high-level user requirements (top-down) and consideration of the available data source systems in the country (bottom-up) are considered in the design and development of the NHDW, as well as the identification of priority use cases for further development of the system.

Why Hybrid Approach?

Hence, where it is possible, the **top-down** approach will be leaned towards to ensure good data management, emphasise data integration, and reap the benefits of consolidated data sources. However, in practice, the NHDW data platform will allow a **bottom-up** approach to allow flexibility to advance continued development of value-adding use cases. Hence, we say that the NHDW will adopt a hybrid approach.

This NHDW project adopts a hybrid approach, combining both approaches described above. In general, the value of the NHDW to the health sector is by its nature the consolidation and integration of data sources, which in the past was not possible. The health sector is in need of a consolidated source of truth that is sector-wide, with linkages between previously unlinked data sources. For this reason, it is recommended that, where feasible, the NHDW should lean towards the **top-down** approach to emphasise data integration first. But in practice, as per examples provided above, there will be many instances where a **bottom-up** approach is used to allow sufficient flexibility in the development process and maximise the benefits of the rapid development in a low-resource environment.

- **A bottom-up** approach will be leveraged for initial proof of concept use cases to support the fastest path to obtaining end-to-end use cases to maximise early learning. This approach will also allow the NHDW to initially focus on data sources that are most mature and more readily available in the short term. This will also allow the NHDW to respond quickly in a flexible way to priority use cases, typically having an initial narrow definition, i.e., not integrating comprehensive sources and subject areas.
- **A top-down** approach will be leveraged as the NHDW matures to design a data model that is unified, consistent, and integrates a broader range of data sets and subject areas in an organised way. It is important to progress in the medium/long term to overcome data silos that will quickly emerge using the bottom-up approach. The purpose and value proposition of the NHDW is to integrate disparate data sources, so unifying the data will be important. It is expected that over time the NHDW will have fewer bottom-up data marts and tend to use a central top-down data warehouse(s) within the NHDW data platform.

The hybrid nature is expected to unfold at least at two levels:

- **Disparate distributed data sources** that will send data into one consolidated data structure at the NHDW level.
    - An example is the supply systems in Ethiopia, which are already consolidated into a subject matter data mart. The NHDW will not extract data from original source systems but rather leverage this subject-level data mart as a data source. This is an example of a bottom-up approach at the data source level, but a top-down approach at the central NHDW level.

- **Multiple data warehouses and data marts are internal** to the NHDW platform for various reasons:
    - An example is a data warehouse that is first built consolidating aggregate data from HIMS systems and other data sources that have related data types. As much care is taken to plan the data models and dimensions to ensure good "town planning" of the data warehouse. This is leaning towards a top-down approach.
    - Later on, there is a requirement for a second accompanying data warehouse to be developed alongside it that focuses on real-time data that must use separate technology from the first because of the performance characteristics required of it. This now results in two data warehouses with inherently different subject matter. However, power users can want to leverage data from both data warehouses for insights. This is an example of a bottom-up approach.
    - Equally, it is a likely scenario that a particular partner in the health sector has funding to develop a specific subject matter, leading to a data mart being built to service that requirement. This may not fully consider all other data sources in the health sector, so inevitably a data mart with specific subject matter will emerge.

2.7.4 Data Vault 2.0 methodology

Data Vault methodology [20] is a hybrid approach that merges elements from the top-down approach, which is widely known as traditional data warehousing, and the agile, business-driven model approach, the bottom-up approach.

Data Vault 2.0 is a methodology and architecture for designing a data warehouse that aims to manage large volumes of data from various sources and handle changes and scalability in a flexible way. Data Vault 2.0 addresses many of the challenges traditional data warehouses face by offering a flexible, scalable, and highly auditable architecture. By separating the core data model from business rules, tracking data lineage, and allowing incremental changes, Data Vault 2.0 is well-suited for the NDWH's complex, rapidly changing, and regulated data environment.

Core concepts:

- Hubs: Represent core business entities (patients, healthcare centres, etc.) and store unique identifiers. Each hub contains only the natural unique identifier and metadata like load date and source information.
- Links: capture relationships between hubs, representing associations between entities. For instance, a healthcare consultation link might connect a patient to a healthcare provider and facility.
- Satellites: Store descriptive data or attributes of hubs and links, capturing history by adding records whenever data changes. For instance, the Patient satellite could track changes in patient attributes over time.

The technical architecture of the NHDW will broadly follow the Data Vault 2. methodology, which provides a comprehensive, scalable, and adaptable approach to data warehousing. The methodology features a 3-tiered modular data warehouse architecture, metadata-driven automation, and agile development for iterative development, making it highly adaptable to changing business needs and big data environments while ensuring full auditability and compliance with regulatory requirements.

Learn more about the methodology from the following official reference [19]: Dan Linstedt_ Michael Olschimke - Building a Scalable Data Warehouse with Data Vault 2.0-…

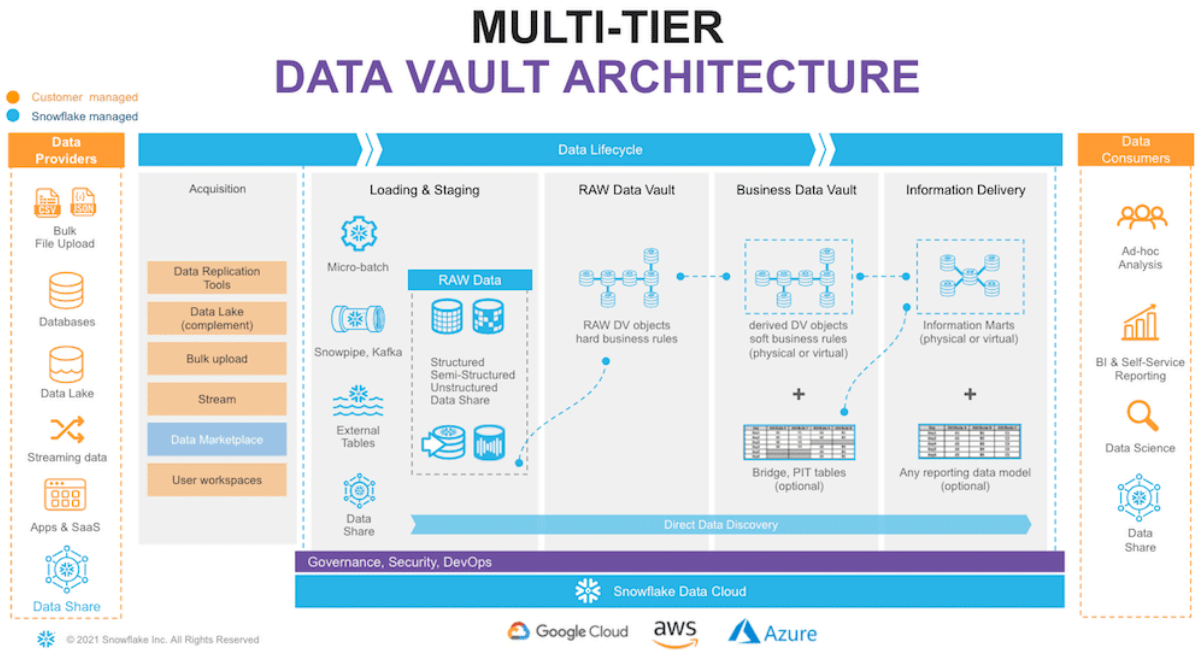


Figure 5: Data Vault 2.0 Architecture

2.7.4.1 Problems Data Vault 2.0 is expected to solve

Data Vault 2.0 is designed to address a range of challenges that traditional data warehousing models (like star schemas and snowflake schemas) often struggle with, especially in complex, large-scale, or frequently changing environments.

As seen in Table 4, the key problems Data Vault 2.0 aims to solve are:

Table 4: Challenges Addressed by Data Vault 2.0

Focus area	Problem	Solution
Managing Complex and Rapidly changing data sources	Health information systems increasingly work with complex, rapidly changing data from multiple sources. In traditional schemas, frequent changes in data requirements can result in complex modifications that are time-consuming and costly.	Data Vault 2.0 provides a highly adaptable structure with hubs, links, and satellites. This allows for simple modifications and additions without disrupting the existing data warehouse structure with minimal impact.
Data Integration from Multiple Sources	Integrating data from multiple sources can lead to inconsistencies, duplicates, and data conflicts, especially if each source has different definitions or identifiers.	Data Vault 2.0 enables a centralised and integrated approach by using hubs to represent unique identifiers and links to capture relationships. It accommodates diverse sources without the need to reconcile every discrepancy upfront, as historical versions are preserved.
Handling Historical Data and Data lineage	Many data warehousing models struggle with capturing and managing historical data and tracking changes over time, making it difficult to understand how data has evolved.	Satellites in Data Vault 2.0 store historical attributes, which makes tracking changes and retaining historical data straightforward. This historical data also supports regulatory compliance by preserving data lineage and audit trails, showing exactly how and when data was added or modified.
Scalability and Performance with Big Data	Traditional data warehouse models often struggle to scale effectively with the high volume, velocity, and variety of modern data environments.	Data Vault 2.0 is designed to scale efficiently in both storage and processing power. Its modular structure allows different parts of the model to be processed and loaded in parallel, supporting high-throughput data ingestion and rapid expansion to accommodate growing data volumes.
Decoupling of business rules from Data Warehouse model	Traditional data warehouses often encode business rules directly into the data model, leading to inflexibility. Changes in business rules can necessitate schema modifications, which are resource-intensive and time-consuming.	Data Vault 2.0 emphasises separating the raw data layer from business rules. Business rules are applied in downstream layers (e.g., data marts), allowing the core Data Vault structure to remain unaffected by changes in rules or calculations. This separation improves adaptability to evolving business requirements without altering the data warehouse structure.
Data Quality and Consistency across systems	Inconsistent or low-quality data can be a major challenge, especially when multiple systems contribute to	Data Vault's structure inherently supports data quality by preserving raw data from source systems in its original

	a data warehouse. Ensuring data quality while maintaining traceability and auditability can be difficult.	form, making inconsistencies traceable. The auditability and historical tracking also facilitate easy identification of data anomalies and corrections without overwriting historical records.
Flexibility in querying and reporting	Traditional star schemas and snowflake schemas are optimised for specific types of queries, but they can be inflexible when new requirements arise.	Data Vault 2.0's separation of concerns provides flexibility by allowing the raw data layer to serve a wide range of data marts for specific reporting needs. This means users can build star schemas or other structures on top of the Data Vault model to support specific reporting or analytical queries without altering the core structure.
Supporting Agile & Incremental development	Building a traditional data warehouse can be lengthy. resource-intensive project that may take years to deliver, which is incompatible with agile, iterative development approaches.	Data Vault 2.0 supports an agile, incremental approach to building a data warehouse, where new data sources and entities can be added gradually as business needs evolve. This accelerates delivery, enabling teams to produce value incrementally and adapt to changes over time.
Ensuring Data Governance and Compliance	Organisations are increasingly subject to strict data governance and compliance requirements, such as GDPR and HIPAA, which require transparency in how data is stored, processed, and accessed.	Data Vault 2.0 is designed with data governance in mind. Its built-in auditability, historical tracking, and separation of raw data from derived information allow for easy compliance with data retention and access control requirements, making it easier to trace data back to its source and understand its lifecycle.

2.7.5 Best Practice Reference Guides from Industry Leaders

- Data Analytics Lens: AWS Well-Architected Framework (amazon.com) [21]
- Big data architecture style: Azure Architecture Centre | Microsoft Learn [22]
- Firat Tekiner, Data Lakehouse Whitepaper (google.com) [23]
- Best practices for operational excellence | Databricks on AWS [24]

2.8 High-Level Business Requirements Approach

Defining the high-level business requirements is a key element of the development of a software system such as the NHDW and is fundamentally important to realising the principle of user-centred design and ensuring the system addresses real needs that will drive uptake and use.

Drivers:

Based on discussions with the Ethiopia MOH and NHDW Technical Working Group (TWG), the following initial drivers for the development of the NHDW were identified.

- Enhanced efficiency and quality health service delivery
- Equitable resource distribution
- Need for improved quality of data and reporting
- Evidence-based clinical and public health practices and research

The Ethiopian Health System Building Blocks, adopted from the World Health Organization:

The Ethiopian health system has six building blocks or key components:

1. Health service delivery (primary to tertiary)
2. Health workforce (ranging from production to retention)
3. Medical products, vaccines and technology Innovations
4. Information system: Surveillance system/M&E/Research
5. Governance and Leadership: Regulatory (facility registry, Healthcare resource management and licensing)
6. Healthcare financing

The Ethiopian and WHO health system building blocks are a framework for structuring a health system and can form the basis for efforts to design and monitor health system strengthening activities [25]. These are the important pillars of any healthcare system. The building blocks were used to guide the development of main subject areas for the NHDW to tie in with international guidance on priority areas for health systems strengthening and serve as a cross-cutting mechanism to improve all priority areas of the health system and monitor the progress.

2.8.1 Process to Define Subject Areas and Example Use Cases

Focus group discussions can be used to define broad use case areas. The corresponding data sources for the subject areas are then identified. An example output of a potential use-case for the NHDW is shown in the NHDW - Requirement Gathering Canvas - Health Service Delivery & Malaria Disease Control [26] below. The MoH Technical Working Group (TWG) will prioritise and define further use cases for the detailed development of the NHDW during subsequent phases of the development of the NHDW. See: Use cases NHDW_Dr Girma.pptx [27].

Use case-based development:

- Focus on specific business needs and analytical requirements
- Allows for quicker demonstrations of value
- Start with high-priority use cases and data sources
- Expand scope based on emerging requirements and demonstrated value

2.8.1.1 Categorization of the subject area according to the WHO health systems building blocks framework

The subject areas and use cases identified in focus group discussions are shown in the table below, classified according to the respective WHO building blocks.

Table 5: Subject Areas Mapped to WHO Health Systems Building Blocks

No.	Ethiopian/WHO Health System building block	Subject area use case examples
1.	Service delivery block (Strengthening as well as Primordial, Primary, Secondary and Tertiary prevention)	• Decreasing the prevalence of diseases (communicable, non-communicable, e.g., malaria and HIV). • Strengthening Service Delivery: e.g., MNCH, laboratory, blood bank, injuries, vaccines and accidents, intensive care. • Supporting Model Households: e.g., hygiene, sanitation and clean water-related indicators, wash, environmental health.
2	Human resource block (healthcare workers population in public, private, and in training)	• Optimising Human Resources: e.g., aligning the training of health professionals and job opportunities.
3	Governance and leadership block (Includes also policy)	• Regulatory: e.g., strengthening the master data registry and licensing (healthcare workers, private clinics, laboratories, medical devices, etc.).
4.	Finance	• Optimising Finances: e.g., community-based health insurance.
5	Medical products and laboratories	• Reducing Wastage: eg triangulating program and commodity data, stock status
6	Health Information Systems	• Surveillance, aggregate data sources (DHIS2, Surveys, SPA Plus, SARA), cuts across subject areas.

2.8.1.2 Criteria for use case prioritisation

Development of the NHDW is envisaged to be a long-term project with multiple levels and iterations in the development process. Therefore, it is important to prioritise the development of use cases to ensure short-term value and maintain buy-in to the project. The following initial list of criteria for prioritising use cases was developed to assist with this process:

1. Coverage: impact on the total population
2. Frequency of the event
3. Data availability
4. Source systems and maturity (quality of data availability)
5. Public health impact

6. Availability of resources (HR, finance, and skills)
7. Perceived benefit of centralisation for decision support.
8. Diseases prevalence

2.8.2 Illustrative use cases

Illustrative use cases are shown in Table 6 below [28]: Further use case identification and use case prioritisation will need to be conducted in subsequent phases of the NHDW development process.

Table 6: High-Level Business Requirements and Use Cases Mapped to Health System Building Blocks

High-Level Business Requirement	Use Case	Health system building block
1. BR1: To build an integrated health data platform where data from various sources is centralised and made accessible for data providers and administrators for use-analytics, triangulation, visualisation, reporting and decision making	Use case 1.1: Enhance data integration and analysis	Health information systems
2. BR 2: To improve the performance, efficiency, and quality of promotional, preventive, curative, and palliative health service delivery by utilising centralised data capable of producing complex queries and analytics for comprehensive data visualisation to inform decisions.	Use Case 2.1: Improve data-driven health service provision	Health service delivery
3. BR 3: To enhance human resource production, distribution, and retention through informed decisions using workforce data, including production, distribution, competency, and training across facilities, and retention by integrating the HRIS, DHIS2, MFR, and external sources data (HSC and universities), providing access to healthcare providers and policymakers.	Use case 3.1: Manage health workforce production, distribution and retention	Health workforce
4. BR4: To manage and monitor the supply chain of medical products, supplies, equipment, and technologies, ensuring effective procurement, distribution, consumption, and maintenance of the equipment across the health system.	Use case 4.1: Enhance pharmaceutical distribution and usage	Medical products and laboratories

5. BR5: Managing financial data related to health financing that includes budget allocation, expenditures, and community-based health insurance claims for transparent and efficient use of resources.	Use case 5.1: Improve healthcare financing management	Healthcare financing
6. BR6: Monitor the health system performance (finance, HRH, HIS, supply chain, and service delivery) and enhance compliance of health facilities and health workers for health policy and standards.	Use case 6.1: Enhance tracking health system performance and comply to protocols	Governance

2.8.3 Example Use Cases

During a NHDW envisioning workshop, members of the TWG developed two example use cases, malaria disease control and health service delivery, using a Business Requirements Development canvas. Further use cases for the NHDW will need to be selected and defined after the high-level requirements have been defined and prioritised.

2.8.3.1 Malaria Disease Control (Health Service Delivery Building Block)

Table 7: Malaria Disease Control

Subject Area	Business Needs	Expected Output / Deliverable
Malaria Disease Control	- Ability to follow the number of suspected Malaria cases tested - Indicate the number of new malaria cases (trend, per region, etc) - Malaria treatment outcomes - Malaria mortality rates - Malaria prevention indicators	A multi-line monthly chart showing: Total number of suspected cases, Suspected cases tested by microscopy, Cases tested by RDT (Rapid Diagnostic Test) Malaria test positivity rate Confirmed malaria cases Confirmed malaria cases rate per 1000 person years In-patient malaria deaths Proportion of population with malaria infection Proportion of population at risk potentially covered by distributed insecticide-treated bed-nets No. and proportion of the population receiving at least one round of indoor residual spraying in the last 12 months

Dimensions or Facts	Business Users / Stakeholders / Decision-makers	Source systems	Data Availability and Ingestion Methods	Meta Data
Dimensions: Time (monthly), Location (health facility, district, region), Test Type (microscopy, RDT) **Facts:** Number of suspected cases, Number of cases tested by microscopy, Number of cases tested by RDT, number of confirmed cases, no. of malaria-related deaths, number of bed nets distributed, number of indoor residual spraying conducted See for example, the Global Fund to fight AIDS, TB, and Malaria indicator guidance sheet Malaria Indicator Guidance Sheets	**Business Users:** Data analysts, Epidemiologists, researchers **Stakeholders:** Health facility staff, Ministry of Health data managers **Decision Makers:** Ministry of Health program managers and policymakers, M&E staff	DHIS2 KPIs EMR (Electronic Medical Records) Malaria Surveillance Systems Laboratory Information Systems Link survey data with regularly collected routine data	**Structured Data:** Numeric counts of malaria tests and results **Individual / Aggregated Data:** Aggregated at the health facility level ETL (Extract, Transform, Load) processes for monthly data consolidation from multiple health facilities Considerations of individual-level vs aggregate level data ingestion	**Backroom Metadata:** **Record Layout:** Attributes include facility ID, date of test, test result, test type **Content (Volumetric):** Monthly data volume per facility **Indexes:** Indexed by facility ID and test date **Scheduling:** Data loaded monthly **Usage & Dependencies:** Utilised in epidemiological analyses and healthcare planning **Referential Integrity:** Linked to facility master data **Process Execution Metadata:** Track ETL run-time, performance issues **Front Room Metadata:** **General Documentation:** Describes the purpose of each dataset (e.g., tracking malaria testing rates) **Business Metadata:** Explains the significance of the data for malaria control strategies **Technical Metadata:** Data types and lineage for audit and traceability

Business Rules	Master data management	Interoperability	Data Lifecycle Management	Data Transformation or Enrichment
Hard Business Rules: - Data must comply with health data privacy regulations - Data must be verified for accuracy before inclusion in reports **Soft Business Rules:** - Recommend regular data quality audits - Suggest monthly data review meetings with stakeholders	Master Facility List integration to harmonise data	ADX Support needed	Data Retention: Data retained for a minimum of 15 years before archival Archiving and Purging: Older data archived off-site, purged as per legal requirements	Potential Integration: Include environmental data (e.g., rainfall, temperature) that could influence malaria transmission rates Integrate malaria household survey data
	Quality Assurance	**Change Management**	**Scalability or Performance**	**Security & Privacy**
	Validation Checks: Ensure data matches reported numbers from facilities Anomaly Detection and Remediation: Identify and correct outliers in test counts	Manage changes through version control and regular updates to the data model and business rules	Data Volume and Load Assessment: Plan for increasing data volumes as testing expands Architectural and Data Ingestion Strategies: Robust infrastructure to handle peak loads	Access Controls: Limited to authorised personnel Encryption and Data Masking: Data encrypted in transit and at rest Regulatory Compliance: Compliant with national health data regulations Audit Trails: Implemented for tracking data access and changes

2.8.3.2 Health Service Delivery

See: NHDW_example1_Requirement Gathering Canvas—Health Service Delivery & Malaria Disease Control

Table 8: Health Service Delivery

Subject Area	Business Needs	Expected Output / Deliverable
Health service delivery canvas: ● This is one of the most comprehensive components of the health system that includes the delivery of promotional, preventive, therapeutic, rehabilitation, and palliative care to the people. ● According to the Ethiopian HSDIP (Health Sector Development Investment Plan), this component has the following use cases: Family planning and reproductive health, maternal health, Neonatal and Child health, Immunization (EPI), Adolescent and youth health, Nutrition, Disease prevention and control (HIV/AIDS, Malaria, TB, NCD and Mental health, NTDs), Community health extension program, engagement and environmental health; Emergency, Injury and critical care; Blood and Tissue services; Laboratory and Diagnostic services. ● Ethiopia provides the above health services in three tiers: primary, secondary, and tertiary health service delivery levels.	● The health service delivery in Ethiopia needs to be efficient, quality, safe, and equitable. There are issues that relate to health service coverage, utilisation, compliance with quality standards, medical errors, shortages of medicine, skilled professionals, and limited accessibility to services. ● To improve the performance monitoring in terms of service coverage and utilisation, minimise medical errors, increase access to specialised and quality care, and ensure equitable distribution of health resources, the health service delivery requires: ● Improved data quality (accuracy/validity, completeness, integration, timeliness, consistency, granularity, security, de-duplicated…) ● Data analytics (descriptive, predictive models, etc.) and Visualization on KPIs and clinical quality indicators (referrer to HMIS indicators reference guide and Clinical quality indicators) ● CDS (patient monitoring, reminder, alert, expert system, prompts, clinical guide or protocol)	● A data mart or warehouse that contains updatable, integrated, quality service delivery data ● Service delivery report (predefined) for KPIs ● Data visualisation for selected quality and performance indicators ● Querying tools for data analytics (for descriptive analysis and predictive models) ● APIs for data extraction ● OLAP cube ● Secured data access

Dimensions or Facts	Business Users / Stakeholders / Decision-makers	Source systems	Data Availability and Ingestion Methods	Meta Data
Identify key dimensions are: • Patient/client • Location • Sex • Facility type • Provider • Service type Facts: • Number of contraceptive acceptor • Number of pregnant mothers with ANC visits • Number of births with skilled birth attendants • Number of C/S done • Number of mothers who received postnatal care • Number of still/live births • Number children with zero vaccine dose • Number of children fully immunised • Number of children screened for nutrition • Number of individuals tested positive for HIV • Number of individuals newly initiated ART • Number of individuals virally suppressed • Number of individuals diagnosed with MDR-TB • Number of individuals who completed TB treatment. • Number of individual RDT test positive • Infant death rate	Users: Care providers, HITs, program managers, data managers and analysts, program heads and directors and executives Stakeholders: users, patients/clients, partners, donors, administrators. Outline who are the decision-makers.	• EMR (individual, but only few hospitals, Standalone) • ART Software (covers >90% of cases on ART, Standalone) • CBS Redcap (individuals, centralised) • Viral load and exposed infant test database (individuals, region level) • eCHIS (individual, in ⅔ of Health posts, centralised at national level) • Dagu/eAPTS (individual pharmaceutical item transaction records) • LIS (individual tests, stand alone) • RIS (individuals, semi-structured and unstructured data, in big hospitals) • DHIS2 (aggregate, centralised, highest geographic coverage) • External data sources (ESS, Disease prevalence estimates, Climate, Agriculture, Education) • Master data registries (client, facility and health workers registries)	• Structured, unstructured, semi-structured • Both Individual and Aggregated Data • Data Ingestion Method: Real-time, periodically	Include information about data governance and metadata management (e.g., data lineage, definitions). 1. HMIS indicators definition 2. NHDD: Terminology definitions 3. ESV ICD 11 4. Text, image, numbers

Business Rules	Master Data Management	Interoperability	Data Lifecycle Management	Data Transformation / Enrichment
Hard Business Rules: List the non-negotiable rules that govern the data processes (e.g., compliance requirements). ● Dates and time dimensions need to be converted to the Ethiopian fiscal year and HMIS calendar. ● Rates, proportion rather than counts (whenever appropriate) Soft Business Rules: Mention guidelines or best practices that are advisable but not mandatory (e.g., data quality thresholds). ● Recommend or suggest appropriate displaying charts	● Data Standardization: date formats, data element mapping, finalise and operationalize NHDD as terminology service, reinforce systems to use common data dictionary and format to capture store and share data ● Data Quality: Content completeness, medical record audits, imposing required fields, validation rules. ● Master Data Governance: No master data governance guide (we have only for MFR). ● Integration with Other Data Systems: Ensuring master data is synchronised across all systems—it is at the early stage.	● Consideration about clinical data translation and/or standardisation (e.g., HL7, FHIR) ● Required terminology mappings	Define policies for data retention, archiving, and purging to comply with legal and business requirements. This should include how long data is retained active before being moved to less costly storage options.	Consider the potential for integrating external data sources that can enrich the analytics capabilities, such as socioeconomic data, environmental data, or data from wearable health devices.
	Quality Assurance	Change Management	Scalability/Performance	Security & Privacy
	Detail the processes and tools that will be used to maintain data integrity, accuracy, and validity throughout the data pipeline. This includes validation checks, anomaly detection, and remediation processes.	Describe how changes to the source system data model, business rules, or any other significant component will be managed. This includes version control of the schema and tracking changes to data	● Any possible challenges to consider in terms of infrastructure and connectivity ● Storage and occupational resources, and there should be different arrangements	● Access Controls: Limited to authorised personnel ● Encryption and Data Masking: Data encrypted in transit and at rest ● Regulatory Compliance: Compliant with national

		structures.	for data pulling from centralised and standalone data sources ● Data Volume and load assessment: Not done so far, ● Architectural and data ingestion strategies to consider:	health data regulations ● Audit Trails: Implemented for tracking data access and changes

Table 9: Implementation Steps for Health Service Delivery

Implementation Steps			
Workshop Facilitation:	Review and validation:	Integration into Project Management:	Documentation and reporting:
Conduct workshops with stakeholders to fill out the canvas and ensure mutual understanding and agreement on the project scope and requirements.	Regularly review the canvas with all stakeholders to validate the requirements and make adjustments as the business environment or objectives evolve.	Integrate the insights from the canvas into project management and development lifecycles to ensure that the data warehouse/data mart solutions meet the specified needs.	Use the canvas as a living document that can be referred to throughout the project lifecycle for guidance and direction.

Table 10: Health Service Delivery Information System

Subject Area	Business Needs	Expected Output / Deliverable
Health Service Delivery Information System: ● This is a cross-cutting component of the health system that includes routine and non-routine health information systems, surveillance, and monitoring and evaluation of the health system that provide information and evidence to improve health outcomes, efficiency of the health system, and enhance quality and equitable health service provision. ● According to the Ethiopian eHealth architecture, this health information system encompasses all digital solutions that collect data from routine and non-routine sources. ● The Health information system includes DHIS2, EMRs, CBHIS, eCHIS, LIS, HRIS, MFR, Supply Chain and logistics management, health-related Surveys, census, and vital event registration	The Ethiopian health information system has several needs related to fragmentation, governance, scale-up, standardised data and exchange formats, data quality, interoperability, infrastructure, analysis, and use of information for decision-making in real time. Hence, the health information system requires: ● Integration and interoperability of system ● Improved data analytics capabilities ● Gross coverage of point-of-care applications ● Standardising health data and exchange formats ● Improved data quality and governance ○ Real-time data analysis and reporting, and triangulation ○ Informed clinical decisions	● A data platform, integrating different data sources ● Improved data governance and quality ● Enhanced data analytics capabilities ● Data visualisation for reporting and triangulation ● Querying tools for data analytics (for descriptive analysis and predictive models) ● APIs for data extraction ● OLAP cube ● Secured data access ● Monitored pharmaceutical consumptions ● Improved clinical data completeness and integration

Dimensions or Facts	Business Users / Stakeholders / Decision-makers	Source systems	Data Availability and Ingestion Methods	Meta Data
Identify key dimensions are: ● Data element/indicator for all health system components ● Location ● Time/periodic and continuous ● Gender ● Age group ● Subject area (disease, service, programs, survey, external source... ● Facility type ● Provider category ● Service type Facts: ● All data elements in HMIS ● Clinical quality indicators	● Users: program managers/experts, HITs, Clinicians, Data managers/analysts, program heads/directors and executives ● Stakeholders: users, patients/clients, partners, donors, health institutions, administrators. ● Outline who are the decision-makers: Lead executive officers, Agencies, higher officials, heads and program managers	● DHIS2 ● EMR ● ART Software ● CBS Redcap (individuals, centralised) ● Viral load and exposed infant test database (individuals, region level) ● eCHIS (individual, in ⅔ of Health posts, centralised at national level) ● Dagu/eAPTS (individual pharmaceutical item transaction records) ● LIS (individual tests, standalone) ● RIS (individuals, semi-structured and unstructured data, in big hospitals) ● External data sources (ESS), Disease prevalence estimates (Climate, Agriculture, Education) ○ Master Facility Registers (client, facility and health workers registries) ○ Vital event registration	● Aggregate or individual data ● Internal and external source ● Structured, unstructured, semi-structured	● HMIS indicators definition ● NHDD: Terminology definitions ● Product catalogue ● ESV ICD 11 ● Text, image, numbers

Business Rules	Master Data Management	Interoperability	Data Lifecycle Management	Data Transformation / Enrichment
Hard Business Rules: ● Dates and time dimensions need to be converted to the Ethiopian fiscal year and HMIS calendar. ● Rates, proportion rather than counts (whenever appropriate) ● Transformed data to a unified format ● Regular data ingestion from the sources Soft Business Rules: ● Recommend or suggest appropriate displaying charts	● Data Standardization: date formats, data element mapping, finalise and operationalize NHDD as terminology service, reinforce systems to use common data dictionary and format to capture store and share data ● Data Quality: pre-analysis data cleaning and preparation; check for content completeness; medical record audits; impose required fields; validation rules; and outlier detection tools. ● Master Data Governance: Develop a master data governance guide (we have only for MFR). ● Integration with Other Data Systems: Ensuring master data is synchronised across all systems (a tan early stage).	● Consideration about clinical data translation and/or standardisation (e.g., HL7, FHIR) ● Required terminology mappings ● Requires standards and shared health services ● Aggregate data exchange format—ADX Support needed		

	Quality Assurance	Change Management	Scalability/Performance	Security & Privacy
	Validation Checks: Ensure data matches reported numbers from facilities Anomaly Detection and Remediation: Identify and correct outliers in test counts		• Any possible challenges to consider in terms of infrastructure and connectivity—storage and occupational resources—and there should be different arrangement for data pulling from centralised and standalone data sources • Data Volume and load assessment: Not done so far, • Architectural and data ingestion strategies to consider:	• Access Controls: Define who can access what data and under what conditions. • Encryption and Data Masking: Methods used to protect data at rest and in transit. • Regulatory Compliance: Ensure adherence to GDPR, HIPAA, or other relevant standards. • Audit Trails: Mechanisms for tracking data access and changes.

2.9 Summary: Chapter 2

Chapter 2 provides an overall perspective of the development and implementation of an NHDW for Ethiopia, which includes a description of users and their characteristics, high-level guiding principles in the development of the NHDW, the documentation that will be delivered and available for the NHDW's users, high-level business requirements for the NHDW, and initial use case examples.

Users of the NHDW will include healthcare administrators, data providers, healthcare providers, data analysts, researchers, policymakers, IT and data management staff, and potentially the general public and patients.

The documentation that will be delivered and available for the NHDW's users will include a data dictionary, API documentation, technical specifications, use cases for the NHDW, a user manual, training materials, a troubleshooting guide, access and security guidelines, a governance framework, an implementation roadmap, change management procedures, a change log, requirements documentation and traceability, and a risk assessment and mitigation plan.

Guiding principles that will be adopted for the development of the NHDW include the Open Group Architecture Framework (TOGAF), Principles for Digital Development, a hybrid methodology that combines the top-down (Bill Inmon) and bottom-up (Ralph Kimball) approaches to development, the FAIR principles, the Data Vault 2.0 methodology, and the Health Data Governance Principles, which bring a human rights and equity lens to the use of data within and across health systems.

Identified drivers for the development of the NHDW include enhancing efficiency and quality health service delivery, equitable resource distribution, a need for improved quality of data and reporting, and enhancement of evidence-based clinical and public health practices and research. For Business Requirements, the Ethiopian Health Systems building blocks were identified as the primary guide for the subject areas for the NHDW, to tie in with international guidance on priority areas for health systems strengthening. These building blocks include six components, i.e. Health service delivery (primary to tertiary); the Health workforce; Medical products including vaccines and technology innovations; Information systems including surveillance systems and monitoring/evaluation and Research; Governance and Leadership including Regulatory aspects; and Healthcare financing.

An initial list of criteria for prioritising use cases for the NHDW include 1) Coverage (the impact on the total population); 2) the frequency of the event; 3) data availability; 4) source systems and maturity (quality of data availability); 5) the public health impact; 6) the availability of resources (HR, finance, and skills); 7) the perceived benefit of centralisation for decision support; and 8) diseases prevalence.

Two example use cases are detailed: Malaria Disease Control, and a Health Service Delivery use case.

Chapter 3: Data Source Requirements

Chapter 3 provides an overall analysis of the existing data sources identified for the development and implementation of an NHDW for Ethiopia. It includes a description of the main data source systems in use, an analysis of the core digital health system currently in use, and an analysis of the data source statistics, including data volume, data update frequency, data ingestion, data type, and data quality and standards. It also includes a section for data ownership and custody.

3.1 Identified Data Sources

The NHDW source systems span various sectors and healthcare domains. Some of these include Electronic Medical Records (EMR) systems, Laboratory Information Management Systems (LIMS), case surveillance systems, and community health management systems. Each of these systems operates at different levels of the healthcare infrastructure, ranging from the health facility level, woreda (district) level, regional level, to the central (national) level. The integration of these varied systems is crucial to creating a comprehensive and unified view of health data across the country.

In addition to the traditional health information systems, the data warehouse will also incorporate data from other critical sources such as national surveys, ESS (Ethiopian Statistical Services) weather services, and other relevant external data providers. This inclusion broadens the scope of the data warehouse, enabling it to capture a wide array of factors that impact public health, population, and disease estimates to compute indicators and identify individuals eligible for targeted health services. By collecting data from these multiple sources, the Ethiopia National Health Data Warehouse will facilitate more robust and nuanced analyses, supporting evidence-based decision-making and improving the overall effectiveness of health interventions and policies.

In addition, sometimes a source system could be a domain-specific Data mart (for example, electronic Regulatory Information System (eRIS), Vitas, mBrana, Dagu/electronic Auditable Pharmaceutical Transaction Services (eAPTS), Antiretroviral Treatment Dispensing Unit (ART DU management), Electronic Patient Management Information System (ePMIS), or from an HIE network such as the HIV NDR). Combining both approaches, “top-down” and "bottom-up,” will facilitate the data load operations.

3.1.1 Core Digital Health Systems in Ethiopia

Although many systems hold datasets that would be ingested into the NHDW, a few of these systems are regarded as core digital health systems. Here is a brief description of these systems. See Annexure Three for further details.

1. **Electronic Community Health Information System (eCHIS):** The eCHIS serves as a key component of the country’s successful health extension program. It captures electronic data on the Health Extension Program (HEP) and other community-level services, as well as utilising this data to improve HEP performance and community health outcomes and also serving as a job aid to the HEWs [29]. eCHIS is built on the CommCare platform (open source) that digitises the existing manual family folder and service workflows to record and report the households and members’ health and related individual-level data.
2. **District Health Information System (DHIS2):** The District Health Information System (DHIS2) is an open-source health information management system used for aggregate level evaluation at the district/woreda level. It has been developed with collaboration led by the

University of Oslo (UiO), and it is currently used as the platform for collecting and analysing health data for more than 20 program areas in separate instances. The MoH has extensively customised the DHIS2 software for various purposes, including the electronic National Health Management Information System (eHMIS), multisectoral nutrition, COVID-19 tracker, COVAX tracker, Woreda-based plan, Woreda transformation, activity budget tracking, and stakeholder mapping, among others [30].

The system captures aggregate-level indicators at health facilities and contains information for measuring primary care performance, using simple analytical methods, at national and regional levels and by facility type. Despite remaining data quality issues, DHIS2 is an important data source for generating health system performance assessment measures on a national scale.

3. **Electronic Medical Records (EMR):** The Ministry of Health is continuing to expand the implementation of the Electronic Medical Record (EMR) through acknowledgement of PulseCare and BahmniEMR as the national systems for scale-up. SmartCare is also an EMR utilised due to its robustness in environments with limited infrastructure [31]. It produces a number of reports and visualisations for reporting at facility level on important HIV program indicators.
4. **Unified Nutrition Information System for Ethiopia (UNISE):** The Unified Nutrition Information System for Ethiopia (UNISE) is a multi-sectoral nutrition data management platform built on DHIS2. UNISE collects, stores, and analyses nutrition data from various sectors at all administrative levels. By analysing this information, UNISE supports evidence-based decision-making and targeted interventions to decrease stunting.

   Key features of UNISE include integration with the national HMIS and DHIS2, capturing both nutrition-specific and nutrition-sensitive data, analysing data and generating reports for decision-making at all levels, enabling data triangulation among sectors for effective multi-sectoral coordination, and informing high-level systems like the National Information Platform for Nutrition (NIPN)
5. **Integrated Human Resource Information System (iHRIS)**: iHRIS is a computerised and integrated Human Resources Information System designed and developed to manage Human Resources for Health (HRH) data. It is a free, open-source information system and includes key functionalities such as management of the HRH data involved in health service delivery, tracking training and capacity building data for both in-service and pre-service sessions, and management of the health professional licensing and certification process.
6. **Supply Chain Management System**: There are currently different systems implemented at the Ethiopian Pharmaceutical Supply Service (EPSS) sites for supply chain management, such as Vitas, Dagu, MEMIS, mBrana, Fanos, NSC, and eAPTS . Below we list the properties of these systems [32].
    a. Vitas is a locally developed, open-source, custom software platform designed to support warehouse management, inventory control, and logistics management information. Vitas is used by EPSS to ensure that each facility has the commodities required to provide high-quality services. Vitas controls inventory of health commodities, from receipt to issue, and manages movement and storage at EPSS centres and hub warehouses. It allows effective demand planning, resource management, warehouse management, and inventory control of health commodities. Vitas is interoperable with downstream supply chain tools such as Dagu, a warehouse management tool at facility level, and mBrana, a vaccine management tool at woreda (district) health office level.

b. Dagu is a compound software solution designed to manage supply chain functions at service delivery points. It includes an inventory management system at the facility store level and a patient service management system at the dispensing unit (eAPTS).

   Over 1294 sites have implemented the Dagu system, and the DAGU 2 Store National Dashboard allows to visualise and monitor stock status reports at the HF level, enabling informed decision-making at the Ministry and Regional levels.

   Dagu enables submission of an electronic report and requisition form (eRRF). The eRRF removes the need to physically visit an EPSS hub to report on stock and order. It is interoperable with electronic auditable pharmaceutical transaction services (eAPTS), a dispensary unit application. The integration enables intra-facility visibility of stock status (e.g., the pharmacist can see what is being prescribed and its availability within the site's store).

c. MEMiS is web-based medical equipment management software that operates online. It is implemented in 140 health facilities with more than 15,000 medical equipment registered by line items and more than 80,000 cold chain equipment imported to the system.

d. mBrana is an open-source mobile software platform designed to manage inventory such as vaccines, bed nets, and other commodities at the district (Woredas) and facility level. It is implemented in 570+ Woredas. It is fully integrated with Vitas at central and regional/hub levels, providing visibility into stock status all the way down to woredas as well as into the quantities of vaccines issued to health facilities. mBrana generates electronic vaccine request forms (eVRF), enabling a rapid submission of supply requests electronically.

e. Fanos is a dashboard used at the EPSS centre and its hubs that allows supply chain data visibility in real time through visual presentation of selected supply chain indicators. It supports procurement, supply, and distribution decision-making for all commodities managed by EPSS. A web-based dashboard that displays Vitas data to guide supply chain performance monitoring. Sample metrics monitored include: stock on hand, months of stock, average monthly consumption, expiries, and pipeline. Fanos is also available as an Android application.

f. National supply chain Dashboard (NSC) is a comprehensive data visibility platform designed to showcase key performance indicators such as Regulatory, Supply chain (EPSS, Facility), Pharmacy Service, Medical Equipment, Local Manufacturer and Traditional Medicine.

g. Currently, MoH is visualising key stock procurement-related data, including procurement status, procurement lead time, stock status, potential to expiry, wastage status, registered products, imported amount, and product registration time.

h. eAPTS is a facility-level dispensary tool designed to facilitate the transaction during medication dispensing and management at the dispensary unit (pharmacy/medical supply site) of a health facility. Prior to the deployment of eAPTS, manual payment processing took longer, shortening patient counselling time. The implementation of eAPTS reduced patient waiting time, enabling longer counselling sessions.

7. **Master Facility Registry (MFR)**: The MFR database captures vital information about each facility, including its name, location, services, infrastructure, and human resources. The MFR serves as a single source of truth for facility data within the national digital health system.

Approximately 80% of the health facilities have had their service domains populated in the MFR.

8. **Laboratory Information Systems (LIS)**: A few LIS are used in Ethiopia to manage and streamline laboratory processes and data [33]. Helping to automate the collection, storage, and analysis of laboratory information. Some of these include:
    a. Polytech LIS (COMP Promed Inc.): At least 25 laboratories in Ethiopia have installed the proprietary electronic Polytech Laboratory Information System (LIS), which is interoperable and networked with DHIS2. The Polytech LIS expedites patient records and automates data collection and reporting. Polytech sends health care providers timely information and simplifies reporting for lab technicians.
    b. Laboratory Data Manager (LDM) (National Technology): LDM is a proprietary laboratory information system that covers all functional areas of multi-speciality laboratories.
    c. LabXpert: LabXpert is a proprietary software solution from Med X International that reports the results of GeneXpert testing for tuberculosis diagnosis and to identify treatment resistance.
9. **AACAHB Central Data Repository for HIV Data**: The Central Data Repository (CDR) installed at the Addis Ababa City Administration Health Bureau (AACAHB) for HIV data management centralises HIV data from >65 HIV treatment facilities in the city. The data centralisation and integration platform uses open-source software, adhering to the Open Health Information Exchange (OpenHIE) architecture specification and employing the Fast Healthcare Interoperability Resources (FHIR™) standard by HL7.
10. **Ethiopia HIV NDR:** The CDR implemented in AACAHB needs to be enhanced and expanded to the other regions and to the national level. Currently there are some efforts in place to reutilise the CDR technology, enhanced and expanded to an HIV NDR system. The priority use case of the HIV NDR is to provide ART reporting for operational tracking of patients and performance of the health centres. Either the AACAHB HIV CDR or the HIV NDR may be ingested as a data source. Ingesting both data sources would lead to data duplication.
11. **Digital Health Project Inventory:** The Digital Health Project Inventory (DHPI) serves as a central digital clearinghouse and certification platform for managing the inventory of digital health systems and solutions in the country. The DHPI acts as a centralised repository for registering and cataloguing all digital health projects.

3.1.2 External data source

The Ethiopian MOH published the Ethiopian National Health ICT Infrastructure, which provides a high-level categorisation for every data source to be used by the ministry. One of the high-level eHealth Architecture eHA categories is called external systems.

Major external data sources to the NHDW include the following:

1. **Ethiopian Statistical Agency:** The Central Statistical Agency of Ethiopia collects various types of data, including:
    - **Demographic and Social Data** (population size, growth rate, age distribution, gender ratio, and household characteristics)
    - **Economic Data** (GDP, employment, inflation, production, and trade statistics)

- **Agricultural Data** (crop areas, yields, livestock numbers, and farming practices)
- **Environmental Data** (natural resources, deforestation, and pollution levels),
- **Health Data** (disease prevalence, health services utilisation, and mortality rates)
- **Education Data** (enrollment rates, literacy rates, and educational attainment),
- **Infrastructure Data** (transportation networks, housing conditions, and access to utilities)
- **Poverty and Living Conditions** (income levels, poverty rates, and living
- **Trade and Industry Data** (export and import volumes, industrial production, and business activities)
- **Social Indicators** (crime rates, social cohesion, and community development)
- **Armauer Hansen Research Institute** collects genomic, clinical training and disease related data.

These data are collected through various methods such as surveys, censuses, and administrative records to support policy-making, planning, and monitoring in Ethiopia.

2. **Ethiopian Agricultural Research Institute:** The Ethiopian Agricultural Research Institute (EARIN) collects various types of data to support its research and development activities in the agricultural sector. Here are the key types of data they collect:
    - **Crop Performance Data**: Information on the yield, quality, and productivity of different crops, including staple grains like maize, wheat, and teff, as well as horticultural crops and vegetables
    - **Livestock Performance Data**: Data on the health, growth, and productivity of livestock, including cattle, sheep, goats, poultry, and fish
    - **Soil and Environmental Data**: Information on soil fertility, pH levels, nutrient content, and other environmental factors that affect agricultural productivity,
    - **Pest and Disease Data**: Data on the prevalence, incidence, and impact of pests and diseases on crops and livestock, including information on pest management and control strategies
    - **Water Resource Data**: Information on the availability, quality, and utilization of water resources for irrigation and other agricultural purposes
    - **Climate Data**: Data on weather patterns, climate trends, and extreme weather events that affect agriculture
    - **Technology Adoption Data**: Information on the adoption and impact of new agricultural technologies, including improved seeds, fertilizers, pesticides, and farming practices
    - **Economic and Market Data**: Data on market prices, demand, and supply of agricultural products, as well as information on farmer income and livelihoods,
    - **Socio-Economic Data**: Information on the social and economic factors that influence agricultural production and rural livelihoods, including education, income, and access to markets
    - **Impact Evaluation Data**: Data on the impact of research projects and interventions on agricultural productivity, food security, and rural development.

    This data is collected using a tool like ODK or KOBOTOOLBOX and collects data on a quarterly and annually basis through various methods such as field experiments, surveys, monitoring programs, and collaborations with other research institutions and stakeholders to support evidence-based decision-making and policy formulation in the agricultural sector.

Rdata, excel and csv are some of the data formats found in this organisation.

3. **The Ministry of Education (MOE):** in Ethiopia's Alumni Information Management System collects various types of data related to alumni of Ethiopian educational institutions. Here are the key types of data they collect, along with information on data format, frequency of data collection, and data collection equipment size:

- **Personal Information**
    - **Data Format**: Structured data (e.g., databases, spreadsheets)
    - **Frequency**: As needed (e.g., when alumni update their information)
    - **Equipment Size**: Varies depending on the size of the database
- **Academic Records**
    - **Data Format**: Electronic transcripts, digital records
    - **Frequency**: Ongoing (updated as new data becomes available)
    - **Equipment Size**: Large datasets depending on the number of alumni and institutions
- **Career Path**
    - **Data Format**: Structured data (e.g., databases, spreadsheets)
    - **Frequency**: As needed (e.g., when alumni update their career information)
    - **Equipment Size**: Varies depending on the size of the database
- **Achievements and Awards**
    - **Data Format**: Structured data (e.g., databases, spreadsheets)
    - **Frequency**: As needed (e.g., when new achievements are reported)
    - **Equipment Size**: Varies depending on the number of alumni and awards
- **Networking and Connections**
    - **Data Format**: Structured data (e.g., databases, spreadsheets)
    - **Frequency**: As needed (e.g., when new connections are formed)
    - **Equipment Size**: Varies depending on the number of alumni and connections
- **Feedback and Surveys**
    - **Data Format**: Structured data (e.g., databases, spreadsheets), unstructured data (e.g., text responses)
    - **Frequency**: Periodic surveys, as needed
    - **Equipment Size**: Varies depending on the number of surveys and responses
- **Donations and Contributions**
    - **Data Format**: Structured data (e.g., databases, spreadsheets)
    - **Frequency**: As needed (e.g., when new donations are received)
    - **Equipment Size**: Varies depending on the number of donations and contributors
- **Alumni Events and Activities**
    - **Data Format**: Structured data (e.g., databases, spreadsheets), unstructured data (e.g., event descriptions, photos)
    - **Frequency**: Ongoing (planned and recorded events)
    - **Equipment Size**: Varies depending on the number of events and activities
- **Reunion and Homecoming Data**
    - **Data Format**: Structured data (e.g., databases, spreadsheets), unstructured data (e.g., event descriptions, photos)

        - **Frequency**: As needed (e.g., when reunions are planned)
        - **Equipment Size**: Varies depending on the number of reunions and participants
- **Impact and Outcomes**
    - **Data Format**: Structured data (e.g., databases, spreadsheets), unstructured data (e.g., impact reports, testimonials)
    - **Frequency**: Periodic assessments, as needed
    - **Equipment Size**: Varies depending on the number of impact measures and outcomes.

      The size of the data collection equipment (e.g., databases, servers) depends on the volume of data collected and stored. The frequency of data collection varies based on the specific needs and updates required for each type of data.

4. **The Ohio State University's Animal Health Survey** collects a variety of data related to the health and well-being of animals. Here are the key types of data they collect:
    - **Animal Health Status**
        - Information on the overall health condition of animals, including the presence of diseases, injuries, and other health issues.
    - **Disease Incidence and Prevalence**
        - Data on the occurrence and spread of diseases among animals, including the number of cases and the prevalence rate of specific diseases.
    - **Treatment and Control Measures**
        - Details on the treatments administered to animals and the effectiveness of these measures in controlling or eradicating diseases.
    - **Farm Management Practices**
        - Information on the practices adopted by farmers and farm managers to maintain animal health, such as vaccination schedules, biosecurity measures, and sanitation practices.
    - **Economic Impact**
        - Data on the economic impact of animal health issues on farms, including costs associated with treatment, loss of productivity, and other financial implications.
    - **Behavioural and Welfare Indicators**
        - Observations and measurements related to animal behaviour and welfare, including signs of stress, discomfort, or other welfare concerns.
    - **Environmental Factors**
        - Information on environmental conditions that may affect animal health, such as temperature, humidity, air quality, and access to clean water and feed.
    - **Veterinary Services Utilisation**
        - Data on the use of veterinary services, including the frequency and types of services accessed by farmers and farm managers.
    - **Biosecurity Measures**
        - Details on the biosecurity practices implemented to prevent the spread of diseases, such as quarantine protocols, movement restrictions, and hygiene practices.
    - **Surveys and Questionnaires**

- Responses from farmers and farm managers to surveys and questionnaires designed to gather information on various aspects of animal health and management practices.

These data are collected through various methods such as surveys, interviews, direct observations, and digital platforms to support the effective management and improvement of animal health by the Ohio State University's project.

5. **The Addis Ababa University Cancer Registry** is a database that collects information about cancer patients treated at the university's hospital. The registry gathers data on various aspects of the patients' health, including their demographics (such as age and gender), the type of cancer they have, and the stage of the cancer at the time of diagnosis. The registry also collects data on the prognosis of the patients, which refers to the likely course and outcome of the disease. This includes information on the treatments received and follow-up data on the patients' health status after treatment. The data collected by the cancer registry is for purposes. It is used for research purposes to study the incidence and prevalence of different types of cancer, as well as to investigate the effectiveness of various treatments. The registry data is also used to monitor cancer trends in the population and to plan and evaluate cancer prevention and control programs.
   The data collected includes:
   - **Patient demographics:** This includes information such as the patient's name, date of birth, gender, and address.
   - **Medical history:** This includes information on the patient's previous illnesses, surgeries, and treatments.
   - **Cancer diagnosis**: This includes information on the type of cancer, the location of the tumour, and the stage of the cancer at the time of diagnosis.
   - **Treatment:** This includes information on the type of treatment received, such as surgery, chemotherapy, radiation therapy, and other treatments.
   - **Prognosis and follow-up:** This includes information on the patient's response to treatment, the progression of the disease, and the patient's overall health status.

   The data is collected in a standardised format using data collection tools such as case report forms. The frequency of data collection depends on the needs of the registry and the availability of resources. In general, data is collected at the time of cancer diagnosis and at regular intervals during follow-up.

   The size of the data collection equipment depends on the volume of data that needs to be collected and stored. The registry may use electronic data management systems to store and manage the data, which can vary in size depending on the needs of the registry.

6. **The Ethiopian National Identification Authority** (NIA) is responsible for collecting and managing biometric data for the issuance of national identification cards. The data collected by the NIA includes personal information such as the individual's name, date of birth, gender, and address, as well as biometric data such as fingerprints, facial recognition data, and iris scans. A digital photograph of the individual is also collected. The data is collected in a digital format using biometric data collection equipment such as fingerprint scanners, facial recognition cameras, and iris scanners. The frequency of data collection depends on the needs of the NIA and the availability of resources. In general, data is collected at the time of application for a national identification card and during periodic updates to the biometric data. The size of the data collection equipment depends on the volume of data that needs to be

collected and the level of security required. The NIA uses high-quality biometric equipment to ensure the accuracy and reliability of the data collected. The equipment may vary in size depending on the specific needs of the registry.

Overall, the Ethiopian National Identification Authority plays a crucial role in providing a secure and reliable system for identifying individuals in Ethiopia. The collection of biometric data ensures that the national identification cards issued by the NIA are secure and difficult to forge, which is important for various applications such as access to government services, various databases, banking, and voting.

7. The **Ethiopian Meteorological Institute** (EMI) is an institute of the Ministry of Health that collects climate data in Ethiopia and provides short, medium and long term weather and climate forecasts. One of EMI's strategic objectives is to diversify climate application products to the basic developmental sectors, such as the Health sector (among others). EMI provides continuous monitoring services of important meteorological and climate data in Ethiopia and stores this data in its data centre in Addis Ababa. This includes data that may affect health, including temperature and humidity indexes, and expected mosquito breeding suitable geographic areas. These data may be combined with health data in a climate-health data integration use case, which may include health risk prediction models. EMI publishes a monthly Health-Meteorology Bulletin that contains essential Climate Information for the Health Sector. The bulletin is an important source of information for Health professionals monitoring Public Health, tourists regarding the comfort conditions of the places in Ethiopia and researchers in the field of Bio-Climatology. The bulletin conveys essential information regarding the monitoring of human comfort conditions based on the analysis of temperature and humidity data and also for the monitoring of Malaria outbreak areas based on the analysis of temperature and precipitation data. Since the monitoring of temperature and rainfall over a given area can be used to assess the likelihood of an outbreak of Malaria with a lag of two months, this information can be an important early warning tool if used judiciously. SSGI could work collaboratively with the MoH in several areas:
    - Development of robust geo-referencing and geo-coding systems to assist with geographic information systems and spatial analysis within the health system.
    - Contribute routine meteorological and climate data to integrate with health data to include weather and climate perspectives in the analysis of data and predictive analytics
    - Collaborate in the development of research methods and conduct reseacrhn on integrated health and climate data sets.
    - Work with health professionals on the representation and visualisation of integrated health and climate data sets.

8. The **Armauer Hansen Research Institute** (AHRI) is a biomedical research institute working in health areas including tuberculosis, HIV, Malaria, Neglected Tropical Diseases, Vaccine development, noncommunicable diseases as well as training and research capacity building. The Institute has a number of research programmes,

including a Knowledge Management Research Directorate (KMRD) and a Biotechnology and Bioinformatics Research Directorate. Its research activities are divided into two parts, one part working in biomedical research and the other part in public health research. AHRI collects and curates a substantial corpus of research data and is presently commissioning a new data centre to assist with managing research data. The KMRD directorate is working to bridge the know- do gap and influence the culture of knowledge translation to enhance evidence-based practices of the health services and health care system of the country and beyond. Within the KMRD there are three divisions, namely Data Science and Management, Knowledge Translation, and One Health. There are many potential collaborations between AHRI and the MoH, including the following:

- Integration of routine and research data sets from routine sources, DHS sites and research databases into the main data warehouse to increase representation and triangulation of data.
- Collaboration on the representation of data elements and potentially use of research datab sets such as the Observational Medical Outcomes Partnership (OMOP) Common Data Model (CDM).
- Training in data science and bioinformatics applied to health data
- Collaborative skills development in data science, standandards and adavanced analytics

9. The **Ethiopian Space Science and Technology Institute** (SSGI) is an autonomous federal institute in Ethiopia that aims to develop multidimensional uses of space science and technology in the space and geospatial sectors and be a robust contributor for national development. It offers essential support, guidance, and leadership in implementing the national space policy and strategies. It also offers research, training, consultancy services and production of space and geospatial products and services as well as the advancement of space and geospatial infrastructures and technologies. SSGI maintains significant data related to its mandate and functions using a data centre in Addis Ababa. SSGI could work collaboratively with the MoH in several areas:
    - Development of robust geo-referencing and geo-coding systems to assist with geographic information systems and spatial analysis within the health system.
    - Collaborate on the development and implementation of spatial analysis of geo-referenced data sets.
    - Potentially host databases and data warehouse within its spacious data centre in Addis Ababa.

3.2 Data Source Statistics

This section gives insight into the nature and properties of the data sources identified during a rapid data source assessment and desk review of source data systems.

Over the course of the NHDW Project's envisioning phase, we identified a list of 83 data sources, which are listed in Annexure 3. The list provided is by no means an exhaustive list. However, it

provides a satisfactory basis for important data sources that the MOH considers essential. For this list of source systems we are able to identify which types of systems are point-of-service systems and may have more immediate data syncs; those that are institution-based services and therefore are necessary for optimal management and governance of healthcare systems; those that are population-based HIS & data sources are essential for understanding health trends and outcome evaluations; and the external systems, which are essentially cross-sectoral and may have unique data sharing requirements.

The bar graph of Data sources by eHA category depicts that amongst the data sources identified, the majority are point-of-service systems. The MOH has also been developing a number of additional shared services, as you can see highlighted as blue in the eHealth Architecture.

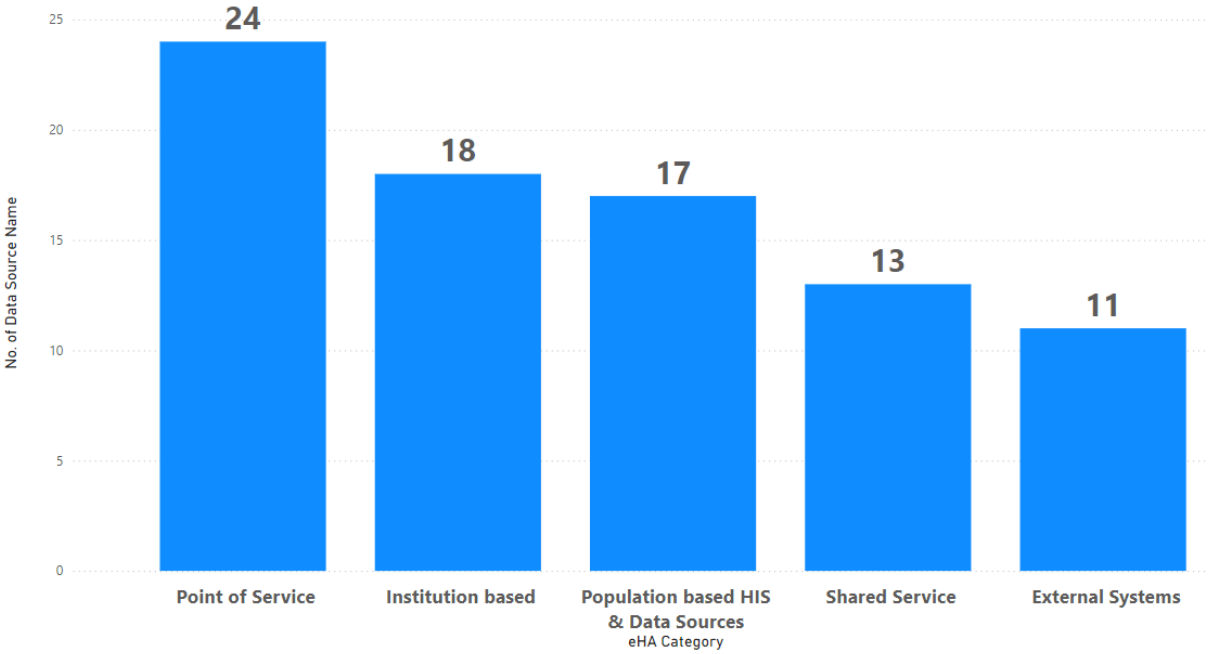


Figure 7: Data sources by eHA Category

From the data gathered, ~50% of the data sources are considered point-of-service and institution-based data sources. Also, eight of the ten core digital health systems highlighted in preceding sections belong to these two categories. This shows how integral these systems are to the eHealth landscape. In Figure 8: Data Sources by eHA Category, the percentages diagram gives a better view of proportions across the five categories.

For the data sources identified, ~45% were identified as individual-level data sources, ~22% were identified as aggregated data sources, and the rest (~38%) of the data sources were unknown. eCHIS is a major data source that is captured at an EMR and community level as individual-level data. Bahmni hospital EMR, SmartCare EMR, the Addis Ababa Central Data Repository for HIV data, the HIV case-based surveillance system using RedCAP, the Polytech Laboratory Information System, the Laboratory Data Manager, and LabXpert are further information systems that capture and utilise individual-level data. The HIV National Data Repository (in development) is also an individual-level system that is centralised and used to provide critical indicators to assess the disease's status nationally. DHIS2 is an aggregate-level reporting system that is well referenced and utilised within the ministry nationally. eCHIS, SmartCare EMR, the HIV case-based surveillance system, and LabXpert are also implemented nationally. Bahmni EMR and the Laboratory Information Systems are implemented in local settings only. A detailed description of each digital health solution, along with its data properties, is described in Annexure Three.

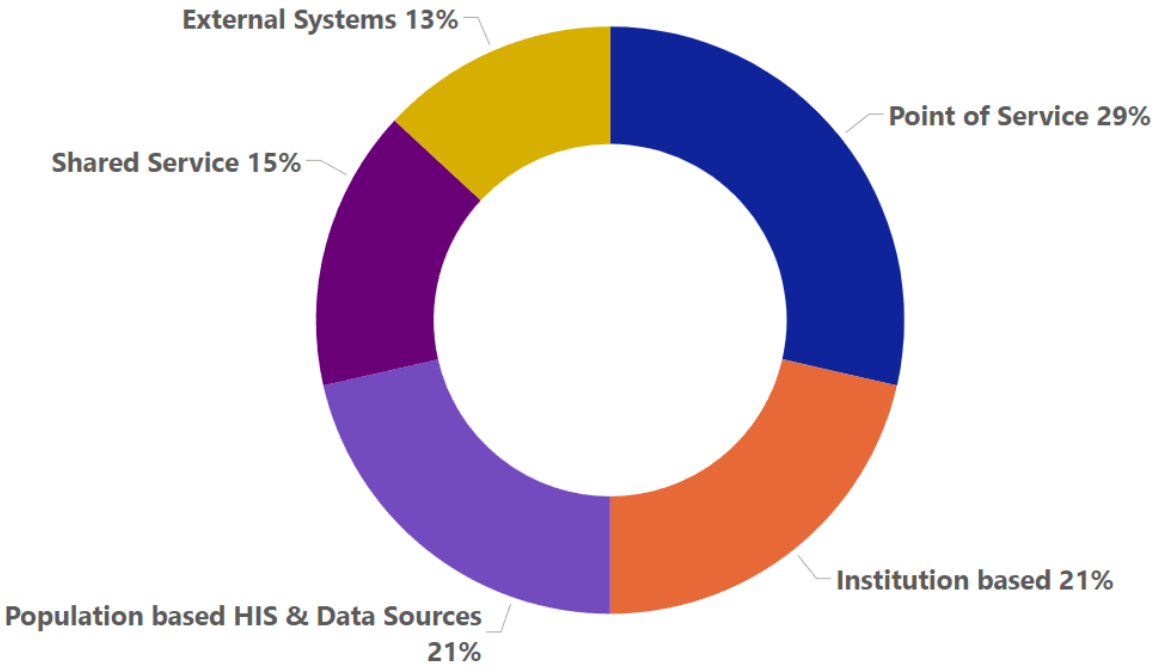


Figure 8: Data Sources by eHA Category In Percentages

3.2.1 Data Volume

During our analysis of the 83 data sources identified, we realised that the largest data volumes came from only a few data sources, some of which include eCHIS and DHIS2. For example, DHIS2 and eCHIS accounted for about 70% of the data volumes coming from the MOH data centre. Considering eCHIS has over 6 million households and 25 million individuals registered and utilises a suite of CommCare applications for Health Extension Workers (HEWs), their supervisors, and health facility-level users, which acts as an integrated primary health care service delivery and reporting tool for the health extension program (HEP), these observations are not unexpected [29]. The total storage size provisioned for eCHIS2 applications across the MOH and Ethiotelecom Data Centres is 55.2 TB. However, the actual data size for eCHIS is estimated to be 2 TB and is expected to double to 4 TB by 2030.

The Ethiopian DHIS2, on the other hand, is widely used within the ministry and at RHBs; it contains regularly submitted data, typically monthly, by up to 30,192 health facilities (which includes health posts) in the country. DHIS2 data volume currently is around one billion records and is estimated to grow to two billion records by 2030. To cater for the current data volumes, a total of 24 TB was provisioned on the MOH Data Centre for DHIS2 applications. The system includes reports on health service activities and the conditions for which people seek care.

For major applications, as seen in eCHIS and DHIS2 data growth trends, data size is expected to double by 2030 for most applications. Hence, the MoH has made significant investments in terms of catering for an increase in data size, including increasing physical server capacity to 7 machines and over 40 virtual servers, deploying converged infrastructure with 8 blade servers and 130+ TB of storage, installing two new air conditioning units for enhanced cooling, and implementing modern monitoring tools for application health.

The ministry has also subscribed to computing resources on Ethio Telecom's Ethiocloud platform, which provides scalable hosting and high availability for these systems, including approximately 1,700 core CPUs, 3,800 memory units, and 150 TB of storage.

3.2.2 Data Update Frequency

Over 60% of the data sources identified have real-time and near-real-time updates from their respective applications. There are some data sources that are updated every ten years, for example, the census data from the Ethiopian Statistics Service, or annually, like DHIS2-WBP; however, the majority of the data source providers also have data in real-time or near-real-time cadences. This means that in considering ETL pipelines that best suit the needs of the Ministry and data source providers, although batch loads may be prioritised as is common within the data analytics and warehousing context, ingestion pipelines that can cater for shorter cadences like real-time and near-time needs should also be considered.

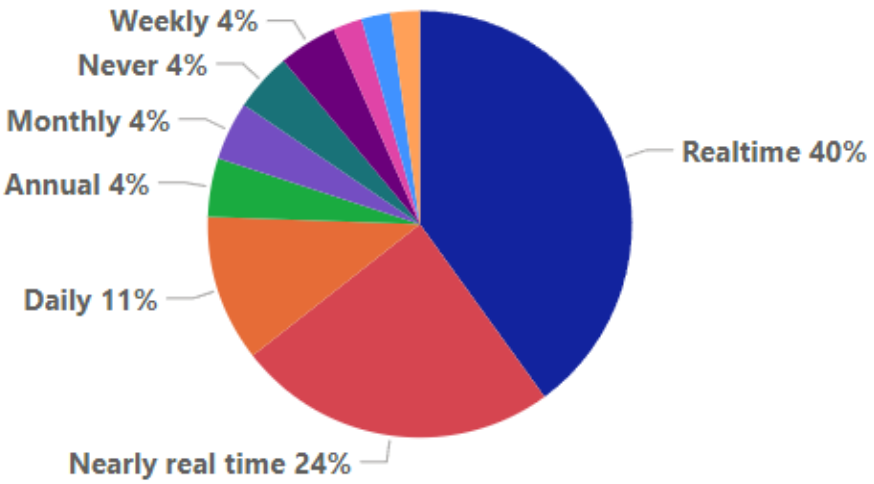


Figure 9: NHDW Data source update frequency

Although the list of 85 data sources is not an exhaustive list, the considerable efforts taken to acquire information on these data sources provide a guiding principle as to the types of data sources to expect and how to cater for them within the NHDW architecture. During implementation, the frequency and rate of data ingestion may change to suit processing needs. For example, data sources previously specified as near-real may be able to send data on a weekly or daily schedule. This is where the information provided can act as the maximum data update frequency requirement for each of the data sources specified. Below we show the illustrative image of an OLAP-cube slice corresponding to DHIS2.

DHIS2 is able to support indicator data across various types of disease indicators. From the illustration image of an OLAP-cube displayed, we can see support for daily data across all disease data from health provider, health centre, primary healthcare unit, district, zonal health department, regional health department, and national. DHIS2 can support weekly and monthly data cadences as well.

3.2.3 Data Ingestion

Regardless of the volume, cadence, or format of the data source, these datasets need to be ingested into the data platform. However, the way these datasets are ingested would differ due to their variations in volume, cadence, format, and transmission requirements. The majority of data sources that are to be ingested into the NHDW are structured data types. These data types easily find home in a relational database; however, considering semi-structured and unstructured datasets as well, the storage types for these sources can not be only relational. Some of the data types considered include flat files, video, images, json-formatted files, and relational data.

Datasets also vary in terms of sensitivity and confidentiality; a high amount of data stored in the warehouse is termed to be highly confidential, with 79% of the dataset anticipated to contain personal information. In addition, the nature of health data is one that requires levels of privacy (both fine-grained and coarse-grained access control) and patient consent before data is shared with third-party institutions. These requirements therefore affect the ingestion of these data sources. Methods for data ingestion, in addition to supporting different cadences, also need to support encryption or data de-identification where needed. The platform tools would also be required to secure data at rest or ensure SSL when data is in transit during ingestion.

Data transmission is also an important consideration during ingestion. We identified two clearly viable options to send data into the data platform: the VPN and the internet. Two VPNs have been identified within Ethiopia for data transmission, SD1 and HealthNet. The most well-known is called HealthNet. MoH and Ethiotelecom [34] are working closely in strengthening the HealthNet and other network-related services. The HealthNet VPN serves as the backbone for connecting various digital health systems and enabling remote access to patient data and clinical applications across healthcare facilities. However, there are limitations in the network's visibility and reliability that pose significant challenges in ensuring uninterrupted service delivery.

It is best practice across industry leaders in cloud and data infrastructure development that any robust data platform should accommodate the diverse communication protocols utilised in public healthcare systems. “They are expected to encompass both bulk data transfers and real-time data streams. This includes secure FTP for bulk transfers, HL7v2 over MLLP, and standard FHIR web services.” Also, legacy protocols that lack encryption must operate over a secure channel, such as a site-to-site VPN.

During the data source assessment, we also identified that 19 applications were able to confirm the existence of an API for data extraction; 16 confirmed not to have this functionality, while the rest were unknown. Majorly, data sources also support data transmission through FTP, via the internet, or a VPN securely. Registry services like the MFR specifically are expected to support data exchange as well.

3.2.4 Data Quality & Standards

Some of the Core Digital Health systems mentioned in the preceding sections implement health data standards necessary for interoperability and data exchange. One such example is the HIV NDR and AACAHB CDR. This system adheres to the Open Health Information Exchange (OpenHIE) architecture specification and employs the Fast Healthcare Interoperability Resources (FHIR) Standard by HL7 [35]. The FHIR standard provides data interoperability across systems. The CDR is among the largest operational data repositories in low- and middle-income countries to use FHIR standards for data interoperability. The CDR receives the majority of its data from SmartCare instances, which ensures that the data that is sent is in the compatible FHIR format. This configuration is also in place within the HIV NDR, ensuring standardisation across SmartCare instances, the AACAHB CDR, and the HIV NDR.

Similar to SmartCare instances that support the FHIR standard, BahmniEMR allows clinicians, researchers, & hospital administrators to use SNOMED CT as a reference for clinical terminology, providing consistency to downstream consumers of these applications. This is realised through standardisation of patient responses, therefore improving exchange with other EHRs. Within BahmniEMR itself, it improves the recording of diagnosis & other clinical data, as well as for reporting and Clinical Decision Support (CDS) use-cases. Bahmni has also integrated open source Snowstorm Terminology Server over FHIR. This allows for directly recording patient clinical data with SNOMED CT terms. Data can be exported in FHIR format with SNOMED CT-coded terms [36].

eCHIS is the largest data source (total application size is 55.2 TB, estimated data size is 2 TB) stored in the MOH data centre. Although it operates offline, it needs connectivity for data synchronisation and to facilitate digital referral linkages. It has the capability to reduce redundant data through the creation of a digital link between unique identifiers and health information about households and individuals, thereby preventing the creation of duplicative household or individual records. Although DHIS2 is also a large and well-utilised system in Ethiopia, it was reported to have some data quality issues [37].

One method for ensuring data quality is through the Master Facility Registry [38]. The system maintains data accuracy, reliability, and robust governance and update processes, assuring data quality through well-defined standards and validation, and improving service delivery by providing accurate facility information. The MFR is successfully interoperable with the DHIS2 platform, enabled by a custom-built connector. Approximately 80% of the health facilities have had their service domains populated in the MFR.

The MOH identified data quality as another important dimension to consider during the data source assessment. Greater emphasis was placed on addressing data completeness and uniqueness as important attributes of data quality. Further investigations of the data sources showed large data sources like eCHIS and DHIS2 to be mostly unique (greater than 90% in rating) and 70% (eCHIS) to 90% (DHIS2) rating in terms of completeness, respectively. Other data sources were described mostly as complete and unique; however, reports mention that HR data sources and the MFR itself are likely 60% complete in data completeness. These observations are likely to impact the ability of shared registry services to ensure data quality and unique references across the data platform.

3.3 Data Ownership and Custody

The concept of data ownership, custody, responsibility, and stewardship is a widely discussed subject across the ministry, its agencies, and its partners. Therefore, the technology that underpins the NHDW should also be able to cater to identity access management, role-based access, fine-grained access control, and other data sharing needs. The Ethiopian MOH should have primary ownership of all health-related data within the country; however, various parties may be responsible for the custody and stewardship of these data sources.

In the MOH Data Centre, the majority of applications are owned, controlled, and managed solely by the MOH. However, there are also some data sources within the Data Centre that are jointly managed by the MOH and its associated agencies, but these are fewer in number compared to those fully owned and managed by the MOH.

As the nature of health data is such that it is cross-sectoral, other departments and ministries like the Ethiopian Statistics Service and the Ministry of Education are also primary owners and custodians of some of the data sources considered for ingestion.

To ensure effective data catalogues and data governance, all data that is ingested into the NHDW data storage layer would require data stewards to provide comprehensive business metadata essential to maintaining FAIR principles on the data.

3.4 Summary

In summary, chapter 3 examines the current data sources essential for developing and implementing the National Health Data Warehouse (NHDW) in Ethiopia. It provides an overview of the main data systems, analyses their characteristics, and addresses data ownership and governance.

We provided descriptions and information on various healthcare systems, including Electronic Medical Records (EMR), Laboratory Information Management Systems (LIMS), case surveillance systems, community health management systems, and many more.

After this, we provided stats on the 83 data sources identified. Highlighting attributes of many important data sources necessary for advancing decision-making during the development phase.

Finally, we address ownership and custody, highlighting the importance of data ownership and stewardship. While ensuring effective data governance through the FAIR principles.

Chapter 4: Functional Requirements (System Features)

Table 11 outlines the functional requirements for the National Health Data Warehouse (NHDW), categorised by the various layers of the system:

Table 11: Functional Requirements for the NHDW

Category	Functional Requirement	Description
FR1 Data Ingestion	FR1.1 Data Source Connectivity	The system must support data extraction from various sources and loading into the data lake and data warehouse. Connectivity should leverage available assets, such as HealthNet VPN, as well as be able to use the public internet for remote connections.
	FR1.2 Data Extraction	The system may support a selection of ETL tools, such as Change Data Capture (CDC), SQL queries, API calls, and web scraping for data extraction, selected as appropriate for the ingestion of a variety of data sources.
	FR1.3 Support for Pull-Based and Push-Based Extraction Techniques	The system may support a selection of event-driven data ingestion, message queues, event streams, webhooks, and data replication tools as appropriate for a variety of data sources and suitable data transmission methods. Mechanisms should be implemented for the data warehouse to actively request data extraction from source systems (pull-based) as well as for source systems to initiate sending data to data warehouses (push-based). Pull-based is preferred for bulk loads of large datasets so that the data warehouse can schedule and manage resource utilisation. Push-bases are preferred for semi-realtime/realtime event data sources.
	FR1.4 File Deposit Capability	The system must provide the ability to handle file deposits via FTP/SFTP or file sync servers for source systems to drop files or database dumps.
	FR1.5 Job scheduling	The system must implement data flow automation tools that are scalable, fault-tolerant, monitored, and have the capability to handle diverse data formats and data source types. The tooling should include task dependency management and efficient workflow customisation.
	FR1.6 Data transmission consistency checks	The system must conduct data consistency checks when data is transferred from data sources into the data warehouse to prevent data loss, duplication, or corruption of data during transmission.
FR2 Data Storage and Management	FR2.1 Storage of Unstructured and Structured Data	The system must store unstructured and structured data derived from various health systems in a centralised repository. Tools that support storage, searchability, querying, and processing of both unstructured and structured data should be available in the system.

	FR2.2 Integration between Data Lake and Data Warehouse	The system must integrate data from the data lake, ensuring that cleansed, enriched, and transformed data is loaded into the data warehouse.
	FR2.3 Support for Real-Time Data Processing	The system must include a database that supports analysing data in realtime; possibly consider a time-series database to store and analyse real-time data.
	FR2.4 Data Vault Modeling	The data storage component must support the Data Vault modelling technique (Hubs, Links, Satellites) for organising data.
	FR2.5 Dimensional Modelling for Data Marts	The data storage component must support dimensional modelling for establishing data marts to support analysis and visualisation.
	FR2.6 Data Archiving	The system should support organised storage of old data. Automated archiving tools and data archive governance (such as retention policies, audit trails, etc.) should be implemented.
FR3 Data Processing and Analytics	FR3.1 Data Ingestion ETL Pipelines	The system must integrate data from various sources, including the data lake, data warehouse, and external sources.
	FR3.2 Access to Diverse Data	The system must provide access to both structured and unstructured data stored in the data lake for machine learning and AI training.
	FR3.3 Data Preprocessing and Feature Engineering	The system must provide tools for data preprocessing and feature engineering, including data cleaning, normalisation, and transformation.
	FR3.4 Data Mining and Exploration	The system must support tools for data mining and exploration to discover patterns and trends.
	FR3.5 Machine Learning and AI Model Development	The system must support the development and deployment of machine learning and AI models using libraries and frameworks like TensorFlow, PyTorch, Scikit-learn, and Keras.
	FR3.6 Prediction and Forecasting	The system must enable predictive models to forecast future trends and outcomes.
	FR3.7 Model Management and Deployment	The system must offer processes for managing and deploying models into production, including version control, evaluation, and CI/CD pipelines.
	FR3.8 Ethical Data Science AI Controls	The system must ensure the integrity, security, and ethical use of data for AI and data science activities.

FR4 Data Visualization and Reporting	FR4.1 User-Friendly Dashboards and Reports	The system must provide intuitive dashboards and reporting tools for business users to access and interact with data.
	FR4.2 Real-Time Data Visualization	The system must support real-time data visualisation capabilities.
	FR4.3 Interactive Data Exploration	The system must offer tools for interactive data exploration, enabling users to drill down, filter, and analyse data.
	FR4.4 Trend Analysis and Forecasting	The system must support tools for trend analysis and forecasting to identify historical patterns and predict future trends.
	FR4.5 Geospatial Analysis	The system must provide geospatial analysis tools to incorporate geographical data into visualisations.
FR5 Data Governance and Master Data Management	FR5.1 Data Dictionary	The system must maintain a data dictionary that defines and describes data elements, attributes, and entities.
	FR5.2 Data Lineage	The system must document data lineage, tracking the flow of data from source to destination and capturing transformations.
	FR5.3 Data Catalog	The system must provide a centralised data catalogue that inventory data assets, including metadata such as data lineage, usage, ownership, and access permissions.
	FR5.4 Data Quality Management	The system must implement processes for data quality management, including data profiling, cleansing, and validation, following Total Quality Management (TQM) principles.
	FR5.5 Business Rules Documentation	The system must document both hard and soft business rules to ensure clarity on data processing and transformations.
	FR5.6 Data Governance Management	The system must enforce data governance policies and security measures to ensure data integrity, data quality, security, and compliance.
FR6 Monitoring	FR6.1 Infrastructure Monitoring	The system must provide real-time monitoring of system performance, tracking key metrics such as power load profile, system uptime, network health, bandwidth, CPU usage, memory utilisation, disc I/O, and storage usage.
	FR6.2 Software System Monitoring	The system must provide real-time monitoring of the software applications that are deployed on the NHDW platform. Application processing performance, user interactions, error monitoring, application uptime, and load-balancing profiling are metrics the system should monitor.

	FR6.3 Anomaly Detection	The system must implement anomaly detection algorithms to identify unusual patterns or deviations from normal behaviour.
	FR6.4 Logging and Log Management	The system must capture detailed logs from various components and provide centralised log management for searching, analysing, and troubleshooting issues.
	FR6.5 Alerting and Notifications	The system must provide configurable alerting mechanisms to notify administrators of threshold breaches or anomalies.
	FR6.6 Performance Reporting	The system must generate regular performance reports summarising system health and performance.
	FR6.7 Security Access of Monitoring Data	The system must implement access controls, data privacy measures, and continuous monitoring to ensure the security of monitoring data.
FR7 Secure Data Access	FR7.1 Centralised User Management	The system must implement centralised user management to streamline user identities and access permissions.
	FR7.2 Role-Based Access Control (RBAC)	The system must establish RBAC to enforce security policies based on user roles and responsibilities.
	FR7.3 Authentication and Authorization	The system must implement robust authentication mechanisms, including multi-factor authentication (MFA), and ensure proper authorisation processes.
	FR7.4 Audit Logging and Accountability	The system must maintain detailed audit logs to track user activities and data access, ensuring accountability and transparency.
	FR7.5 Secure Data Sharing	The system must facilitate secure data sharing between authorised users and systems while maintaining strict access controls and data privacy.
	FR7.6 Secure API endpoints	The system should provide API endpoints for external systems to consume data for secondary data use. Data sharing protocols must be implemented to restrict data sharing via API for authorised data use only. Typically, API endpoints will be built as a component of the business intelligence layer, drawing from data mart structures that are prepared for sharing with external data consumers.

Chapter 5: Non-functional Requirements

5.1 Tabulated non-functional requirements

The following table outlines the non-functional requirements for the National Health Data Warehouse (NHDW), ensuring that the system meets performance, scalability, security, and compliance standards.

Table 12: Non-functional Requirements for the NHDW

Category	Non-Functional Requirement	Description
NFR1 Performance	NFR1.1 System Response Time	The system must provide quick response times for data queries, loading, and processing, ensuring timely access to information.
	NFR1.2 Throughput	The system must handle a high volume of data transactions per second without degradation in performance.
NFR2 Scalability	NFR2.1 Horizontal Scalability	The system must support horizontal scaling to handle increasing data volumes and user loads by adding more servers or nodes.
	NFR2.2 Vertical Scalability	The system must support vertical scaling to enhance the capacity of existing hardware resources as needed.
	NFR2.3 Elasticity	The system must automatically or manually scale resources up or down based on current demand to optimise performance and cost-efficiency.
NFR3 Availability	NFR 3.1 High Availability	The system must ensure high availability with minimal downtime, employing redundant systems and failover mechanisms.
	NFR 3.2 Disaster Recovery	The system must have disaster recovery plans and backup mechanisms in place to restore functionality quickly in case of failure.
	NFR 3.3 Reliability	The system must consistently perform its intended functions without failures, ensuring reliable access to data and services.
NFR4 Security	NFR 4.1 Data Encryption	The system must use strong encryption methods to protect data at rest and in transit, ensuring data confidentiality and integrity.
	NFR 4.2 Access Controls	The system must enforce strict access controls, including RBAC, to ensure only authorised users have access to sensitive data and system functions.
	NFR 4.3 Audit and Compliance	The system must maintain detailed audit logs and comply with healthcare regulations, ensuring accountability and legal compliance.

	NFR 4.4 Incident Response	The system must have procedures in place to detect, respond to, and mitigate security incidents promptly.
NFR5 Usability	NFR5.1 Intuitive user Interface	The system must provide an intuitive and user-friendly interface for all user roles, including data scientists, analysts, and business users.
	NFR5.2 Documentation	The system must provide comprehensive documentation, including user manuals, API documentation, and operational guides, to assist users and administrators.
	NFR5.3 Training and Support	Training programs and support services should be designed to ensure users can effectively utilise the system.
NFR6 Maintainability	NFR 6.1 Modularity	The system must be designed with a modular architecture to facilitate easy maintenance, updates, and enhancements.
	NFR 6.2 Automated Testing	The system must support automated testing to ensure the quality and reliability of code changes and updates.
	NFR 6.3 Proactive Monitoring and Alerts	The system must provide robust monitoring tools and alert mechanisms to identify and address issues proactively.
NFR7 Interoperability	NFR 7.1 Standards Compliance	The system must comply with relevant data and communication standards to ensure seamless integration with other healthcare systems.
	NFR 7.2 API Support	The system must offer comprehensive API support to enable integration with external applications and systems.
	NFR 7.3 Data Integration	The system must support integration with various data sources and formats, including structured, semi-structured, and unstructured data.
NFR8 Flexibility	NFR 8.1 Configurability	The system must allow configuration changes without requiring significant code modifications, enabling adaptation to evolving requirements.
	NFR 8.2 Customizability	The system must support customisation to meet specific organisational needs and preferences.
NFR9 Compliance	NFR 9.1 Regulatory Compliance	The system must adhere to all relevant regulatory requirements, including healthcare data privacy and security standards.
	NFR 9.2 Data Governance	The system must implement data governance policies and procedures to ensure data quality, consistency, and security across the organisation.
NFR10 Data Quality	NFR 10.1 Data Accuracy	The system must ensure high levels of data accuracy through validation, cleansing, and reconciliation processes.

	NFR 10.2 Data Consistency	The system must ensure data consistency across all layers and components, maintaining synchronised and harmonised datasets.
	NFR 10.3 Data Completeness	The system must ensure that all necessary data is captured and stored, avoiding gaps and missing information.
NFR11 Efficiency	NFR 11.1 Resource Utilisation	The system must optimise the utilisation of hardware and software resources to provide efficient performance.
	NFR 11.2 Cost Efficiency	The system must provide cost-efficient solutions, balancing performance and resource usage to minimise operational costs.
NFR12 Auditability	NFR 12.1 Comprehensive Logging	The system must maintain detailed logs for all critical operations, ensuring that all actions can be audited for compliance and troubleshooting purposes.
	NFR 12.2 Audit Trail	The system must provide an audit trail that records all access and modifications to data, ensuring transparency and accountability.

5.2 Quality Assurance Requirements

Quality assurance (QA) requirements are essential for ensuring that the final product to be delivered as NDWH and all related features meet the defined specified standards and satisfy all the stakeholders and users expectations.

For the development of the NDHW, the expectation is to use the national and international standards and guidelines to be agreed upon within the technical working group. As part of the roles and responsibilities of the stakeholders and TWG, there is a need to select a team that will be responsible for the QA and who will be working on setting measurable quality objectives aligned with the defined goals to help guide the QA process. Identifying and addressing defects early in the process will help reduce waste and lower costs.

The QA team will be responsible for overseeing the implementation of QA processes, conducting audits, and ensuring compliance with the defined standards. In order to have a better performance, this team must be engaged in comprehensive training on QA principles and procedures.

5.2.1 Data Accuracy and Integrity

- Source Data Verification: Ensure that the data ingested into the EDW from various sources is accurate and consistent with the source systems.
- Data Transformation Validation: Validate that all ETL (Extract, Transform, Load) processes are correctly transforming data as per the business rules.
- Referential Integrity: Ensure that foreign key relationships and other referential integrity constraints are maintained within the EDW.
- Data Consistency Checks: Perform consistency checks across different tables, schemas, and systems to ensure data is uniformly integrated.

5.2.2 Performance Testing

- Load Testing: Simulate large volumes of data and user queries to assess how the EDW performs under peak load conditions.
- Query Performance Optimisation: Ensure that complex queries run efficiently without impacting system performance. Indexing, partitioning, and query optimisation strategies should be tested.
- Batch Processing Performance: Test the performance of nightly or periodic batch jobs, ensuring they complete within the required time windows.

5.2.3 Security and Compliance

- Data Access Control: Verify that only authorised users have access to sensitive data, with appropriate role-based access controls (RBAC) in place.
- Data Masking and Encryption: Ensure sensitive data is masked or encrypted both at rest and in transit.
- Audit Trails and Logging: Implement and test logging mechanisms to track data access and changes for compliance and auditing purposes.
- Regulatory Compliance Testing: Ensure the EDW adheres to relevant regulations such as GDPR, HIPAA, or other industry-specific standards.

5.2.4 Data Quality Management

- Data Profiling: Continuously profile data to identify and correct anomalies, such as missing, duplicate, or incorrect data.
- Data Quality Metrics: implement metrics to monitor data quality, including accuracy, completeness, timeliness, and consistency.
- Error Handling and Resolution: Test the system's ability to handle errors during data processing and ensure that errors are logged, reported, and resolved.

5.2.5 Scalability and Capacity Planning

- Scalability Testing: Assess the system's ability to scale horizontally and vertically to accommodate increasing data volumes and users.
- Capacity Planning: Develop and test scenarios for future growth, ensuring the EDW can handle increased data loads and user activity over time.

5.2.6 Backup and Recovery

- Backup Procedures: Verify that data backup procedures are reliable and that backups occur within the specified windows.
- Disaster Recovery Testing: Test the disaster recovery plan to ensure data can be restored quickly and accurately in case of a system failure.

5.2.7 End-to-End Testing

- Integrated System Testing: Perform end-to-end testing to ensure that all components of the EDW, including data sources, ETL processes, and reporting tools, work together as expected.
- User Acceptance Testing (UAT): Involve business users in testing to ensure the EDW meets all business requirements and expectations.

5.2.8 Automation of Testing

- Automated ETL Testing: Implement automated testing scripts to validate ETL processes regularly.
- Automated Regression Testing: Use automation to perform regression testing whenever changes are made to the EDW, ensuring existing functionality is not impacted.

Chapter 6: Technical Approach

6.0.1 Introduction

This section covers the technical approach taken for the implementation of the National Health Data Warehouse. Herein we discuss the architectural considerations, explaining the reasoning behind technology choices and the benefits it brings to ingestion, storage, processing, analytics, security, and governance of the data platform.

The term “Data Warehouse” in National Health Data warehouse is used generically to define a data repository that supports analytics and decision-making. Data warehousing itself is not a new concept; however, many efforts have led to the maturity of Decision Support Systems (DSS) to cater to the growing needs and larger amounts of data being produced by today’s source systems. Modern analytics requirements still require that structured data is processed as is done traditionally; in addition, there are needs to process and analyse semi-structured and unstructured data like text, audio, and video, allowing for advanced analytics and machine learning. Hence, a modern data architecture approach is taken to the development of the NHDW.

6.0.2 Approach & Frameworks

A modern data architecture is one that incorporates the FAIR principles, which focus on Findability, Accessibility, Interoperability and Reuse. To achieve this, we propose a hybrid approach of Top-Down and Bottom-Up towards the creation of data marts that aid the decision-making process. A brief summary of these approaches is that a Top-Down allows users to first consolidate and unify data in a data warehouse before creating data marts that can serve multiple subject areas. The Bottom-Up approach, on the other hand, prioritises acquiring value from the data marts first before consolidating data from these subject-area-specific data marts into the data warehouse. For more details on this, kindly refer to the development framework section of this document.

6.0.3 Scope

The scope of this architecture includes the components, technology, tools, and data sources required to meet the functional and non-functional requirements of the NHDW.

6.1 NHDW High-Level Architecture

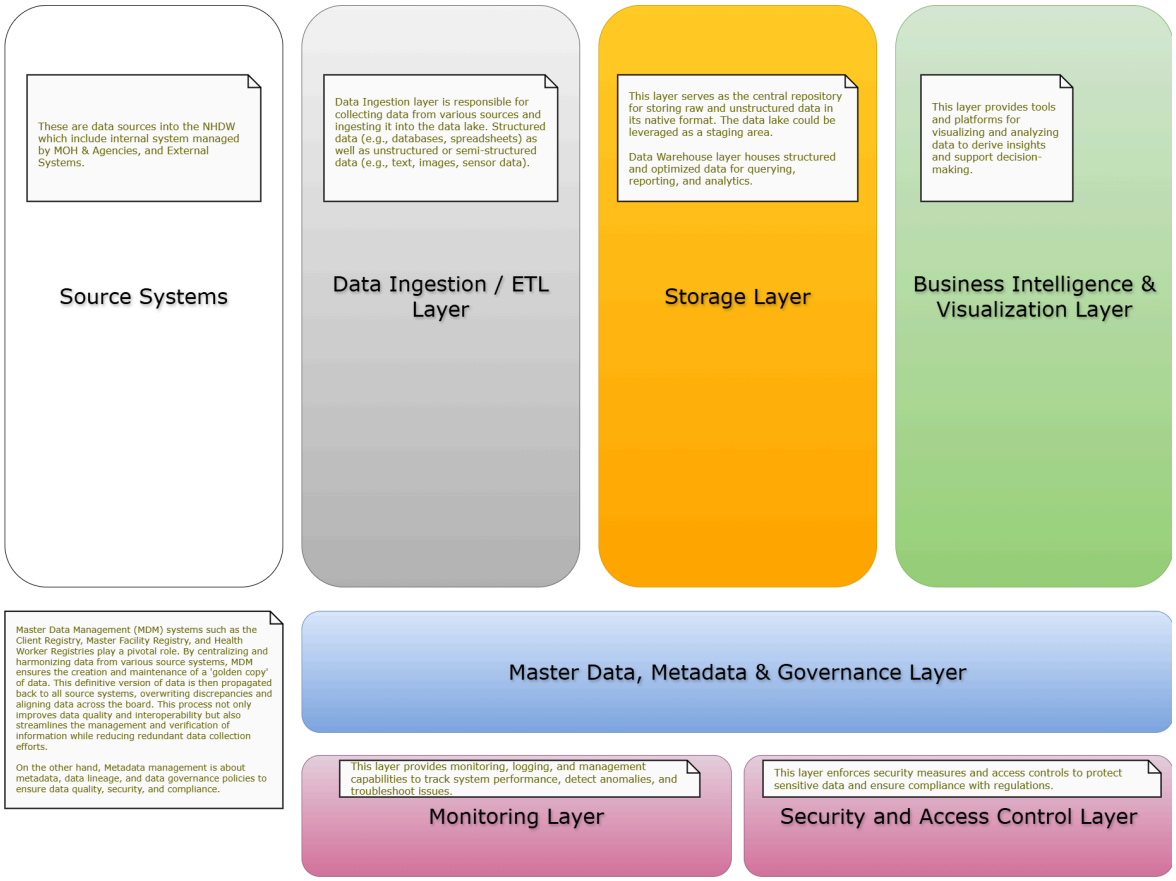


Figure 11: Proposed NHDW high-level architecture

6.1.1 Description

This section covers the layers essential for the implementation of certain functions of the National Health Data Warehouse.

As data is received from the source systems, much effort is needed to ensure the support of a wide variety of communication protocols used by healthcare systems including bulk data feeds and real-time data streams (e.g., FTP, HL7v2, or FHIR). Data storage can be done on low-cost storage with high availability, and then security becomes functional in this area through encryption and data lifecycle management. Data processing generally happens throughout the architecture; however, some areas involve more processing than most, which is in the ingestion, storage, and intelligence layers. Metadata layers then help to ensure compliance. End users then interact with the data and insights across all the normalised healthcare data.

6.1.2 Data Flow and Integration

Integrating data sources is crucial to building a scalable, secure, and efficient data warehouse that supports analytical use cases and complies with governance standards. The process starts with data mapping to assess sources, followed by ETL/ELT data ingestion to streamline data flows. Validation checks ensure data integrity, while governance measures secure access and compliance. Continuous monitoring supports performance tuning, proactively addressing issues, and ensuring quality.

6.1.3 Layers Description & Relationship

Each layer has a description of its function. Data moves from the source systems into the ingestion layer before it is stored in the storage layer and utilised in the Business Intelligence layer. The governance layer, monitoring layer, and security layer are mostly non-functional layers that support the functioning of components that exist within the Data Warehouse, ensuring appropriate functioning of the data warehouse. Details of the components within each layer are in the following sections.

The architecture layers are as follows:

1. **Source Systems:**

   These are data sources into the NHDW, which include internal systems managed by MOH & agencies and external systems. The data source categories listed in this layer are informed by the eHealth Architecture data source categories.

2. **Data Ingestion/ETL Layer:**

   The Data Ingestion layer is responsible for collecting data from various sources and ingesting it into the data lake. Structured data (e.g., databases, spreadsheets) as well as unstructured or semi-structured data (e.g., text, images, sensor data). It also informs the ETL jobs that are used to move data between applications in the cluster.

3. **Storage Layer:**

   This layer serves as the central repository for storing raw and unstructured data in its native format. The data lake could be leveraged as a staging and storage layer for raw, processed, or archived data. The Data Warehouse within this layer houses structured and optimised data for querying, reporting, and analytics.

4. **Business Intelligence & Visualisation Layer:**

   This layer provides tools and platforms for visualising and analysing data to derive insights and support decision-making. It also provides tools for data mining, AI, and machine learning.

5. **Master Data, Metadata & Governance Layer:**

   This layer provides tools such as the client registry, master facility registry, and health worker registry. These tools play a pivotal role by centralising and harmonising data from various source systems, ensuring the creation and maintenance of a 'golden copy' of data. It also contains metadata management tools such as metadata catalogues, data lineage, and data governance policies to ensure data quality, security, and compliance.

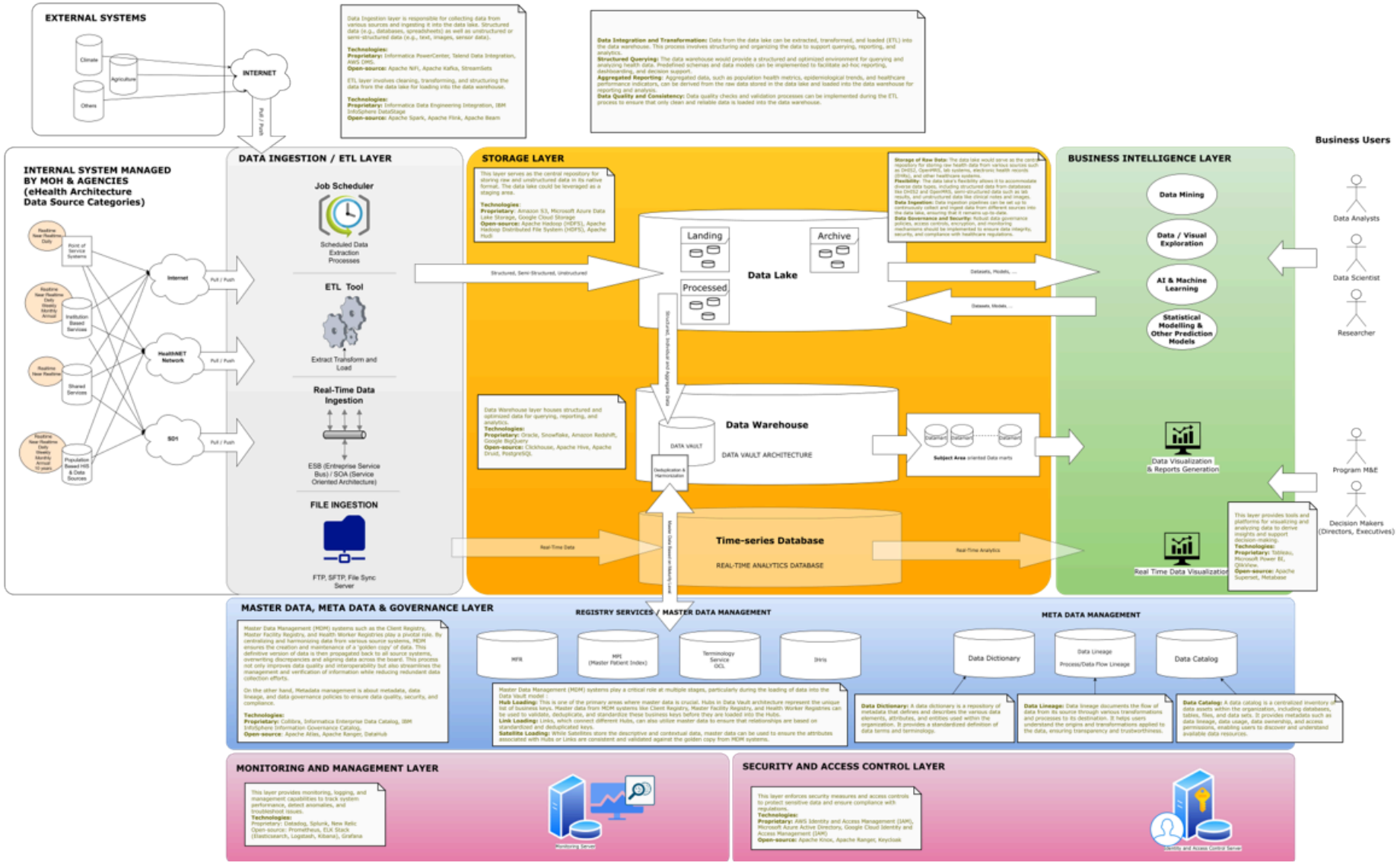
6. **Monitoring Layer:**

   This layer provides monitoring, logging, and management capabilities to track system performance, detect anomalies, and troubleshoot issues.

7. **Security and Access Control Layer:**

   This layer enforces security measures and access controls to protect sensitive data and ensure compliance with regulations.

6.2 Detailed Architecture



6.2.1 Architectural Pattern, Guidelines and Alignment with the Enterprise Architecture

The architectural pattern followed the implementation of the top-down approach to decision support systems, as highlighted in previous sections of this document. However, we implement this flow according to more modern principles. In our flow, the data lake is where the initial staging and storage for all data is done, for further processing of data written into the Data warehouse where required.

This modern Data Architecture setup allows us to make all datasets available for big data processing and machine learning. It supports the implementation of FAIR principles as the data lies on the platform. It also supports the implementation of the Lakehouse architecture (i.e., support for interactive querying of all data types using Presto and Apache Drill).

The benefits of the Architecture are that it is implementable in both open-source and proprietary cloud environments, following the guidelines specified in this document. Refer to the preceding section on guiding principles for alignment of this architecture to the TOGAF Enterprise architecture and other frameworks.

Finally, Data sources have been grouped according to eHealth categories and may be transmitted into the platform through a VPN, either the HealthNET network, the SD1 network, or the Internet. These transmission types support both push- and pull-based approaches. Further details are in the following sections.

6.3 Source Systems

In this section and the sections that follow, we dive into each layer of the architecture, detailing each component and its function. We also discuss some technology options and the approach to ingestion, storage, modelling, analytics, governance, monitoring, and security.

The diagram below depicts data source categories as input into the data platform and the cadences at which they support being ingested using the three different transmission methods.

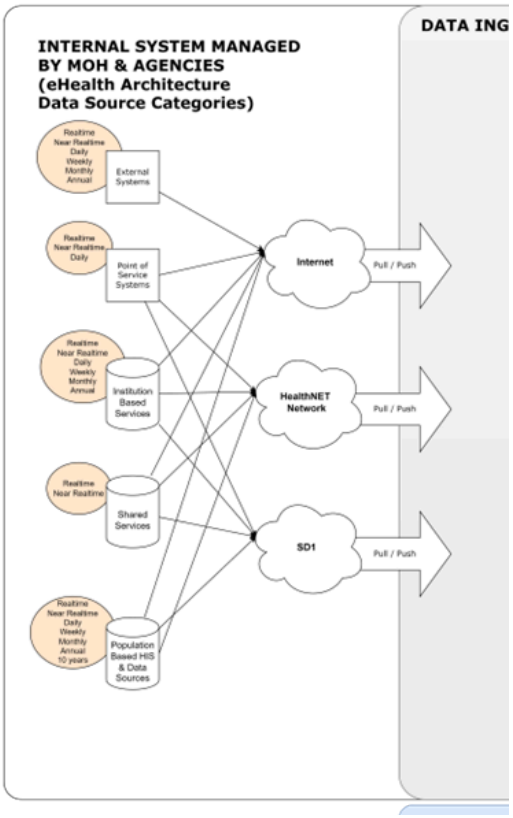


Figure 13: NHDW Architecture: Data Sources Categories

6.3.1 Grouping in Alignment with eHealth Architecture

The definitions utilised for each eHealth category are below:

1. **Point of Service Systems:**

   Real-time Data Capture: These systems allow for immediate data entry and access at the point of care, which enhances patient management and service delivery.

   Patient tracking facilitates the tracking of individual patient data, leading to improved health outcomes and continuity of care.

   Decision Support: Provides health professionals with timely information that aids in clinical decision-making.

2. **Institution-Based Services:**

   Operational Efficiency: These systems are designed to streamline and optimise internal processes within healthcare institutions, such as HR, finance, and supply chain management.

   Performance Monitoring: They provide critical data for evaluating the performance of health services, enabling better decision-making and resource allocation.

   Compliance and Accountability: Helps institutions meet regulatory requirements and standards for quality assurance and reporting.

3. **Shared Services:**

   Interoperability: Systems in this category often facilitate data sharing across different platforms, enhancing collaboration and communication between various stakeholders.

Standardisation: They promote the use of standardised data formats and definitions, improving the quality and consistency of data across health systems.

Resource Optimisation: Centralised resources can lead to cost savings and more efficient use of data across multiple organisations or programs.

4. **Population-Based HIS & Data Sources**

   Epidemiological Insights: Systems in this category are essential for understanding health trends and outcomes in specific populations, informing public health interventions.

   Resource Allocation: Data from these systems help policymakers allocate resources effectively based on the needs of the population.

   Research and Evaluation: They support research initiatives by providing critical data for analysis and evaluation of health programs and policies.

5. **External Systems**

   Cross-sector Collaboration: These systems facilitate data integration between health and other sectors (e.g., agriculture, education), promoting a holistic approach to health.

   Policy Development: Data from external systems can inform health policies by highlighting the broader social determinants of health.

   Risk Management: They help identify and mitigate risks associated with environmental and socioeconomic factors impacting health.

6.4 Data Ingestion

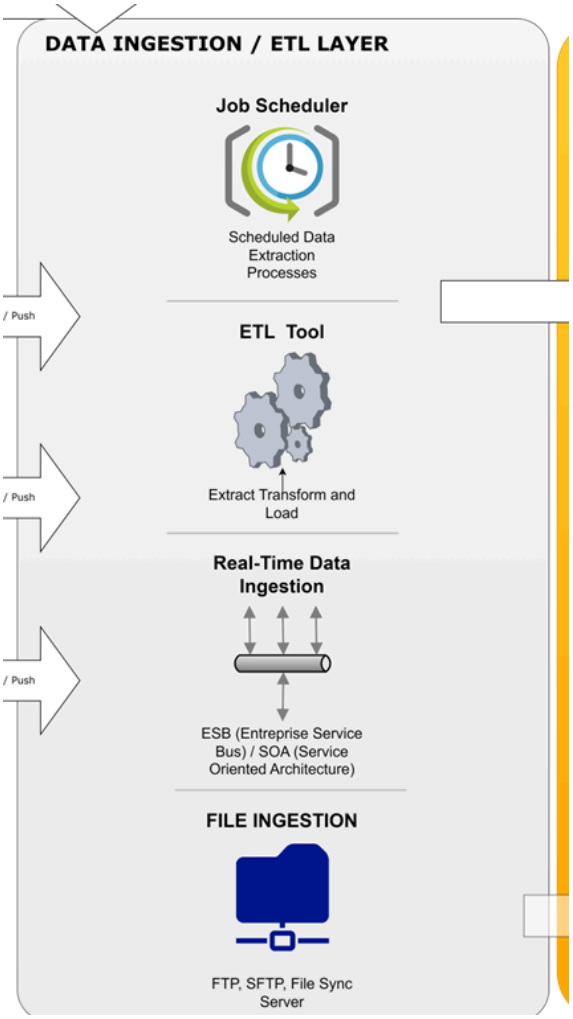


Figure 14: NHDW Architecture: Data Ingestion / ETL Layer

The Data Ingestion/ETL (Extract, Transform, Load) Layer is a critical component of the Ethiopia National Health Data Warehouse, responsible for the efficient and reliable extraction of data from various source systems. Data extraction can be achieved through two primary approaches: pull-based techniques and push-based techniques, each with its own set of methods and tools suited to the specific nature and requirements of the data sources and the data warehouse environment.

6.4.1 Job Scheduler & ETL

Pull-Based Techniques

1. **Batch Processing using scheduled jobs & ETL Tools**

   Data can be extracted at regular intervals using scheduled jobs. Specialised ETL tools like Apache Nifi, Talend, or Apache Airflow can automate the extraction process, ensuring data is pulled from source systems efficiently. Licensed technologies include Microsoft SQL Server Integration Services.

2. **Query-Based Extraction**:

    1. **SQL Queries**: Direct SQL queries can be used to extract data from relational databases.
    2. **API Calls**: Periodic API calls to web services or cloud-based applications enable the extraction of data as per the defined schedule.

6.4.2 Real-Time Data Ingestion

Pull-Based Techniques

1. **Change Data Capture (CDC)**:

    1. **Log-Based CDC**: By monitoring database transaction logs, this method captures and extracts only the changes made since the last extraction, reducing the load on source systems.
    2. **Trigger-Based CDC**: The database triggers record changes in a separate table, which can then be extracted periodically.

Push-Based Techniques

Event-Driven Data Ingestion:

1. **Message Queues**: Systems like Apache Kafka or RabbitMQ facilitate real-time data push from source systems to the data warehouse.
2. **Event Streams**: Platforms such as Apache Kafka allow continuous pushing of data as it is generated.

Webhooks:

1. **HTTP Callbacks**: Webhooks can be configured in source systems to push data to specified endpoints in real time upon the occurrence of certain events.
2. **Data Replication**: Tools like GoldenGate, DbVisit Replicate, or native database replication features enable the push of data changes directly to the data warehouse.

According to the Data Vault 2.0 architecture [18], [19], it's necessary to apply the **hard business rules** before storing the data in the staging area.

6.4.3 File Ingestion (Push-Based Techniques)

To accommodate various data formats and sources, the National Health Data Warehouse will also offer a file deposit capability, allowing source systems to drop files or database dumps using FTP/SFTP or a file sync server. This functionality ensures the warehouse can handle both structured, semi-structured, and unstructured data, as well as individual and aggregate data, encompassing both clinical and non-clinical information. The "Data Ingestion/ETL Layer" is thus designed to be versatile and robust, capable of integrating data through both push and pull mechanisms to support comprehensive data analysis and decision-making in the health sector.

6.4.4 Protocols and Standards

Throughout the Data Ingestion process, data can be aligned with standard clinical terminologies (such as SNOMED, ICD, CPT, and NDC) to ensure consistency for downstream users. Datasets should be cleaned, monitored, and versioned to uphold their quality. Utilise a central data catalogue for registering and discovering these datasets. Automate master data management processes to lessen the workload and enhance adoption. Clearly label sensitive datasets with enforced authorization. Monitor licences for relevant datasets. Standard healthcare ontologies are preferred over proprietary ones to facilitate reuse and data sharing.

6.4.5 OpenHIM’s role: Information Mediator Service

OpenHIM may also play a significant role in the data ingestion layer as an information mediator, particularly by ensuring:

1. **Data standardisation:** OpenHIM can be used for data standardisation by ensuring incoming data conforms to the HL7 FHIR standards while queuing transactions from online data sources like EMRs.
2. **Real-Time Data Processing:** It can allow facilities or other real-time data sources to send data to the data platform in real-time or near real-time.
3. **Monitoring and Logging:** may be used to monitor and log incoming transactions to ensure data quality and check API metrics from source systems.
4. **Security and Compliance:** OpenHIM can enforce security protocols. The newer version will support role-based access to data within the OpenHIM console.

6.5 Data Storage

6.5.1 Data Lake

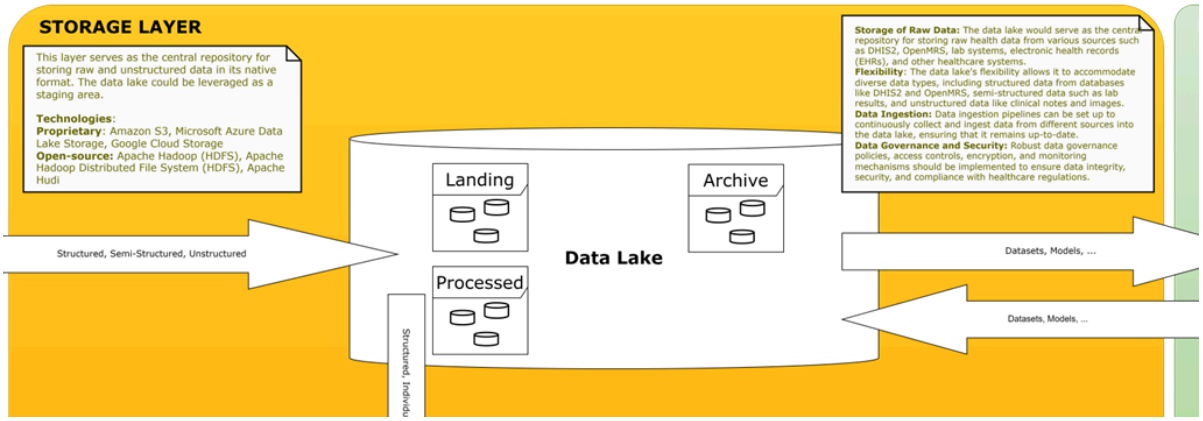


Figure 15: NHDW Architecture: Data Lake / Big Data Layer

The Data Lake/Big Data Layer serves as the central repository for storing raw and unstructured data in its native format. This layer is designed to provide a scalable and flexible environment that can accommodate the vast amounts of data generated by various health systems, ensuring that it is readily available for further processing and analysis. The data lake will transition data as it goes through ETL pipelines from landing to processed and archived data.

Technologies
To implement the Data Lake/Big Data Layer, open-source technologies can be utilised. Some options include the below, but a more extensive technology selection process should be conducted by the implementer of the NHDW:

- **Apache Hadoop (HDFS)**: A distributed file system designed to store large data sets across multiple machines, ensuring high availability and reliability.
- **Apache Hudi**: An open-source data lake framework that simplifies incremental data processing and provides near real-time data ingestion.

Storage of Raw Data
The data lake will serve as the central repository for storing raw health data from various sources, including DHIS2, OpenMRS, lab systems, electronic health records (EHRs), and other healthcare systems. By storing data in its native format, the data lake ensures that all data, whether structured, semi-structured, or unstructured, is preserved with its original context and detail.

Flexibility
One of the key advantages of the data lake is its flexibility, allowing it to accommodate diverse data types. This includes:

- **Structured Data**: Data from relational databases like DHIS2 and OpenMRS, which are typically organised in rows and columns.
- **Semi-structured Data**: Data such as lab results, which may not fit neatly into traditional database structures but still contain some organisational properties.

- **Unstructured Data**: Data like clinical notes, images, and other forms of media that do not follow a specific format.

Data Ingestion

To ensure the data lake remains up-to-date, data ingestion pipelines will be established to continuously collect and ingest data from various sources. These pipelines can be configured to handle real-time data streams as well as batch processing, ensuring timely and accurate data capture. This process will be automated using tools and frameworks that support continuous integration and delivery.

Data Lake as a Staging Area

In the context of a hybrid Data vault Architecture (a type of Data vault architecture that integrates both raw and processed data), the data lake may function as a staging area for the data warehouse. This approach allows for the initial storage of raw data in the data lake, where it can be transformed and prepared for loading into the data warehouse. The staged data in the data lake can be enriched, cleansed, and integrated with other data sources, ensuring that the data warehouse receives high-quality, well-structured information for analysis and reporting.

6.5.2 Data Warehouse & Data Marts

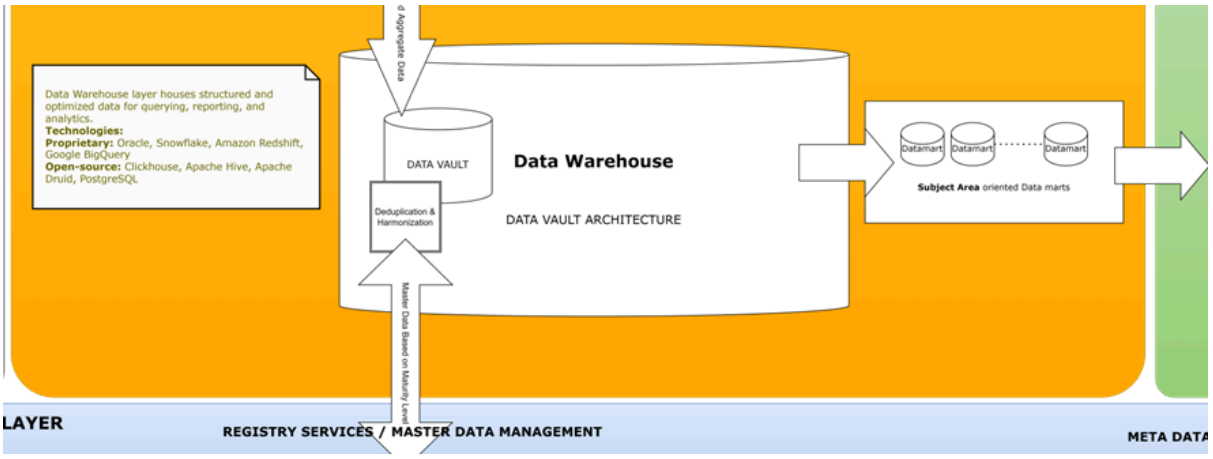


Figure 16: NHDW Architecture: Data Warehouse & Time-Series Database

The Data Warehouse Layer serves as the structured and refined central repository for structured and relational, individual, and aggregate data, whether it's clinical or non-clinical data, enabling advanced analytics, reporting, and decision-making. While the Data Lake/Big Data Layer handles raw and unstructured data, the Data Warehouse Layer focuses on transforming this data into an organised, accessible format. Leveraging the data lake as a staging area, the Data Warehouse Layer integrates various data types into a cohesive, analytical structure following the Data Vault modelling technique. This section outlines the technologies, data storage methodologies, and governance practices essential for the Data Warehouse Layer.

Technologies to consider
To support the Data Warehouse Layer, a combination of proprietary and open-source technologies can be employed, ensuring flexibility and scalability. Key technologies include:

- **Proprietary Solutions**:
    - **Oracle:** Oracle's data warehousing solutions offer robust performance, advanced analytics capabilities, and comprehensive security features.
    - **MS SQL Server:** Microsoft's SQL Server provides a powerful and integrated data warehousing solution with advanced analytics, security, and management tools.
- **Open-Source Solutions**:
    - **PostgreSQL**: An advanced, open-source relational database system with robust features and performance optimisation for data warehousing.
    - **Apache Hive**: A data warehouse infrastructure built on top of Hadoop for providing data summarisation, query, and analysis.
    - **ClickHouse**: A columnar database management system for online analytical processing (OLAP) that allows for real-time data processing and analysis.

Storage of Structured Data
The Data Warehouse Layer will store structured data derived from various health systems, such as DHIS2, OpenMRS, lab systems, electronic health records (EHRs), and other healthcare systems. By structuring this data, the warehouse enables efficient querying and analysis, supporting diverse use cases in the health sector.

Flexibility and Integration
The Data Warehouse Layer's design will accommodate diverse data types by integrating data from the data lake, which serves as a staging area. This integration ensures that the data warehouse receives cleansed, enriched, and transformed data, ready for analysis and reporting. The flexibility of this approach allows for the inclusion of structured, semi-structured, and unstructured data, ensuring comprehensive data coverage.

Data Ingestion and Real-Time Processing
Data ingestion pipelines will be established to continuously collect and transform data from the data lake into the data warehouse. Additionally, the Data Warehouse Layer will include the ability to analyse real-time data. This capability is crucial for applications requiring timely insights, such as monitoring disease outbreaks or managing emergency responses. Real-time data should be ingested and streamlined to Real-time Data Visualisation Tools directly, ensuring that decision-makers have access to the most current information.

Data Vault Modeling Technique
The Data Warehouse will be modelled using the Data Vault methodology, which divides the data model into hubs, links, and satellites:

- **Hubs**: contain unique business keys and represent core business concepts.
- **Links**: Capture relationships between business keys and reflect associations between different entities.

- **Satellites**: Store descriptive attributes and context, allowing for historical tracking and auditability.

This modelling approach ensures a scalable and maintainable data warehouse structure. When applying soft business rules to establish data marts, dimensional modelling is recommended. This technique facilitates data analysis and visualisation by organising data into fact and dimension tables, supporting comprehensive and intuitive reporting.

6.6 Business Intelligence

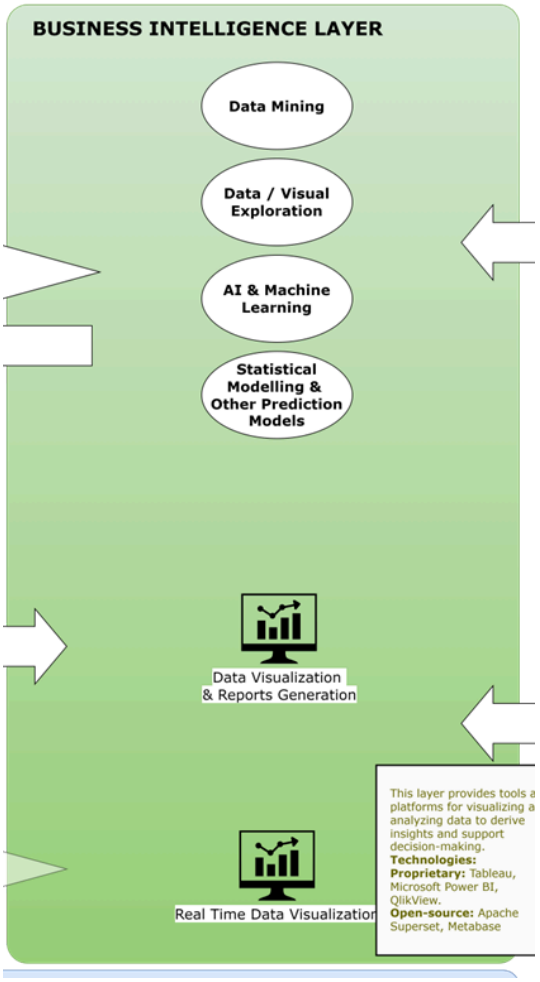


Figure 17: NHDW Architecture: Business Intelligence Layer

The Business Intelligence Layer is designed to empower data scientists, data analysts, medical researchers, and other stakeholders to leverage the extensive data within the Data Lake for advanced analytical and AI-driven applications. It is also designed to serve business users, including program managers, monitoring and evaluation (M&E) specialists, directors, and executives.

6.6.1 AI & Data Science

Part one of the layer facilitates data mining, machine learning, predictive modelling, and other data science activities, transforming raw data into actionable insights that drive healthcare innovations and improvements.

Key Features and Capabilities

1. **Access to Diverse Data**: The Data Lake stores a comprehensive collection of both structured and unstructured data from various health systems, including EMRs, lab systems, EHRs, and other healthcare databases. This extensive data repository serves as a rich source for machine learning and AI training datasets, enabling a wide range of analytical applications.
2. **Data Mining and Exploration**: Data scientists and analysts can perform data mining to discover patterns, correlations, and trends within the data. Tools and frameworks such as Apache Spark, Hadoop, and SQL-based querying will be available to facilitate deep data exploration and analysis.
3. **Machine Learning and AI Models**:
    - The AI/Data Science Layer supports the development and deployment of machine learning and AI models. Popular libraries and frameworks such as TensorFlow, PyTorch, Scikit-learn, and Keras will be integrated into the environment, providing robust tools for building and training models.
    - These models can be used for various applications, including disease prediction, early appropriate interventions, patient risk stratification, personalised treatment recommendations, and resource optimisation.
4. **Interactive Data Analysis and Visualisation**: Interactive data analysis tools and visualisation platforms like Jupyter Notebooks, RStudio, Tableau, and Power BI, Clickhouse, Superset, Metabase, ElasticSearch, and Kibana. will be provided to enable intuitive and dynamic exploration of data insights. These tools support collaborative analysis, allowing researchers and analysts to share findings and collaborate on projects effectively.

6.6.2 Data Presentation

Part two of the layer provides essential tools and platforms for visualising and analysing data to derive actionable insights, supporting evidence-based decision-making and effective policy-making.

Key Features and Capabilities

1. **User-Friendly Dashboards and Reports**: This layer offers intuitive, user-friendly dashboards and reporting tools that enable business users to easily access, understand, and interact with data. Customisable dashboards allow users to focus on key performance indicators (KPIs) and metrics relevant to their specific roles and responsibilities.
2. **Real-Time Data Visualisation**: Real-time data visualisation capabilities provide up-to-date insights, allowing decision-makers to respond promptly to emerging trends and issues. These tools help in monitoring ongoing programs and initiatives, ensuring timely and informed interventions.
3. **Interactive Data Exploration**: Interactive data exploration tools enable users to drill down into data, filter by various dimensions, and explore different data sets to uncover underlying patterns and trends. This interactivity enhances the ability to perform detailed analysis and generate comprehensive reports.

4. **Trend Analysis and Forecasting**: Trend analysis tools help in identifying historical patterns and predicting future trends. This capability is crucial for planning and resource allocation, enabling managers and executives to make proactive decisions based on predictive insights.
5. **Geospatial Analysis**: Geospatial analysis tools provide visualisations that incorporate geographical data, helping to analyse and interpret spatial relationships and patterns. This is particularly useful for public health initiatives, allowing for targeted interventions based on regional data.
6. **Data Integration**: Integration with various data sources ensures that the visualisations and analyses are comprehensive and up-to-date. This includes data from the data lake, data warehouse, and external sources, providing a holistic view of the health data landscape.

Technologies

To support the Data Analysis and Visualisation Layer, a combination of proprietary and open-source technologies can be utilised. These technologies offer a range of features to cater to the diverse needs of business users:

- **Proprietary Solutions**:
    - **Tableau**: A powerful data visualisation tool that offers interactive dashboards, real-time data analysis, and extensive customisation options.
    - **Microsoft Power BI**: An integrated business analytics solution that provides data visualisation, ad-hoc reporting, and real-time dashboards.
    - **QlikView**: A data discovery and visualisation tool that enables dynamic dashboards and interactive analysis.
- **Open-Source Solutions**:
    - **Apache Superset**: An open-source data exploration and visualisation platform designed to be highly scalable and intuitive.
    - **Metabase**: An open-source business intelligence tool that provides an easy way to generate and share dashboards and reports.

Data Governance and Security

Ensuring the integrity, security, and compliance of data within the Data Analysis and Visualisation Layer is paramount. Key aspects include:

- **Access Controls**: Implementing role-based access controls to ensure that users have appropriate permissions based on their roles, safeguarding sensitive information.
- **Data Privacy**: Ensuring compliance with healthcare regulations to protect patient privacy and sensitive information.
- **Secure Data Transmission**: Using encryption and secure protocols for data transmission to prevent unauthorised access and breaches.
- **Audit Trails**: Maintaining audit trails to monitor and record data access and usage, ensuring accountability and transparency.

By providing robust tools and platforms for data analysis and visualisation, the Data Analysis and Visualisation Layer will empower business users to make data-driven decisions. enhance program effectiveness and support strategic planning and policy-making. This layer is essential for

transforming raw data into meaningful insights that drive improvements in healthcare delivery and outcomes.

6.7 Data Governance

6.7.1 Master Data Management

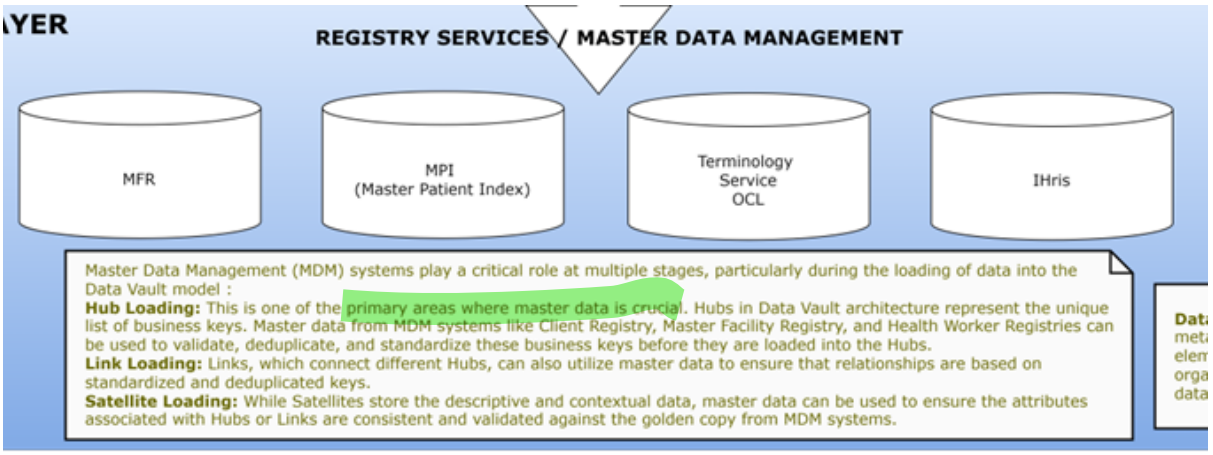


Figure 18: NHDW Architecture: Master Data Management

The registry services in the OpenHIE Architecture can be leveraged as the NHDW Master Data Management (MDM). MDM is a critical component of the NHDW architecture since it ensures the consistency, accuracy, and accountability of core data entities across the healthcare ecosystem. The implementation of MDM facilitates the effective management and harmonisation of key data entities such as clients, healthcare facilities, and healthcare workers. This harmonisation is essential for ensuring interoperability and operational efficiency within the health sector.

Core Components

- **Client Registry:** This centralised registry will maintain a unique, authoritative source of all client records. It will manage demographic and clinical data, ensuring consistency across all healthcare systems.
- **Master Facility Registry (MFR):** The MFR will serve as the definitive source of all health facility data, providing essential details such as facility name, location, services offered, and operational status. This registry is crucial for logistical and operational planning, enabling efficient resource allocation and management.
- **Master Patient Index (MPI):** Establishing a centralised MPI to uniquely identify patients across all health systems may be implemented as required. The MPI will link patient records from different sources, providing a comprehensive view of patient data and avoiding duplication. An option for this includes integration with the National ID.
- **Health Worker Registry (iHris):** This registry will contain up-to-date information about all health workers, including their qualifications, certifications, and employment history. It is vital for workforce planning, training, and certification processes.
- **Terminology Service (OpenConceptLab):** This registry allows mapping and reconciling of different concepts from different standards, such as SNOMED, LOINC, and CIEL.

Functional Requirements

- **Data Reconciliation:** Implement processes to reconcile data from various source systems to create a 'golden copy' of master data. This involves selecting the most accurate, complete, and relevant data attributes from disparate sources.
- **Data Synchronisation:** Develop mechanisms to feed the reconciled data back to all source systems, ensuring that each system reflects the most current and accurate version of the data. This synchronisation must support both batch and real-time updates to accommodate different system capabilities.
- **Data Quality Improvement:** Establish continuous data quality improvement processes that include validation, deduplication, and standardisation workflows. These processes will enhance the reliability and usability of data across healthcare applications.
- **Interoperability Facilitation:** Ensure that the MDM systems adhere to national and international data standards (e.g., HL7 FHIR, LOINC, SNOMED, etc.) to facilitate data interoperability across different health information systems. This includes the use of common identifiers and structured data formats. Seamless integration with other national and international health data systems, support for data exchange with non-health systems (e.g., social services, environmental data), and interoperability testing and validation processes.
- **Information Processing and Checking:** Create efficient processes for receiving, processing, and checking information as it enters the MDM system. This includes the implementation of automated checks for data integrity and compliance with predefined rules.

6.7.2 Metadata management

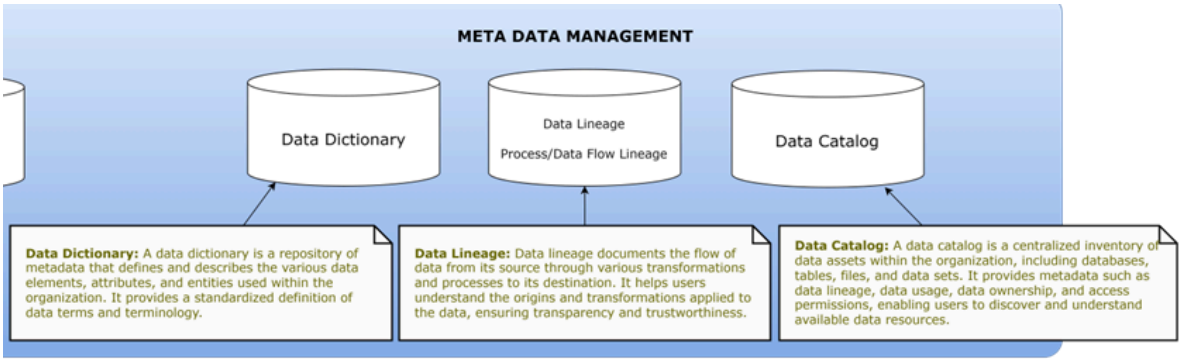


Figure 19: NHDW Architecture: Metadata Management

The Metadata and Data Governance Layer is a crucial component of the National Health Data Warehouse (NHDW) architecture. It manages metadata, data lineage, and data governance policies to ensure data quality, security, and compliance. This section outlines the technologies, processes, and governance practices essential for this layer.

Technologies

To support the Metadata and Data Governance Layer, a combination of proprietary and open-source technologies can be utilised. These technologies offer robust features for managing metadata and enforcing data governance policies.

- **Proprietary Solutions**:
    - **Collibra**: A data governance platform that provides tools for data cataloguing, data stewardship, and data quality management.

    - **Informatica Enterprise Data Catalogue**: An AI-powered data catalogue that provides comprehensive metadata management, data lineage, and data quality capabilities.
    - **IBM InfoSphere Information Governance Catalogue**: A solution that offers data governance, metadata management, and data lineage tracking.

- **Open-Source Solutions**:

    - **Apache Atlas**: An open-source metadata management and data governance platform that provides data lineage and data classification.
    - **Apache Ranger**: A framework to enable, monitor, and manage comprehensive data security across the Hadoop platform.
    - **DataHub**: An open-source metadata platform that provides metadata ingestion, search and discovery, data lineage, and more.
    - **OpenMetadata**: An open-source metadata management tool that integrates with various data sources to provide a unified view of metadata.
    - **Hive Metastore** is a critical component of Apache Hive, which is a data warehousing and SQL-like query engine built on top of Hadoop. The Metastore serves as an open-source centralised repository that stores metadata information about the data stored in Hadoop Distributed File System (HDFS) and other data sources.

Key Components

1. **Data Model Standards**: Using standards such as SNOMED, ICD, CPT, or NDC would help align to known healthcare vocabularies, enabling better understanding of clinical concepts and eliminating ambiguity in identifying patients or individuals. This includes ensuring that data elements conform to these standard terminologies and coding systems.
2. **Data Dictionary**: A data dictionary serves as a repository of metadata, defining and describing the various data elements, attributes, and entities used within the organisation. It provides standardised definitions of data terms and terminology, ensuring consistency and clarity across the NHDW.
3. **Data Lineage**: Data lineage documents the flow of data from its source through various transformations and processes to its destination. This transparency helps users understand the origins and transformations applied to the data, ensuring trustworthiness and facilitating troubleshooting and impact analysis.
4. **Data Catalogue (Data Lake)**: A data catalogue is a centralised inventory of data assets within the organisation, including databases, tables, files, and datasets. It provides metadata such as data lineage, data usage, data ownership, and access permissions, enabling users to discover and understand available data resources. The catalogue enhances data governance by providing a clear view of data asset management.
5. **Data Quality Management**: Data quality management involves defining data quality standards and implementing processes to measure, monitor, and improve the quality of data. Techniques such as data profiling, data cleansing, and data validation are employed. Following Data Vault 2.0 principles, Total Quality Management (TQM) practices are integrated to ensure high data quality and reliability. Automated data quality checks and validation processes, regular data cleansing and deduplication, and mechanisms for continuous data quality improvement are also employed.
6. **Business Rules Documentation**: Both hard and soft business rules are documented to provide clarity on data processing and transformation rules. Hard business rules are strict,

non-negotiable rules applied to ensure data integrity and compliance, while soft business rules are more flexible guidelines that support data processing and analysis.

6.8 Monitoring

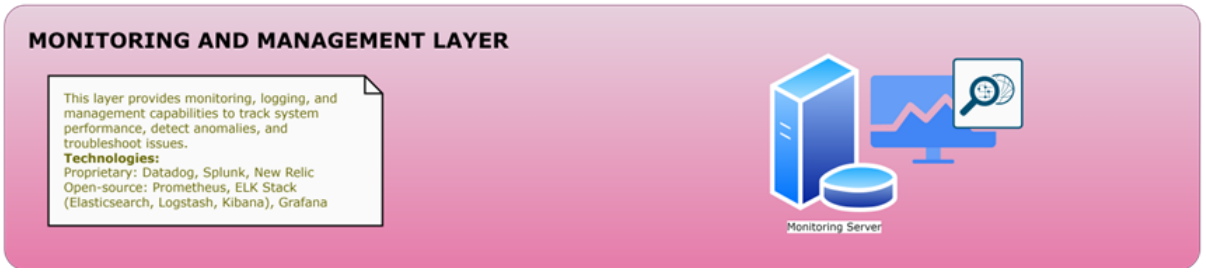


Figure 20: NHDW Architecture: Monitoring and Management Layer

The Monitoring Layer is an essential component of the National Health Data Warehouse (NHDW), providing comprehensive monitoring, logging, and management capabilities to ensure the system's health, performance, and reliability. This layer tracks system performance, detects anomalies, and facilitates troubleshooting across all layers of the NHDW architecture. By implementing robust monitoring solutions, stakeholders can ensure that the data warehouse operates efficiently and meets the required service levels.

Technologies
To support the Monitoring Layer, a combination of proprietary and open-source technologies can be employed. These technologies offer extensive features for real-time monitoring, alerting, and logging:

- **Proprietary Solutions**:
    - **Datadog**: A cloud monitoring service that provides real-time monitoring, alerting, and visualisation of metrics and logs from various sources.
    - **Splunk**: A powerful platform for searching, monitoring, and analysing machine-generated data in real time, providing insights into system performance and security.
    - **New Relic**: A comprehensive monitoring and observability platform that offers real-time analytics, performance monitoring, and alerting capabilities.
- **Open-Source Solutions**:
    - **Prometheus**: An open-source monitoring and alerting toolkit designed for reliability and scalability, ideal for collecting and querying metrics.
    - **Grafana**: An open-source platform for monitoring and observability that integrates with Prometheus and other data sources to provide powerful visualisations and dashboards.

Key Features and Capabilities

1. **Real-Time Monitoring**: The Monitoring Layer provides real-time monitoring of system performance, tracking key metrics such as CPU usage, memory utilisation, disc I/O, network

latency, and application performance. This continuous monitoring helps identify performance bottlenecks and ensures the system operates within acceptable parameters.

2. **Anomaly Detection**: Advanced anomaly detection algorithms analyse the collected metrics to identify unusual patterns or deviations from normal behaviour. This proactive approach enables early detection of potential issues, allowing for prompt intervention before they impact system performance or data integrity.
3. **Logging and Log Management**: Comprehensive logging capabilities capture detailed logs from various components of the NHDW architecture, including data ingestion processes, ETL jobs, database queries, and application interactions. Centralised log management solutions aggregate these logs, making it easier to search, analyse, and troubleshoot issues.
4. **Alerting and Notifications**:
    - Configurable alerting mechanisms notify administrators and relevant stakeholders when predefined thresholds are breached or anomalies are detected. Alerts can be sent via email, SMS, or integrated with incident management tools, ensuring timely response to critical issues.
5. **Dashboarding and Visualisation**:
    - Interactive dashboards and visualisation tools provide a comprehensive view of system performance and health. Technologies like Grafana offer customisable dashboards that display key metrics and trends, enabling stakeholders to make informed decisions based on real-time data.
6. **Performance Reporting**:
    - Regular performance reports summarise system health and performance over specific periods. These reports help identify recurring issues, track performance improvements, and support capacity planning efforts.

6.8.1 Performance Monitoring Tooling

1. Query response times
2. Data throughput
3. Data storage capacity
4. System load: CPU, memory, I/O, network of different nodes
5. Alerting to aid system administrators to identify degradation of performance and optimisations

6.8.2 Cost Efficiency Management

Cost management tools are especially important where infrastructure resources are provided by a service that is billed per usage. Need a framework to monitor and balance the costs vs. performance of the system. The framework should be able to justify essential costs by demonstrating key performance degradation implications.

6.9 Security



Figure 21: NHDW Architecture: Security Layer

Ensuring the integrity, security, and compliance of data within the NHDW is paramount. Key aspects include:

- **Access Controls**: Implementing role-based access controls to ensure that users have appropriate permissions based on their roles, safeguarding sensitive information.
- **Data Privacy**: Ensuring compliance with healthcare regulations to protect patient privacy and sensitive information. This includes compliance with international data protection regulations (e.g., GDPR, HIPAA).
- **Secure Data Transmission**: Using encryption and secure protocols for data transmission to prevent unauthorised access and breaches.
- **Audit Trails**: Maintaining audit trails to monitor and record data access and usage, ensuring accountability and transparency.
- **Monitoring and Auditing:** Continuous monitoring and auditing of data access and usage to detect and prevent unauthorised activities and ensure compliance with governance policies.
- **Ethical AI Practices**: Adherence to international ethical guidelines for AI usage, ensuring transparency, fairness, and accountability in AI-driven decisions and recommendations.
- **Encryption**: Encryption mechanisms to protect data both at rest and in transit, safeguarding against unauthorised access and breaches. Implementation of advanced encryption methods and secure data access protocols. This includes identity and access management, detective controls, infrastructure protection, data protection, and incident response.

The Data Access and Security Layer is a critical component of the National Health Data Warehouse (NHDW), responsible for enforcing security measures and access controls to protect sensitive health data and ensure compliance with regulatory requirements. This layer establishes a centralised approach to managing users and role-based access control (RBAC) across all systems and applications within the NHDW architecture. By implementing robust security and access management practices, the NHDW can safeguard data integrity, confidentiality, and availability.

Technologies

To support the Data Access and Security Layer, a combination of proprietary and open-source technologies can be employed, providing comprehensive and scalable solutions for managing access and ensuring security:

- **Proprietary Solutions**:
    - **AWS Identity and Access Management (IAM)**: A service that helps manage access to AWS resources securely by controlling permissions and defining roles for users and services.

    - **Microsoft Azure Active Directory**: A cloud-based identity and access management service that provides secure access to resources and applications, including user authentication and RBAC.
    - **Google Cloud Identity and Access Management (IAM)**: A service that allows fine-grained control over who can access specific Google Cloud resources, ensuring secure and compliant access management.

- **Open-Source Solutions**:

    - **Apache Knox**: A gateway for providing secure access to Hadoop and other big data ecosystems, offering authentication, authorisation, and auditing services.

    - **Apache Ranger**: A framework to enable, monitor, and manage comprehensive data security across the Hadoop platform, providing centralised security administration, fine-grained authorisation, and auditing.
    - **Keycloak**: An open-source identity and access management solution that provides single sign-on (SSO), user federation, strong authentication, and RBAC.

Key Features and Capabilities

1. **Centralised User Management**: Implementing centralised user management to streamline the administration of user identities and access permissions. This approach ensures consistent and efficient user provisioning, de-provisioning, and role assignments across all layers of the NHDW architecture.
2. **Role-Based Access Control (RBAC)**: Establishing RBAC to enforce security policies based on user roles and responsibilities. RBAC ensures that users have access only to the data and resources necessary for their roles, minimising the risk of unauthorised access and data breaches.
3. **Authentication and Authorisation**: Implementing robust authentication mechanisms, such as multi-factor authentication (MFA), to verify user identities. Authorisation processes ensure that users can access only the data and resources they are permitted to, based on their roles and access levels.

    Data Governance and Security Policies

    Implementing comprehensive data governance and security policies is essential to maintaining the integrity and security of the NHDW.
    Access Control Policies: Defining and enforcing access control policies to ensure that only authorised users can access sensitive data and resources.
    Data Privacy Policies: Establishing data privacy policies to protect personal health information and comply with regulatory requirements.
    Incident Response Policies: Developing incident response policies to effectively manage and respond to security incidents, minimising their impact, and preventing future occurrences.
    Regular Security Audits: Conducting regular security audits to assess the effectiveness of security measures and identify areas for improvement.

6.10 Summary

In summary, Chapter 6 outlines the technical strategy for implementing the NHDW, focusing on scalability, interoperability, and alignment with enterprise architecture principles. The NHDW

integrates diverse data sources through a multi-layered architecture, with each layer designed to ensure efficient data flow and processing.

Data ingestion is a key component, employing real-time ingestion, scheduled ETL, pull, and push-based techniques supported by secure protocols. We also highlight the role of OpenHIM's as an information mediator service. Data is then stored in a data lake as raw data and in warehouses or marts for analytics-ready uses, optimising flexibility and performance.

The NHDW emphasises business intelligence through AI and data science, paired with user-friendly presentation tools for actionable insights. Strong governance frameworks, including master data and metadata management, ensure data quality, compliance, and traceability.

Finally, robust monitoring tools and a security framework safeguard system performance, manage costs, and protect data integrity, providing a comprehensive technical foundation for a scalable and reliable health data system.

Chapter 7: Infrastructure Requirements assessment

In determining suitable requirements for the hosting infrastructure of the NHDW. We consider industry standards and configurations that support the functional and non-functional requirements from the ministry of health. Hence, the infrastructure needs to be performant, robust, scalable, reliable, available, and secure. This section outlines the key infrastructure requirements necessary to support the NHDW effectively.

To manage feasibility, usability, and sustainability of the infrastructure, we consider a phased approach to the requirements of the data centre, its buildings, and compute infrastructure requirements.

7.1 Data Center Assessments

To assess the ability of a data centre and ensure maximum availability and performance in line with the functional and operational requirements of the NHDW. We recommend the UpTime Institute Tier classifications. The Uptime Institute's Tier Certification is by far the most referenced industry standard for data centre reliability. The tier classification system is the globally recognised standard for data centre performance, with over 1,000 Certifications for Data Centre Design, Construction, and Operational Sustainability issued in over 85 countries [39].

7.1.1 Approach

To assess that a data centre infrastructure meets requirements for the NHDW, we recommend that an Uptime certified professional services company is in charge of the site assessments. We recommend that UpTime Institute Tier II-certified Data centres in design and construction should be considered for hosting the NHDW. For data centres that are not Tier II certified yet, we recommend that an SIR take place to ensure an upgrade that minimises risk.

A Site Infrastructure Review (SIR) is an in-depth evaluation of the physical infrastructure that supports an organisation's data centre or critical IT environment. The purpose of a SIR is to assess the current state of a site's infrastructure to ensure that it is resilient, reliable, and capable of meeting both operational and business continuity requirements. The SIR provides a comprehensive report that provides the Tier Gap Analysis and a risk assessment of the facilities.

The Tier Gap Analysis is a critical evaluation process used to assess the difference between an organisation's current data centre infrastructure and the requirements needed to meet a specific UpTime Institute Tier Standard. Key components of the Tier Gap analysis are the current Tier level of the facility (based on existing infrastructure), a list of gaps that need to be addressed to meet a desired Tier level, prioritised recommendations for improvements, and an action plan for closing those gaps, including a roadmap for infrastructure upgrades, project costings, process changes, and any necessary certifications.

Although the Tier Gap analysis highlights gaps and the risks that come with these gaps, the risk assessment provides the data centre with those direct problem areas that, when addressed, enable the data centres to meaningfully reduce operational failures. A risk assessment is a systematic process used to identify, evaluate, and prioritise potential risks that could impact the reliability, security, and performance of a data centre's infrastructure. It contains identified risks and risk mitigation strategies.

7.1.2 Parties Involved

The SIR process, particularly in alignment with UpTime Institute's Tier standards, involves a range of internal and external stakeholders who collaborate to assess the current state of a data centre's physical infrastructure.

A list of internal stakeholders involved in the process will include MOH executive leadership, facilities management, Data centre operations teams, and IT teams. These stakeholders help ensure alignment of the SIR with strategic directions, ensure resource allocation to support the project, and provide critical inputs on facility readiness for assessment.

A list of external stakeholders involved includes UpTime Institute-certified professionals and engineers. These UpTime Institute's Tier standards and best practices professionals lead the SIR process, conducting detailed assessments of the data centre's infrastructure and providing independent evaluations of its readiness for Tier certification.

7.1.3 Gap Analysis Requirement

To estimate the timeline and budget required for a SIR. We require the desired Tier Standard Certification (Tier I, Tier II, Tier III, or Tier IV), facility size (in m/sq), and the base load of the data centre, among other requirements.

7.2 Compute Resource Requirements

Given both functional and non-functional requirements, industry standards, and estimated workloads. We estimated the demands of the data platform for the short and long term. These requirements have been documented in Table 13 below:

Table 13: Data Platform Requirements

Resources	Expectation	Criteria
vCPUs	1500	cores
RAM	2	TB
SSD Storage	0.5	PB
HDD Storage	1.5	PB
Total Storage	2	PB
GPUs	20	units
Linux-based OS	GNU/Linux debian ubuntu	name and distributions
Windows OS	Windows server 2019 data centre or greater	versions and editions

The primary objective of the data warehouse is the centralisation of all healthcare-related data into one data platform, aiding decision-making through value-driven analytics and insights mined from various subject areas. We recommend a phased procurement and provisioning of hard discs and compute capacity to meet the resource requirements of the data warehouse. Hence, we reduce the project initiation requirements by prioritising the resource capacities needed for the initial phases of the project while giving the ministry time to plan for a phased upgrade of the environment to meet platform needs.

The phases outlined follow the Implementation Roadmap phases. Here are the objectives and assumptions per phase.

- **Phase 1: Prioritise centralisation of raw data**. Before the centralisation of raw data, we will prioritise high-value data sources while considering the data sizes of these data sources. Small to medium-sized data sources may offer ease and less problematic uploads during data transfer in comparison to the largest data sources. Hence, an initial 100 TB storage specification is recommended to start with, which is roughly calculated as a sum of current storage requirements of major data sources listed in Chapter 3. The recommendation starts on the lower end to also ease initial procurement requirements. To process this data, we recommend 500 vCPU cores and 500 GB of RAM.
- **Phase 2: Operationalize sample narrow business use cases (MVP)** To adequately process data for data analytics and warehousing, additional storage is recommended, as duplication of data will happen during data warehousing and analytics. An additional 500vCPU and 500 GB of RAM will help support these workloads, along with the provisioning of GPU cores to support graphics processing use cases.
- **Phase 3: Develop master & metadata components**. By the third phase, the platform will have most data sources centralised, and the data volume will be near anticipated volumes. Hence, we recommend the provisioning of 500 TB of SSD storage to support quicker reads and writes to disc, while adding an additional 900 TB of HDD storage would be required to

support future data analytics and storage needs of the platform's use cases. An additional 500 vCPU cores and 1 TB of RAM are also required to ensure longer-term platform stability.

To review timeline estimates for each phase, kindly refer to the Implementation Roadmap and Project Timeline sections of this document.

In the table below, we present the resource recommendations for each phase of procurement and provisioning

Table 14: Compute Resource Requirements

Resources	Phase 1	Phase 2	Phase 3	Total	Criteria
vCPUs	500	1000	1500	1500	cores
RAM	0.5	1	2	2	TB
SSD Storage	0	0	0.5	0.5	PB
HDD Storage	0.1	0.6	1.5	1.5	PB
Total Storage	0.1	0.6	2	2	PB
GPUs	0	10	20	20	units
Linux-based OS	GNU/Linux debian ubuntu				name and distributions
Windows OS	Windows server 2019 data centre or greater				versions and editions

7.2.1 Storage

Therefore, the storage requirements of such a system need to cater for the large disparate data sources relating to health that exist within the country. The requirements also include the processing and storage of cleaned, transformed, and aggregated data that would be stored and utilised for further processing across the platform. Considering the source system data volumes, we estimate a 2PB storage to cater for data storage and duplication during the processing.

Implementing RAID (Redundant Array of Independent Discs) storage technology ensures data redundancy and reliability. RAID configurations help protect against data loss in the event of hardware failures by distributing data across multiple discs.

7.2.2 Processing

High-performance computing systems are required for processing large amounts of data through batch and real-time ETL workloads, as well as for data warehousing analytics, data visualisation, and machine learning. CPUs and GPUs are necessary for distributed data processing and accelerated deep learning workloads for Intelligence use cases. The number of vCPU cores specified for the initial requirements are those necessary for running necessary ETL and analytics workloads in the initial maturity phases of the NHDW. The number of GPUs specified caters to experimental deep learning workloads, which we anticipate will grow in demand after stability and productionization of the NHDW.

7.2.3 Recommended OS

The Linux operating system specified is a well-known stable operating system for big data clusters. The set of GNU/Linux packages in the debian ubuntu distribution is a good balance between usability and stability. It is a lightweight yet stable compilation of libraries that allows clusters to run

efficiently. Windows Server 2019 data centre or greater has been specified to support Windows applications that may be used for data warehousing (like SQL Server), data visualisations (like Power BI), and other workloads that require Microsoft software. Windows Server is optimised for the Microsoft ecosystem.

7.2.4 Other Compute Considerations

In addition, the data centre should be able to support the following resource scalability requirements:

1. **On-Demand Resource Allocation:** The ability to add virtual servers or storage servers quickly and seamlessly is critical. This ensures that the NHDW can handle increased data loads without significant downtime or performance degradation.
2. **Cloud Environment:** a local cloud (private cloud) that supports the flexibility to scale resources on demand. This environment should offer elasticity, allowing the NHDW to adjust resources based on current needs efficiently.
3. **Resource Estimation:** While it is challenging to precisely define the hardware resources required for data storage and computing power, a cloud-based infrastructure enables the NHDW to start with a baseline configuration and scale up as data volume and processing needs grow.

7.2.5 Catering for Infrastructure Growth Demands: Rule of Thumb

To meet the growing demands of the NHDW. The data centre is required to be able to support a scalable and flexible architecture that anticipates data volume increases or newer processing requirements without precise growth rates over the next five years. Elastic storage solutions may also allow for on-demand scaling, and data tiering strategies based on data access frequency may help manage costs and performance, while archived data can be stored in areas of lower cost and slower storage access. Effectively managing less predictable scaling requirements will be crucial to maintaining the stability and functional requirements of the NHDW within the chosen data centre.

7.3 Buildings Requirements

The data centre building requirements for the NHDW are guided by the UpTime Tier II design, construction, and operationalisation standards [40]. Local Regulation Compliance is also a necessary undertaking to ensure the data platform operates within legal frameworks and industry standards, influencing geographical location choices.

UpTime Tier II Design and Construction Standards and ISO 27001 have been selected as necessary certifications for the NHDW. The UpTime Tier II standards focus on redundant components that reduce disruptions from planned and unplanned activities. Some of the general considerations include at least a tenant building type, raised floor, UPS, and engine generator, which we go into further details about in this section. However, note that although Tier II provides 99.741% infrastructure availability, maintenance of the critical power path and other parts of the site infrastructure will require a processing shutdown.

7.3.1 Load Profile

The load profile determines the amount of available load that the data centre can provide vs. the amount that has been allocated to the running services and applications. This is important for capacity

planning to ensure the system can handle the expected load while ensuring performance and avoiding overprovisioning or underprovisioning.

Table 15: Load Profile Requirements

Load Profile	Initial Requirement	Criteria
Total Available / Assigned	1.2:1	MW
Total Actual load	0.6	MW
Peak Load	1	MW
Total Available Power	1.2	MW

A good load profile ensures that the available power is 10–30% more than the highest recorded peak load value and the actual load is 40% less than the peak load. The long-term demands may scale to 5X to cater for the increased load through increased services running on the cluster, concurrent runs, IOPs, and compute capacity. PUE (power usage effectiveness) target is expected to be between 1.2 and 2.0.

7.3.2 IT Floor Space

UpTime Tier II specifications require the IT floor space to be raised by 18 inches and that the floor load capacity is 4.79 kilowatts per metre squared. Ensuring that potential fire hazards can be suppressed through water sprinkler systems is also at least necessary.

7.3.3 Facilities: Backup Power

All backup power should be able to provide redundancy beyond the peak load to ensure reliability and availability of the infrastructure comfortably, even if a UPS module or standby power fails or is undergoing maintenance. Hence N+1 has been selected for both UPS and standby power redundancy and configuration. This means we have one more Gen Set than is technically needed to serve the peak load.

7.3.4 Connectivity and latency

High-speed networking is essential for efficient data ingestion from multiple source systems and for facilitating quick data access and transfer within the NHDW. The infrastructure should support high-bandwidth, low-latency network connections to ensure smooth data flow.

Table 16: Connectivity and Latency Requirements

Connectivity and latency	Initial Requirement	Criteria
Bandwidth	10 Dedicated connection Gbps	Gbps
Latency Between Zones	100 ms if across regions	ms
Cross-Zone Data Replication	0	Y/N
Inter-Zone Bandwidth	1 Dedicated connection Gbps	Gbps

The connectivity and latency affect the data transfer rate across the platform. Dedicated connections are important to ensure the rate of data transfer does not fluctuate widely, ensuring that we have enough bandwidth available to handle peak usage without congestion. Latency between zones influences distributed applications during compute cycles; for example, map-reduce applications when using the Hadoop framework send data to different nodes when distributing data and collect data at a single node.

Cross-Zone Data Replication is not necessary at a data centre level during the short-term phase; however, for long-term needs, it is necessary to ensure data synchronisation. Inter-zone bandwidth that is on a dedicated connection reduces the time to transfer large datasets and enables faster disaster recovery and load balancing across zones.

7.3.5 Security

A multi-layered approach to security has been adopted to protect against data breaches and to ensure data privacy. The following four layers of on-premise security have been set as a requirement for the NHDW Data Centre:

1. **Physical Mantrap Entry**: This is a physical security feature to ensure only authorised access by ensuring a door that provides access into the facility has to close before another door opens.
2. **Human 24/7 on-site security**: Continuous on-site security personnel presence increases the data platform's security against physical threats and intrusions. Tier II necessitates 1 shift for staffing.
3. **Biometric**: Biometric security involves automating recognition of vetted individuals' physical access to the data centre through physical characteristics, thereby safeguarding the hardware that runs the data platform.
4. **CCTV surveillance**: Provides monitoring and incident record capabilities through CCTV cameras for enhanced security and auditing purposes.

7.3.6 ISO Certifications

A wide range of ISO certifications are recommended for data centres; however, we have focused initially on ISO 27001, which is the Information Security Management System (ISMS), as an initial requirement for the data centre. In the long term, the list of ISO certifications required for the infrastructure is as follows:

- ISO 22301
- ISO 27001
- ISO 9001
- PCI DSS
- SOC 1 Type I
- SOC 2 Type II
- ISO 14001
- ISO 50001
- ISO 45001

7.3.7 Disaster Recovery and Business Continuity

To minimise downtime, data loss, and preserving business operations. Appropriate recovery from disasters through a well-formulated DR plan is crucial. The UpTime Tier II specifies an annual

site-caused IT downtime of 22 hours. However, for each system downtime, an RPO and RTP are necessary. Below are the requirements for the disaster recovery:

Table 17: Disaster Recovery and Business Continuity Requirements

Disaster Recovery and Business Continuity	Initial Requirement	Criteria
Disaster Recovery (DR) Plan	Y	Y/N
Recovery Point Objective (RPO)	24 hours	Time
Recovery Time Objective (RTO)	4 hours	Time
Backup Systems	Y	Y/N

The RPO (Recovery Point Objective) impacts the maximum acceptable data loss in terms of time, affecting the data platform's recovery strategies. Hence, for the NHDW, there is an acceptable maximum 24-hour period in which data might be lost due to major incidents and specifies the necessary backup frequency.

The RTO (Recovery Time Objective) determines the targeted duration within which services should be restored following a disruption affecting the continuity of platform services. For the NHDW, there is an acceptable 4 hours of downtime at any instance of data centre failure. Therefore, the capacity and reliability of backup systems directly affect the data platform’s resilience against data losses and service interruptions.

7.3.8 Infrastructure Project Plan

A comprehensive project plan is essential for ensuring the systematic deployment and scaling of the data platform within defined timelines. This plan should include regular testing of disaster recovery and business continuity procedures to guarantee data availability and resilience in the event of an emergency. Sharing the project plan with key stakeholders is crucial for managing expectations, fostering transparency, and minimising service disruptions related to building schedules. By establishing clear goals, timelines, and contingency measures, the project plan supports a smooth and reliable implementation process.

Key goals of an infrastructure project plan are to:

1. **Deployment Plans and Timeliness:** Ensure that the project follows a structured timeline, facilitating efficient deployment and scaling of the platform.
2. **Disaster Recovery and Business Continuity:** Highlight the need for regular testing of disaster recovery and business continuity procedures to maintain data availability and protect against potential emergencies.
3. **Stakeholder Communication:** Underscores the importance of sharing the project plan with relevant stakeholders to set clear expectations, keep them informed of the project’s progress, and minimise any service interruptions associated with build schedules.

7.4 Summary

In summary, Chapter 7 defines the infrastructure requirements essential for supporting the NHDW, focusing on scalability, reliability, and adherence to international standards.

The chapter begins by assessing the data centre requirements, emphasising Tier II design and construction to ensure robust operations. We highlight the necessity of a gap analysis that reveals

areas for improvement. We also highlight that the involvement of key stakeholders ensures alignment with technical and operational needs.

For compute resources, the chapter details phased requirements for storage, processing, and operating systems, alongside guidelines for accommodating infrastructure growth. It emphasises the need for flexible, scalable resources to meet future demands.

The building requirements address critical aspects such as IT floor space, backup power, connectivity, and security. Standards like ISO certifications and disaster recovery plans are underscored to ensure resilience and business continuity.

Hence, Chapter 7 provides a comprehensive framework to establish secure, scalable, and efficient infrastructure for the NHDW.

Chapter 8: Governance

The Ethiopian NHDW must establish robust governance and security, which is crucial to ensuring data integrity, privacy, and compliance with national and international legal standards [41].

The NHDW will need to comply with the following Ethiopian laws and legislation that govern access to health and related data.

- The mass media proclamation (Proclamation No. 590/2008).
- E-commerce Law, Computer Crime Proclamation (Proclamation No. 958/2016).
- The 2005 criminal code of the Federal Democratic Republic of Ethiopia.
- The 1960 Ethiopian Civil Code, Access to Genetic Resources and Community Knowledge and Community Rights Proclamation (Proclamation No. 482/2006).
- Proclamation concerning inventions.
- Minor Inventions and Industrial Designs (Proclamation No. 123/1995).
- Copyright and Neighbouring Rights Protection Proclamation (Proclamation No. 410/2004).
- Electronic Signature Proclamation. Proclamation No. 1072/2018.
- Personal Data Protection Proclamation No. 1321/2024, July 24, 2024 [42].

During the course of the project, it is required to form a multi-disciplinary committee comprising stakeholders from government agencies, healthcare providers, researchers, and IT experts to oversee the project and establish specialised groups focusing on specific areas such as data quality, security, and user engagement. This multi-disciplinary committee will be responsible for creating the data governance framework defining the ownership and responsibilities for data and its management and access, and provide guidance to oversee data quality, integrity, and compliance with policies and establish standards for data accuracy, completeness, consistency, and timeliness.

The NHDW must be monitored and evaluated by defined metrics to evaluate the effectiveness of governance practices and data quality and defined metrics for periodic audits to assess compliance with governance frameworks and identify areas for improvement, in compliance with the Information Revolution Roadmap, the draft revised health policy, the draft health service proclamation of Ethiopia Guidelines implemented by EPHI, CSA HDSS, and the IRB guideline from the MoST. The health data governance principles developed and recently launched by Transform Health can be used to provide more guidance [43].

The NHDW must allow expanding analytic content and user adoption and be ready to receive other systems from other sectors, identify additional data sources by continuously assessing new data needs and sources and system updates, and allow for iterative updates to accommodate new requirements.

Chapter 9: Risk Management

The risks for the implementation of a national data warehouse include factors that can be beyond the control of the implementer and can negatively impact the organisational ability to achieve the established goals. A risk assessment of a data warehouse is a process of identifying, evaluating, and prioritising the potential threats and vulnerabilities that may affect the data quality, security, performance, and usability of the data warehouse and its related applications.

The risk assessment process includes the following:

- Identify hazards: analyse the data sources, data flows, data transformations, data models, data storage, data access, data governance, and data usage of the data warehouse and identify possible sources of errors, inconsistencies, breaches, failures, or inefficiencies that may compromise the data quality or security.
- Determine who is at risk: Identification of the stakeholders who may be affected by the hazards, such as data warehouse developers, data analysts, data consumers and owners such as physicians and MoH program staff, or data regulators such as funders.
- Evaluate risks: assessments of the likelihood and severity of each hazard, as well as the existing controls or mitigations that are in place to prevent or reduce the risks.
- Record observations: documentation of the findings and recommendations of the risk assessment, such as the identified hazards, the affected stakeholders, the evaluated risks, the proposed actions, and the responsible parties.
- Review and update protocols: monitoring and reviewing the implementation and effectiveness of the risk management actions, as well as updating the risk assessment periodically to reflect the changes in the data warehouse environment, requirements, or regulations.

The draft risk matrix below aims to prioritise the risks based on their probability and impact on the successful implementation of the NHDW project in Ethiopia, along with some proposed mitigation strategies.

Table 18: Draft Risk Matrix

Risk categories	Significant risk factors identified	Actions to mitigate the risks
External Dependency	1. Overlapping scope with other systems developments 2. Change in governmental policies on data management 3. Sustainability of infrastructure service provider 4. Lack of sustainable funding for NHDW development and lack of attaining sufficient business value to attract long-term non-donor funding or commercialisation.	1. Clear identification of scope and socialisation in MOH governance and associated stakeholders. 2. Participate in governmental policy-making to advise non-disruptive regulations. 3. Assess sustainability of infrastructure service provider 4. Ensure business case clearly defined and mapped to planned funding sources

Organisational risks	1. Number of user areas/decision-makers 2. Unavailability of stakeholders 3. Non-existence of organisational and change management plans 4. Budgeting limitation	1. High level buy-in to the objectives and activities of the NHDW project 2. Well-defined budget 3. Well-defined change management plans
Planning risks	1. Unrealistic deadlines 2. Number of change request that come out from the validations sessions 3. Informal control procedures 4. Planned resources available	1. The lifecycle of the project will be conducted in phases 2. Change requests will be identified, tracked, approved, and prioritised
Business case risks	1. Reporting discontinuity 2. Missed data and data loss, irretrievable data	1. Conduct gap analysis and requirement validation 2. Data migration mapping specifications
Technological risks	Environmental: 1. New/unfamiliar technology 2. Use of development methods/standards 3. Project-related: 4. Complexity of functions 5. Complexity of database 6. Shared databases with undefined applications 7. Number of physical systems interfaces 8. Package solution available 9. Multi-level hardware implementation 10. Data security 11. Operational: 12. Failover internet connections and infrastructure downtime 13. potential disruptions to the production environment.	1. Use load balancers or routers that can automatically switch to backup connections in case of primary connection failure. 2. Implement a testing environment

Other	1. Environmental changes and Natural disasters 2. Armed Conflicts	Assess likelihood of of large-scale disasters and develop appropriate mitigation plan

Chapter 10: Implementation roadmap

The NHDW is a large-scale enterprise vision, and knowing where to start and how to prioritise and sequence activities is important. It is important to appreciate what the most determinant business drivers and constraints are. While there are many very important business drivers and constraints, it is necessary to appreciate which of them are the most determinant factors in project planning. A summarised presentation of this content can be found at this reference [44].

10.1 Prioritising constraints

A. Budget

- The single most determinant factor is the budget constraint, which will help shape project team staffing, infrastructure decisions, development capacity, use case complexity, data completeness efforts, and the degree of collaboration that can be centrally sustained. The budget factor should be. For this reason, a structured approach to cost estimation is recommended. By default, early project milestones should be defined that are broken down into minimum viable products associated with a minimal cost to demonstrate value. Scaling up costs should be associated with a related scale up in investment.

B. Legal framework

- This is almost equally important because it is a constraint that is not easily solved by adding more budget. Data jurisdiction, security, and privacy laws place constraints on how the project can be implemented. These can be solved by the appropriate investments but may incur project delays that should be planned for. In addition, data sovereignty in some cases is embedded in legal frameworks that may impede the MOH from centralising required data from other institutions or sectors.

C. Inter-organisational collaboration

- **Governance:** a centralised system that services a plurality of stakeholders from different activities, organisations, and sectors will require a robust governance structure. It will take time for governance processes to build traction and efficiency and for interested organisations to understand how to engage the governance structure appropriately. The early stages of the project will involve repeated negotiations and meetings to build alignment and understanding. The policy framework (e.g., data sharing agreements) needs to rapidly be in place to clear the way for participating organisations to send data sources to the NHDW.
- **Shared interest and buy-in:** The MOH will initially rely on the buy-in of other institutions to participate in the NHDW project by sending data sources and collaborating on use case development and other initiatives. The MOH will need to consider what will drive the buy-in of the affected organisations and structure the collaboration around those shared interests, such that participating organisations agree on the importance of expanding their human resources in advancing this project.

D. **Technical constraints:**

- **Data transmission limitations**: There are technical and logistical constraints to the movement of some data sources to a central server. For example, lack of internet connection, lack of technical staff to perform the data movement, etc. The NHDW project team should accommodate offline data submissions but also anticipate that data completeness will be a major challenge in the early stages of the project.

10.2 Prioritising business drivers

The approach to defining business requirements is summarised in Section 2.6 The objectives are broadly defined and specific use cases must be identified and prioritised.

Let us highlight the guiding principle of a hybrid approach combining top-down and bottom-up approaches:

A. **Bottom-up:** More precise planning will result in reduced development costs. Without good planning and strong pre-analysis, a potentially high-cost development effort will be introduced to respond to unplanned data discoveries or user requirements:

   - **Unplanned data discoveries:** The NHDW integrates data from sources that are disparate across multiple organisations and even sectors. This can present a challenge to the project team to conduct effective implementation planning. Late discoveries of the details of a data source can incur development costs. The NHDW project team and its collaborators will plan much more effectively should the FAIR data management principles be prioritised [17].
   - **Unplanned user requirements:** Considering that the NHDW will be a new system, it is natural that in early phases the business use case requirement definitions may lack specificity and precision until initial use cases become operational and precedence is introduced. However, unplanned user requirement changes will incur development costs. Hence, it is advisable to start with *initial use cases that are simple and narrow (specific)* to reduce the impact of changes in user requirements. Once precedence is set, larger user requirements can be accepted, with a higher standard of specificity expected from use case requirements documents, leveraging learnings from initial use cases implemented.

   A maturity model is proposed in the next section to address a strategy to mitigate the above challenges.

B. **Top-down:** It will be important for the MOH to demonstrate the business value of the NHDW as early as possible to attract further investment, which is necessary for the sustainability of the project. A major concern of various business stakeholders in the sector will be how project costs will be curtailed and a clear plan on how to ensure a high return on investment. The MOH should prioritise early use cases that provide a rapid path to demonstrating operational business value. End-to-end implementation of early use cases will enable the project team and broader stakeholders to rapidly learn and refine planning and development processes to ensure the most efficient use of resources for the development of further use cases.

10.3 Phase 1: Prioritise centralization of raw data

Goal:

- As far as it depends on MOH, to have **at least one complete copy** of every data source that is eligible for the NHDW.
- Enable detailed data discovery of all data sources in central place, which is the raw staging area of the NHDW

The NHDW project team will implement the data ingestion layer and the staging area of the data lake. This staging area will contain a copy of all the raw as-is data sources in a secure perimeter. The NHDW in Phase 1 just needs to store at least one copy of each data source, which can be the latest version of that data source sent. If the NHDW has more storage capacity, then it can store older copies of the data sources as well, but to limit the budget required, it is recommended to store at least one copy of each data source.

After the multi-mode data ingestion facilities are ready and the infrastructure store space of the staging area is prepared to store a copy of all eligible data stores, then the MOH will mobilise data owners of the data sources to send the data by whatever mode is most feasible for them.

For the MOH to successfully mobilise different data owners to send their data to the NHDW, barriers of entry should as much as possible be removed beforehand:

- Implement mechanisms to allow all basic forms of data ingestion, i.e., make it as simple as possible for data owners to send data in the mode they are most comfortable with.
    - Batch ingestion
    - API ingestion
    - File ingestion by FTP
    - Data replication tools
    - Upload tools for data owners who need to send data of systems that are without internet
- Prepare the staging area as a secure zone
    - Infrastructure security must be well configured
    - Limit staff access to staging areas to minimise security exposure; for example, only the data ingestion team and limited MOH authorised personnel have access to this zone.
- Provide evidence of the necessary compliance standards to allow the storage of PII data in the secure staging zone. PII data will not be moved out of the secure staging zone, but centralising data sources as-is at the NHDW enables comprehensive data profiling and enables the NHDW to conduct effective deduplication procedures. Data is deidentified before being moved into the data lake and data warehousing components of the NHDW.

MOH can initiate a strategy to maximise the buying of participating organisations:

- Removal of barriers to entry must be in place as mentioned above, so as not to give reasons for participating organisations to defer sending of their data.
- Enlist data sources that are loaded in the NHDW for early consideration for integration in Phases 2, 3, and 4, so that participating organisations can be motivated to load their data sources as soon as possible.

An important principle to follow in the maturity progression of the NHDW is to build an **inclusive forum** where any participating organisation can collaborate, but in parallel to that, MOH should have a plan to **advance without dependency** on participating organisations. So applying the principle to this phase:

- MOH fosters inclusivity by removing barriers to entry and availing different tools for data ingestion. In addition, a broad set of stakeholders are invited to participate in the data loads.
- Simultaneously, MOH advances without waiting for organisations to participate and continues to complete data loads from sources over which it has direct jurisdiction. MOH should not wait for other organisations to participate before advancing to Phase 2, because other inter-organisation engagement will take time, and buy-in will increase when MOH successfully demonstrates progress on its own data sources.

The main objective of Phase 1 is to enable the discovery of data sources in a centralised location. A team of data profilers should analyse incoming data sources upon successful data load and compile the assessments into a comprehensive, detailed data map.

The structure of the team of data profilers should be designed to allow rapid progress, for example by selecting a smaller focused group of expert, authorised individuals to mitigate data security concerns of access control to the secure staging zone. Enterprise security programs will take more time to mature, so in the interim a limited group of authorised users can be given access to profile the data, with strict security and auditability around accessing the data to ensure that only authorised users can access the secure zone. .

10.4 Phase 2: Operationalize sample narrow business use cases (MVP)

Goal:

- Demonstrate the early business value of the NHDW through low-hanging fruit sample use cases that are narrowly defined to maximise feasibility and speed-to-implementation but that are varied to demonstrate the potential diversity of features the NHDW can support.
- Establish the essential foundations for end-to-end implementation to maximise early learning, which will allow formulation of standardised implementation processes for introducing new use cases to the NHDW.

Considering that the NHDW is a new component in Ethiopia, upfront business requirement specification, comprehensive data analysis, and modelling will take a lot of time and resources and demand high levels of collaboration between data producers and data consumers. Many stakeholders will be unsure what the early capabilities of the NHDW will be and may struggle to provide specific requirement specifications. It will be important for the NHDW to demonstrate sample use cases to stakeholders to help tangible envisioning of what can realistically be achieved as a means to guide the development of more comprehensive use cases. The MOH should spearhead implementation of sample use cases as early as possible and then disseminate these as practical examples to broader stakeholders, who can then specify their own business requirements with more precision.

Again, a principle to follow in the early stages of implementation is to build an **inclusive forum** where any participating organisation can collaborate, but in parallel to that, MOH should have a plan to **advance without dependency** on participating organisations:

- MOH fosters inclusivity by opening a forum for broad stakeholders to start envisioning and documenting business requirements to the best degree of detail possible. This is an iterative process and will take time to mature, so best to start early. To build buying into this process, stakeholders can be informed that the forum will be used as a repository to inform implementation decisions, and good quality specifications could earn a use case an earlier place in the implementation queue. It should be encouraged for business requirements to be shared with the whole forum so that a community of practice develops and different organisations and teams can leverage learnings from the others. MOH should give stakeholders feedback as to what presents a higher quality and more effective business requirement definition.
- highly feasible, MOH advances without depending on participating organisations by defining use case requirements of its own that are specifically designed to be highly feasible, demonstrate early value (quick win), and present a variation of use case types to help inform broader stakeholders as to what is possible with NHDW.
  The broader forum can be used as a source of concepts but should not be a dependency for the MOH to move forward on selecting the early use cases because the forum will require a lot of consultation and Phase 2 requires the MOH to implement quickly. Use case prioritisation should emphasise feasibility, including rapid availability of data sources. Use cases in this phase should be narrowly defined, without extensive requirements, so that the time to completion can be short.

Successful and rapid implementation of early use cases in an end-to-end manner will set a pattern for implementation that can then be standardised in subsequent implementations. The MOH can then leverage this pattern to document guides for future implementations, which can be disseminated to broader stakeholders.

10.5 Phase 3: Develop master & metadata components

Goal:

- Consolidate data discovery and profile data sources
- Establish foundational master data
- Enable high-quality data standards and mapping that enable detailed use case planning and modelling

The cost of implementation in the long-term existence of the NHDW will depend on well-structured data architecture and detailed planning. However, the major obstacle to achieving this will be a lack of detailed profiling of data sources, organisation of master data, and establishment of data quality processes.

Data in the health sector is currently stored in disparate locations; there is a lack of a centralised data map of what data exists, precise definitions and allocations of data elements, terminology mappings, and de-duplicated master data. Regular errors and rework will be encountered if data is not profiled thoroughly. Building an organised data warehouse will be challenging without having a precise map of the data that must be organised, which will undoubtedly lead to substantial and regular rework.

This phase should establish a team of data stewards who collaborate with the data profilers and who compile and manage the central data map in an organised and systematic way. This central data map becomes a reference point to greatly accelerate use case planning leveraging integration of disparate

data sources, which previously would have been a laborious error-prone process. The central data map will be maintained by the use of meta data storing the profiling of data source elements.

A guiding standard for this phase is to achieve implementation of FAIR data management principles [45].

This phase should also perform the systems development of master data registers to allow accurate and unique data mappings and ensure deduplicated data sets. Examples of master data registries:

- Terminology registry
- Patient registry (MPI)
- Facility registry (MFL)
- Health worker registry (iHIRS)

The master data registries may already exist at varying levels of maturity in the health sector, and the NHDW systems team would need to implement an architecture to integrate the existing master registries with NHDW processes.

This allows the NHDW to set standards for data quality and sets a foundation for effective data integration from disparate sources. Lays the foundation for data quality standards and processes, validation rules, data cleaning, etc. Also lays foundations for high-quality data warehouse modelling and datamart modelling in the next maturity phases. This work of data profiling and master data management.

10.6 Phase 4: Operationalize a data research unit

Goal:

- Expand access to the data lake for broader stakeholders to participate in data profiling, analytics and innovation of use cases
- Enable innovation and seed thinking, helping to shape use case definitions, identifying areas of strengthening
- Build capacity across a broader user base from participating organisations, especially universities.

The data lake would already have been implemented in Phase 1 and can be made available to a broader set of users with minimal technical effort.

The primary challenge to enable this phase is the implementation of an enterprise secure access management system that can support uses from multiple organisations. This will include the formalisation of data sharing agreements and regulations such that organisations that have provided the NHDW with data that they own would be comfortable with who is able to access their data in a secure way. Systems to manage partial access to datasets must be well implemented.

This phase will require the NHDW engineers to construct secure ETL processes to move data from the secure zone of the staging area to a more widely accessible zone of the data lake with data that has been approved for data analysis. A matrix of data producers (owners) and data consumers (analysts) must be created and managed with strict protocols to ensure that the relevant legal agreements are in place prior to any given user obtaining access to a particular data set.

The benefit of operationalising this collaborative data research unit is that it is leveraging the varied skill sets and resources of participating organisations that may not be housed within the MOH core

team. Another benefit is that participating organisations can more thoroughly understand data from other data sources they previously had poor visibility of, enabling them to envision use cases and seed innovation. This research unit can also be a beneficial feedback loop for the MOH to identify areas of strengthening in the NHDW and the health sector at large. Ethiopian universities can be engaged to participate in this research unit, as well as institutions with analytical capabilities such as EPHI. Through this mechanism, universities can be encouraged to build skill sets and human resource capacity that are more directly relevant for the NHDW operations, which becomes a long-term pool of resources for the NHDW sustainability.

The NHDW team can consider availing tools to the research unit for more advanced profiling and analytics, including enabling predictive analytics and AI/ML tools. The responsibility of the NHDW engineering team is to ensure data security and integrity are maintained in a well-defined workspace within the data lake.

10.7 Phase 5: Operationalize a major business subject area and expand data use

Goal:

- Expand the business layer of the NHDW with focus on a major business subject area
- Enable data-driven operations in a major unit of the MOH and its partners

Now that strong foundations have been laid in building a centralised data map with functional master and metadata and have derived learnings from early use case implementations, the NHDW would now have matured enough to expand operations to support a major business problem of the MOH and its partners.

This phase will begin with a thorough use case business requirement specification. Data warehouse modelling can be conducted in a well-planned and structured manner, preventing constant rework due to a lack of understanding of the data and requirements. Data Vault would be considered the preferred architecture for the data warehousing to enable an extensible and flexible approach. This is important considering the diverse use cases that the NHDW will need to service over time that are not all known up front.

An emphasis in this phase will be data completeness, meaning that the MOH will need to conduct drives with partnering stakeholders to ensure that data sources are fully loaded and up to date. Data completeness is an important principle to make the NHDW an important reference point for operationalising regular data use.

The MOH can consider progressively and incrementally expanding collaboration as data sharing policies and robust data governance controls . The technical team should implement in this phase functionality for segmented data accounts in the data warehouse, distinguishing between data producers and consumers, as well as organisation-level workspaces in the business layer. Data marts in the business layer can be restricted to an organisation's workspace or alternatively shared across multiple workspaces with the relevant access control and data sharing agreements.

Business areas can be further expanded to consolidate data producers and data consumers from the private sector, with the relevant governance controls in place. Inter-sectoral use cases, such as consolidating use cases with meteorology, nutrition, water and sanitisation, and agriculture, can also be supported in this phase.

As the NHDW continues to mature, advanced capabilities can be implemented, including real-time processing and other emerging advanced capabilities.

10.8 Project Timeline

NHDW Roadmap

Table 19: Draft Roadmap

Activity	2025-Q1	2025-Q2	2025-Q3	2025-Q4	2026-Q1	2026-Q2	2026-Q3	2026-Q4	2027-Q1	2027-Q2	2027-Q3	2027-Q4
Phase 0: Project inception												
Project planning and setup	x	x										
Planning workshops	x	x										
Phase 1: Prioritise centralization of raw data												
Goal: • As far as it depends on MOH, to have at least one complete copy of every data source that is eligible for the NHDW. • Enable detailed data discovery of all data sources in central place, which is the raw staging area of the NHDW												
Determine initial data centre for data storage and data discovery purpose	x	x	x									
Implement raw data staging and landing zone	x	x	x									
Implement data lake baseline	x	x	x									

Activity	2025-Q1	2025-Q2	2025-Q3	2025-Q4	2026-Q1	2026-Q2	2026-Q3	2026-Q4	2027-Q1	2027-Q2	2027-Q3	2027-Q4
Implement data profiling tools	x	x	x									
Implement metadata management tool	x	x	x									
Implement baseline of data security with limited access	x	x	x									
Implement baseline of audit and compliance monitoring tooling	x	x	x									
Implement essential data ingestion mechanisms (push/pull database replication via VPN/internet)	x	x	x									
Mobilise data owners of different data sources to send data to NHDW			x									
Profile data sources to collate information to build central data profile map (ongoing)		x	x	x								
Document the initial assessment of the data profile map to inform: ● Potential linkages of related data ● Data completeness and quality measures ● Data format assessment ● Recommendations for master and metadata			x									

Activity	2025-Q1	2025-Q2	2025-Q3	2025-Q4	2026-Q1	2026-Q2	2026-Q3	2026-Q4	2027-Q1	2027-Q2	2027-Q3	2027-Q4
Phase 2: Pilot MVP use cases Operationalise sample narrow business use cases												
Goal: • Demonstrate the early business value of the NHDW through low-hanging fruit sample use cases that are narrowly defined to maximise feasibility and speed-to-implementation but that are varied to demonstrate the potential diversity of features the NHDW can support. • Establish the essential foundations for end-to-end implementation to maximise early learning, which will allow formulation of standardised implementation processes for introducing new use cases to the NHDW.												
Establish regular forum for stakeholders to envision and document business requirements (on-going)			x	x	x	x	x	x	x	x	x	x
Select 3-4 varied sample narrow use cases with emphasis on immediate feasibility			x									
Establish a software baseline: • Establish Data Warehouse tooling • Establish ETL tooling • Establish BI visualisation and analytics software tooling		x	x	x	x							
Implement sample use cases end to end			x	x	x	x						
Quality assurance testing					x	x						

Activity	2025-Q1	2025-Q2	2025-Q3	2025-Q4	2026-Q1	2026-Q2	2026-Q3	2026-Q4	2027-Q1	2027-Q2	2027-Q3	2027-Q4
Evaluation of development and operational processes and lessons learnt report						x						
Document guideline for new business use case development based on lessons learnt						x						
Phase 3: Organise Data, Develop Master & Metadata Components												
Goal: • Consolidate data discovery and profile data sources • Establish foundational master data • Enable high-quality data standards and mapping that enable detailed use case planning and modelling												
Establish data management team with structured operational process					x	x						
Consolidate data profiling of source data and populate a metadata bank						x	x	x				
Assess linkage potential of related data by leveraging the meta data bank and data profiling tools in data lake						x	x	x				
Assess the completeness and data quality of data sources						x	x	x				
Recommend/implement initiatives to strengthen data quality at source							x	x				

Activity	2025-Q1	2025-Q2	2025-Q3	2025-Q4	2026-Q1	2026-Q2	2026-Q3	2026-Q4	2027-Q1	2027-Q2	2027-Q3	2027-Q4
Conduct data quality improvements within data lake to make data more usable					x	x	x	x				
Assess data duplication across data sources and document recommendations for deduplication						x	x					
Conduct deduplication of critical data entities in the NHDW, such as patients, facilities, terminologies, etc						x	x					
Strengthen existing master data registries with deduplication output data							x					
Implement master data registers that do not yet exist						x	x	x	x			
Integrate NHDW ETL pipelines with master data registries						x	x	x	x			
Recommend/implement data quality standards & guidelines to improve								x	x			
Phase 4: Operationalize a data research unit												
Goal: • Expand access to the data lake for broader stakeholders to participate in data profiling, analytics and innovation of use cases • Enable innovation and seed thinking, helping to shape use case definitions, identifying areas of strengthening • Build capacity across a broader user base from participating organisations, especially universities.												
Implement Enterprise Identity Access Management				x	x	x						

Activity	2025-Q1	2025-Q2	2025-Q3	2025-Q4	2026-Q1	2026-Q2	2026-Q3	2026-Q4	2027-Q1	2027-Q2	2027-Q3	2027-Q4
Formalise data sharing policies, regulations and operational processes				x	x	x						
Assess short-term opportunities and objectives for data research unit				x	x	x						
Establish data research governance						x	x					
Establish partner onboarding process and governance						x	x					
Establish a knowledge repository for research outcomes with process definition						x	x					
Assemble research unit team						x	x					
Implement tools for data sharing with audit and compliance features												
Implement tools for advanced profiling and analytics, predictive analytics and AI/ML tools						x	x	x				
Conduct MVP research projects (conducted by research unit) to demonstrate value						x	x	x	x	x		
Assess research output and document lessons learnt										x	x	

Phase 5: Expand national data use, operationalise a major business subject area and develop data platform functionality												
Goal: • Expand the business layer of the NHDW with focus on a major business subject area • Enable data-driven operations in a major unit of the MOH and its partners												
Assess business requirement opportunities and prioritise		x	x		x	x						
Business user requirements elicitation for prioritised business subject area			x	x	x	x	x					
Software Requirement Specification							x	x				
Implement data completeness as per minimum requirement for identified business requirement								x	x			
Conduct data quality and master data strengthening as per specification									x	x		
Design data modelling, data flows and analytics user interfaces as per requirements								x	x			
Implement data warehouse and/or data vault engine to drive business requirements									x	x	x	
Implement business data mart to drive data use										x	x	
Implement data visualisation and analytics interfaces										x	x	
Expand data sharing and export tooling for non-technical staff											x	x

Quality assurance testing											x	x
Document data use guideline												x
Capacity building for technical staff		x	x	x	x	x	x	x	x	x	x	x
Capacity building for data analytics staff				x	x	x	x	x	x	x	x	x
Capacity building for data use								x	x	x	x	x

Table 20: Staffing Plans

Role	LOE (Mths)	LOE Calculation	Competency
ICT Infrastructure & Operation			
** Assume outsourcing to Ethiotelecom			
Lead Focal Official / Manager	36	1 x Full-time for all 3 years	Expertise and deep understanding of the network infrastructure, cloud services, and data centres. Skills in managing timelines and resources. Ability to lead and mentor teams, provide accurate communication with the MoH leaders, and promote collaboration in order to increase system performance efficiency. High-level analytical skills to troubleshoot and resolve the infrastructure issues and the ability to effectively communicate with third-party service providers, if needed.

Enterprise Security Architect	36	1 x Full-time for all 3 years	Full Knowledge of security frameworks and other legal instruments to comply with data security and protection as well as best practices. Skills in assessing and mitigating security risks across the enterprise and effectively communicating security concepts to non-technical stakeholders and MoH. Expertise in security technologies such as firewalls, intrusion detection systems, and encryption. Knowledge of regulatory and governance requirements and ability to keep a good tracking system.
IT Support / System Admin	72	2 x Full-time for all 3 years	Expertise in troubleshooting hardware and software issues. Strong interpersonal skills to assist users effectively and empathetically and document issues/solutions. Strong skills in support services tracking systems. Familiarity with IT service management frameworks in place. Skills in creating and maintaining technical documentation, SOPs, and user manuals. Flexibility to work under pressure and to learn new technologies quickly and adapt to changing environments.
Database Admin	72	2 x Full-time for all 3 years	High-level skills in database design, implementation, and maintenance. Expertise in optimising database performance and managing high-level volume storage. Skills on backup strategies and disaster recovery plans for a national database. High-level skills in database security practices and access controls. Proficiency in writing scripts for automation and data manipulation.
Software Engineering			
Lead Focal Official / Manager	36	1 x Full-time for all 3 years	Expertise in software development methodologies according to needs. Proficiency in planning, executing, and overseeing high-level data volume projects. Ability to clearly communicate with MoH, stakeholders, and NHDW users. Strong analytical skills to troubleshoot and resolve issues. Strong knowledge of software architecture and design patterns.

Data Warehouse Architect	36	1 x Full-time for all 3 years	Expertise in dimensional modelling and data warehousing concepts. Proficiency in designing and implementing ETL processes. high level knowledge of databases and data warehousing tools. Skills in optimising database performance and query efficiency and great understanding of governance aspects such as data quality, data access, security, and compliance standards.
Systems Architect	36	1 x Full-time for all 3 years	Proficiency in designing complex systems and understanding system integration. High level understanding of the cloud services in use, needed redundancy, and on-premises solutions. good understanding of in-country security protocols and best practices. Knowledge of the APIs used/proposed for the NHDW and all the architecture and capacity of creating clear related documentation.
Data Engineer	144	4 x Full-time for all 3 years	High level knowledge of the languages and big data technologies needed for the NHDW. Experience in building and maintaining data pipelines. Skills in databases and the ability to implement data validation and cleansing processes.
Software Engineer			Strong skills in programming languages relevant to the NHDW project. Understanding the system life cycle, its processes, and methodologies. Familiarity with version control tools. Skills in writing tests and debugging code.
Data Management			
Lead Focal Official / Manager	36		Proven abilities to lead teams and manage high-level volume data. Great understanding of data policies and compliance applicable in the country. Proficiency in data management tools and databases. Skills in developing data strategies aligned with business goals and proven abilities in communicating with MoH and other stakeholders.

Data Analyst	144		Strong ability to understand, interpret, and analyse health-related data with proficiency in statistical methods and tools. Proficiency in data visualisation tools (Tableau, Power BI, etc.). Ability to query databases effectively. Skills to identify trends and solve business problems using data.
Data Scientist	72		Proficiency in programming skills. Knowledge of algorithms and model development for machine learning and artificial intelligence. Strong foundation in statistics and probability. Skills in cleaning and preparing health-related data for analysis and the ability to translate the findings and present them to the MoH and stakeholders as required.
Data Quality Analyst	144		Strong focus on data accuracy and data quality. Ability in analytical skills with ability to assess data quality metrics and issues. Familiarity with data quality tools and software. Skills in process improvement such as identifying and implementing data quality processes. Skills in creating clear documentation for data quality standards and communicating with the MoH and stakeholders.
Data Governance Specialist	72		Understanding of data-related regulations and compliance. Proficiency in policy development and skills in creating and implementing data governance policies. Ability to assess risks related to data management and skills collaboratively with the MoH and the various stakeholders. Knowledge of data stewardship principles and practices.
Health Terminologist	36		In-depth knowledge of medical terminology and coding systems. Ability to analyse and standardise healthcare data. Strong focus on accuracy in terminology usage. Ability to train and communicate with healthcare professionals. Proficiency in researching and updating medical terms.

Health Informatician	72		High level knowledge of the health information systems and technologies in use in Ethiopia. Ability to analyse health data for insights. Understanding of healthcare practices and workflows of all the health care-related systems in Ethiopia. Skills in working with healthcare providers and IT professionals and familiarity with healthcare regulations and standards.
Governance			
Lead Focal Official / Manager	36		High-level leadership skills and capacity to influence stakeholders. Understanding organisational goals and aligning governance practices accordingly. Proficiency in identifying and managing risks related to governance. Strong verbal and written communication to convey governance policies effectively. Familiarity with relevant laws and regulations affecting governance. Ability to analyse data and make informed decisions for the MoH and stakeholders. skills in project management and stakeholder engagement.
Legal officer	36		In-depth knowledge of laws, regulations, and compliance related to country governance. proficiency in critical thinking and negotiation skills. Meticulous in reviewing legal documents and ensuring compliance. Strong understanding of ethical standards in governance. High-level research and advisory skills as well as interpersonal skills.

11. References

[1] The Federal Democratic Republic of Ethiopia Ministry of Health, “Health Sector Transformation Plan 2015/16 - 2019/20 (2008-2012 EFY).” Aug. 2015. [Online]. Available: file:///Users/chrisseebregts/Downloads/HSTP%20Final%202015-10-19.pdf

[2] Ministry of Health of Ethiopia, “Health Sector Transformation Plan II (HSTP II); 2020/21-2024/25, (2013 EFY - 2017 EFY).” Feb. 2021. [Online]. Available: https://p4h.world/app/uploads/2021/07/ETH_Ethiopia_Health-Sector-Transformation-Plan-II_2021-2026.x23411.pdf

[3] Federal Republic of Ethiopia Ministry of Health, “Information Revolution Strategic Plan (2018-2025).” Jun. 2018.

[4] Ethiopia Ministry of Health, “Information Revolution Roadmap.” 2016.

[5] Ministry of Health of Ethiopia, “Ethiopian-Digital-Health-Blueprint.” Accessed: Oct. 17, 2024. [Online]. Available: http://repository.iphce.org/bitstream/handle/123456789/1658/Ethiopian-Digital-Health-Blueprint.pdf?sequence=1&isAllowed=y

[6] Ministry of Health of Ethiopia, “Ethiopia eHealth Architecture, Version 1.5.” Jul. 2019.

[7] Ethiopia Ministry of Health Data Warehouse Technical Working Group and Data Use Partnership, “Data Warehouse Requirement Documentation.” Dec. 2019.

[8] W. Shiferaw and M. Kifle, “A Data Exchange Interoperability Framework for eHealth Applications in Ethiopia”, Accessed: Sep. 11, 2024. [Online]. Available: https://core.ac.uk/reader/162043660

[9] Ministry of Health of Ethiopia, “Health Sector Medium-Term Development and Investment Plan.” Accessed: Oct. 17, 2024. [Online]. Available: https://arm.moh.gov.et/wp-content/uploads/2023/10/HSDIP_final.pdf

[10] Ministry of Health of Ethiopia, “Health Sectors National Data warehouse. Development and implementation roadmap.” Sep. 2023.

[11] Ethiopia Ministry of Health, “Data Warehouse Technical Working Group: Requirement Documentation.” Dec. 2019.

[12] The Institute of Electrical and Electronics Engineers Computer Society, “Software Requirements Specifications.” [Online]. Available: https://www.computer.org/resources/software-requirements-specifications

[13] The Open Group, *The Open Group Architecture Framework (TOGAF) Standard © 2005-2022 The Open Group, All Rights Reserved*, vol. 10. 2022. [Online]. Available: https://publications.opengroup.org/

[14] Coalition, “Principles for Digital Development.” [Online]. Available: https://digitalprinciples.org/

[15] W. H. Inmon, *Building the Data Warehouse*, Fourth.

[16] R. Kimball, *The Data Warehouse Toolkit*, Third. Wiley, 2013.

[17] “The FAIR principles of data management,” Software solutions for Life Science Data management and integration. Accessed: Oct. 24, 2024. [Online]. Available: https://genestack.com/resources/library/the-fair-principles-of-data-management/

[18] M. D. Wilkinson *et al.*, “The FAIR Guiding Principles for scientific data management and stewardship,” *Sci. Data*, vol. 3, no. 1, p. 160018, Mar. 2016, doi: 10.1038/sdata.2016.18.

[19] S. Pantano, “Complete Guide to Data Vault 2.0: A Revolutionary Approach to Data Architecture,” *Medium*, May 24, 2024. Accessed: Sep. 29, 2024. [Online]. Available: https://medium.com/@srpantano/complete-guide-to-data-vault-2-0-a-revolutionary-approach-to-data-architecture-9fd0d4125f67

[20] D. Linstedt and M. Olschimke, *Building a Scalable Data Warehouse with Data Vault 2.0*. Elsevier, 2016.

[21] “The Principles: Health Data Governance Principles.” Accessed: Nov. 08, 2024. [Online]. Available: https://healthdataprinciples.org/principles

[22] "Data Analytics Lens - AWS Well-Architected Framework".
[23] Asergaz, "Big data architecture style - Azure Architecture Center." Accessed: Oct. 08, 2024. [Online]. Available: https://learn.microsoft.com/en-us/azure/architecture/guide/architecture-styles/big-data
[24] R. Levy, S. Thill, and F. Tekiner, "building-a-data-lakehouse." Accessed: Oct. 08, 2024. [Online]. Available: https://services.google.com/fh/files/misc/building-a-data-lakehouse.pdf
[25] "Best practices for operational excellence." Accessed: Oct. 08, 2024. [Online]. Available: https://docs.databricks.com
[26] World Health Organization, *Monitoring the building blocks of health systems: a handbook of indicators and their measurement strategies*. Geneva: World Health Organization, 2010. Accessed: Aug. 26, 2024. [Online]. Available: https://iris.who.int/handle/10665/258734
[27] G. Tadesse, "NHDW Requirement Gathering Canvas - Health Service Delivery & Malaria Disease Control." 2024.
[28] Dr Girma Tadesse, "National Health Data Warehouse High level business requirement with use case approach," Oct. 18, 2024.
[29] "Electronic Community Health Information System Three Years Strategic Plan 2016-2018 EFY (2023/24-2026/27)."
[30] Ministry of Health of Ethiopia, "DHIS2." Accessed: Oct. 25, 2024. [Online]. Available: https://www.moh.gov.et/en/projects-3-col/dhis2?language_content_entity=en
[31] Tilahun, Binyam and Fritz, Fleur, "JMIR Medical Informatics - Comprehensive Evaluation of Electronic Medical Record System Use and User Satisfaction at Five Low-Resource Setting Hospitals in Ethiopia." Accessed: Sep. 03, 2024. [Online]. Available: https://medinform.jmir.org/2015/2/e22/
[32] JSI, "Product sheet, supply chain management systems." [Online]. Available: https://imlive.s3.amazonaws.com/Federal%20Government/ID154338048054776417302282591939002747 96/J.6_Appendix_6.0_-_Current_Platform_Overview.pdf
[33] Ethiopian Public Health Institute, "Strategic-Plan-for-Health-Laboratory-System-in-Ethiopia-1." Jun. 2023. Accessed: Sep. 11, 2024. [Online]. Available: https://ephi.gov.et/wp-content/uploads/2023/08/Strategic-Plan-for-Health-Laboratory-System-in-Ethiopia-1.pdf
[34] Ethio Telecom, "Modular Data Centers." Oct. 2022. [Online]. Available: https://docs.google.com/presentation/d/1AQq5a52tEQQ89l3UlrJIxtFHyrQgv0ZY/edit#slide=id.p1
[35] "FHIR® - Fast Healthcare Interoperability Resources. eCQI Resource Center." Accessed: Oct. 28, 2024. [Online]. Available: https://ecqi.healthit.gov/fhir?qt-tabs_fhir=about
[36] "Bahmni Open Source EMR & Hospital Information System (HMIS)," Bahmni™. Accessed: Oct. 28, 2024. [Online]. Available: https://www.bahmni.org
[37] C. Arsenault, A. Derseh Mebratie, S. Kassahun Gelaw, and D. Shamebo, "Routine data in a primary care performance dashboard, Ethiopia," *Bull. World Health Organ.*, vol. 102, no. 07, pp. 465-475A, Jul. 2024, doi: 10.2471/BLT.23.291122.
[38] Ministry of Health of Ethiopia, "MFR - Master Facility Registry." Accessed: Nov. 06, 2024. [Online]. Available: https://mfrv2.moh.gov.et/#/home
[39] "Tier Classification System," Uptime Institute. Accessed: Nov. 05, 2024. [Online]. Available: https://uptimeinstitute.com/tiers
[40] "Tier Certification Overview," Uptime Institute. Accessed: Oct. 28, 2024. [Online]. Available: https://uptimeinstitute.com/tier-certification
[41] Transform Health, "Model Law on Health Data Governance." May 2024. [Online]. Available: https://healthdataprinciples.org/pdfs/Model-Law-on-Health-Data-Governance-English.pdf
[42] Federal Democratic Republic of Ethiopia, *Personal Data Proclamation No. 1321 2024*.
[43] "Stronger Health Data Governance," Transform Health. Accessed: Oct. 28, 2024. [Online]. Available: https://transformhealthcoalition.org/wp-content/uploads/2022/11/TH-Logo-Alt-Fill-03.png
[44] Jembi Team, "Phased Implementation Roadmap; National Health Data Warehouse Meeting," Oct. 17, 2024. Accessed: Oct. 25, 2024. [Online]. Available: https://docs.google.com/presentation/d/1nO24PHSSp2C8qWcXzUXL3srcNK4uoz3o4Xi1JmHR

GJE
[45] J. Delgado and S. Llorente, “FAIR Aspects of a Health Information Protection and Management System,” *Methods Inf. Med.*, vol. 61, no. S 02, pp. e172–e182, Dec. 2022, doi: 10.1055/s-0042-1758765.
[46] Jembi Health Systems, “National Health Data Warehouse Desk review.” Oct. 2024. Accessed: Oct. 28, 2024. [Online]. Available: https://docs.google.com/document/d/1r6vZgaF7baGN8P6xG-QtJ9qVxoKyWHS9/edit?usp=drive_link&ouid=108293876975947734066&rtpof=true&sd=true
[47] Gomez, Gibril, “Nigeria National Data Repository Experience Sharing Session,” Jul. 03, 2024.
[48] Mokete, Baboloki, “Botswana National Data Warehouse,” Jul. 04, 2024.

12. Annexures

Annexure One: Acknowledgements

The development of this report was led by the Ethiopian Ministry of Health and the National Health Data Warehouse (NHDW) Technical Working Group (TWG). Technical support was provided by Jembi Health Systems through the CDC HQ Technical Assistance Platform (TAP), funded by the United States President's Emergency Plan for AIDS Relief (PEPFAR).

Table 21: NHDW Technical Working Group

Name	Position	Organisation
Mr Mesoud Mohammed Ahmed (TWG Chair)	Statistics and Information Team Lead / Deputy Director	Strategic Affairs Executive Office, Ethiopia Ministry of Health
Oli Kaba Eba	TWG Member/Digital Health Services Desk Head	Digital Health Lead Executive Office/MoH
Mr Yakob Wonderad	TWG Member/HIS & M&E Advisor	Strategic Affairs, Ethiopia Ministry of Health
Dr Girma Tadesse	TWG Member/	eCHIS Project Manager, JSI Ethiopia
Dr Adane Letta	TWG Member	General Manager, HabTech
Kalechristos Abebe	TWG Member/Digital Health	Associate Digital Health Expert
Eyasu Ashebir	TWG Member/Snr Supply Chain Informatics Advisor/Global	Palladium Group/Informatics: Digital advisor
Bantalem Yeshanew	TWG member/Data use Director	JSI/Digital Health Activity
Biruhtesfa Abere	TWG Member/Senior Technical Director	JSI/DAH
Antensay Amare	TWG member/DH program manager	ICAP/MoH

Table 22: CDC Ethiopia

Name	Position
Dr. Dereje Woldehanna	Country OfficeFTE Lead, LES
Dr. Jemal Aliy Hassen	Subject Matter Expert (SME)
Dr. Aklilu Asrat	Subject Matter Expert (SME)

Table 23: Jembi Team

Name	Position
Dr. Chris Seebregts	CEO
Adrian Floor	Senior Technical Expert
Bami Oni	Data Solutions Architect
Zainabe Dada	Projects Coordinator: Health Information Systems Specialist
Ryan Crichton	Head of Engineering
Dr. Geoff Fatti	Head of Strategic Information
Anteneh Teka	In-country Coordinator

Table 24: Contributors of the work phases

Organisations	Other Partners
Ethiopian Public Health Institute	John Snow International Digital Health Activity
Clinton Health Access Initiative	International Center for AIDS Care and Treatment Program (ICAP)
Ethiopian Food and Drugs Administration (EFDA)	Palladium
Haramaya University	Health Information Systems Program (HISP) Ethiopia
Mekelle University	Ethio telecom
Hawassa University	
Addis Ababa University	
University of Gondar	
Addis Ababa City Administration Health	

Bureau (AACAHB)	
Ethiopian Pharmaceutical Supply Service (EPSS)	
Armauer Hansen Research Institute (AHRI)	
Ethiopian Meteorology Institute (EMI)	
Ethiopian Health Insurance (EHI)	

Annexure Two: Abbreviations and Acronyms

Abbreviations and Acronyms	Description
AI	Artificial Intelligence
API	Application Programming Interface
ART	Antiretroviral treatment
AWS	Amazon Web Services
BI	Business Intelligence
CBMP	Capacity Building and Mentorship Project
CDC	Centers for Disease Control and Prevention
CDM	Common Data Model
CDR	Central Data Repository
CI/CD	Continuous Integration and Continuous Deployment
CPT	Current Procedural Terminology
CPU	Central Processing Unit
CRVS	Civil Registration and Vital Statistics
DHIS2	Health Management Information systems
eAPTS	electronic Auditable Pharmaceutical Transaction Services
eCHIS	Electronic Community Health Insurance System
EFDA	Ethiopian Food and Drug Authority
eHA	Ethiopia eHealth Architecture
EHIS	Ethiopian Health Insurance Service
EHR	Electronic Health Records
EMR	Electronic Medical Records
ePMIS	Electronic Patient Management Information System
EPHI	Ethiopia Public Health Institute
eRIS	electronic Regulatory Information System
EQA	External Quality Assurance System
Ethio telecom	Ethiopian Telecommunications Corporation

ETL	Extract, transform, and load
Fayda	Ethiopian National Digital Identification system
FHIR	Fast Healthcare Interoperability Resources
FR	Functional Requirement
FTP/SFTP	Secure File Transfer Protocol
GDPR	General Data Protection Regulation
GUI	Graphical User Interface
HCI	Hyper-converged Infrastructure
HIE	Health Information Exchange
HIPAA	Health Insurance Portability and Accountability Act
HIS	Health Information System
HIV	Human Immunodeficiency Virus
HL7	Health Level-7
HR	Human Resources
HSDIP	Ethiopian Health Sector Development mid-term Investment Plan
HMIS	Health Management Information System
HSTP	Health Sector Transformation Plans
IAM	Identity and Access Management
ICD	International Classification of Diseases
ID	Identification Document
IEEE	Institute of Electrical and Electronics Engineers
IFMIS	Integrated Financial Management Information System
iHRIS	Integrated Human Resource Information System
HDSS	Health and Demographic Surveillance Sites
IT	Information Technology
LDM	Laboratory Data Manager
LIMS	Laboratory Information and Management System
LIS	Laboratory Information System

M&E	Monitoring & Evaluation
MDM	Master Data Management
MFA	Multi-Factor Authentication
MFR	Master Facility Registry
MNCH	Maternal Newborn and Child Health
MOFE	Ministry of Finance Ethiopia
MOH	Ethiopia Ministry of Health
MPI	Master Patient Index
MVP	Minimum Viable Product
NDC	Nationally Determined Contributions
NDR	National Data Repository
NHDW	National Health Data Warehouse
NFR	Non-Functional Requirement
OMOP	Observational Medical Outcomes Partnership
OpenHIM	Open Health Information Mediator
OpenMRS	Electronic Medical Records platform
PEPFAR	The U.S. President's Emergency Plan for AIDS Relief
PII	Personal Identifiable information
RAID	Redundant Array of Independent Disks
RBAC	Role-based Access Controls
RDT	Rapid Diagnostic Test
SARA	Service Availability and Readiness Assessment
SNOMED	Systematized Nomenclature of Medicine
SPA+	Service Provision Assessment Plus
SRS	Software Requirements Specification
SQL	Structured Query Language
TB	Tuberculosis
TDWI	The Data Warehousing Institute

TQM	Total Quality Management
TWG	Technical Working Group
UAT	User Acceptance Testing
UI	User Interface
WHO	World Health Organization

Annexure Three: Stakeholder List

Stakeholder Type	Organization	Departments	Job Function
Ministry	MOH	Strategic Affairs Executive Office	Program leaders (HIV, Malaria, TB, PHEM, Regulatory, Logistic or supply, HR) All Directorates in MOH and Agencies
Ministry	MOH	Strategic Affairs Executive Office	Planning, Monitoring and Evaluation Leaders
Ministry	MOH	Strategic Affairs Executive Office	HMIS Focal Point
Ministry	MOH	All Data Analysts of the MOH and Agencies	Data Analysts (HIV, Malaria, TB, PHEM, Regulatory, Logistic or supply, HR)
Ministry	MOH	Digital Health Lead Executive Office	Infrastructure (server and database engineering) expertise
Ministry	MOH	Digital Health Lead Executive Office	Governance expertise
Ministry	MOH	Electronic Community Health Information System (eCHIS) programme	Planning, Monitoring and Evaluation Leaders Community Health Workers
Ministry	MOH	Addis Ababa City Authority Health Bureau	Planning, Monitoring and Evaluation Leaders
Ministry	MOH	MoH Maternal, Neonatal and Child Health	Planning, program management
Ministry	MOH	HRIS (Human Resource)	Human Resource Planning

Collaborating Institutions	Health and Demographic Surveillance Sites (HDSS) Universities (e.g. University of Gondar Addis Ababa University Hawassa University Mekelle University Jimma University Haramaya University)	Capacity Building and Mentorship Project (CBMP) Project	CBMP Directors and researchers
Donor	CDC Ethiopia	Strategic Information	Donor/Funder, monitoring and evaluation
Donor	WHO	HMIS Departement	HMIS Focal Point
Donor	PEPFAR/CDC/Global Fund	Health system strengthening	Health system strengthening Focal
Agency	AHRI (Armeur Hansen Research Institute)	Planning, Monitoring and Evaluation Directorate	Monitoring and Evaluation , HIS Focal
Implementation Partner	Ethio Telecom	-	Infrastructure (server and database engineering) experts
Implementation Partner	ICAP	Information Systems	Implementing district and regional data collection systems
Agency	Ethiopia Public Health Institute (EPHI)	Laboratory, PHEM, Research (Surveys) (National Data Management Center)	Leader, Experts and Data Analysts
Agency	National Health Insurance	Planning, Monitoring and Evaluation Directorate	Monitoring and Evaluation Experts, HIS Focal
Agency	Ethiopian Health Insurance Services (EHIS)	-	Health insurance services
Agency	Ethiopian Food and Drug Authority	Planning, Monitoring and Evaluation Directorate	Monitoring and Evaluation Experts, HIS Focal

Agency	Ethiopian Pharmaceutical Supply Services EPSS (Fanos, Vitas, mBrana)	Planning, Monitoring and Evaluation Directorate	Monitoring and Evaluation Experts, HIS Focal
Agency	INSA (Information Network Security Agency)	Security Department	Security Experts
Agency	AI Institute (Artificial |Intelligence Institute)	Software Development	Derive Machine learning and AI outputs
Office of the Prime Minister	FAYDA (National ID Program)	System integration department	System integration expert
Agency	National Blood Bank Services	Planning, Monitoring and Evaluation Directorate	Monitoring and Evaluation Experts, HIS Focal
Donor	Central Statistics - Civil Registration and Vital Statistics (CRVS), Census)	Planning, Monitoring and Evaluation Directorate	Monitoring and Evaluation Experts, HIS Focal
Ministry	Black Lion (Cancer Registry)	-	Surveillance and research
Private Health Sector	Private Health Association	-	Planning
Ministry	Ministry of Finance Ethiopia (MOFE)	Integrated Financial Management Information System (IFMIS)	Financial planning

Annexure Four: Review of NHDWs in Other Countries

As part of the requirements envisioning process, a desk review of national data warehouses in other countries was performed [46]. Presentations of NHDW implementations from other countries, including Kenya, Nigeria [47], Zambia, and Botswana [48], were also considered part of the Ethiopian NHDW development planning process.

The desk review aimed to document the development and implementation experience of health data warehouses in other countries and their uses and benefits to public health, drawing from published literature. A number of health data warehouses have been developed in both high-income and low- and middle-income countries. Increasingly, countries have been developing repositories for patient-level data in contrast to aggregate-level data repositories.

In Africa, these warehouses have primarily been focused on HIV data. In a well-resourced province in South Africa (Western Cape), a data warehouse has been in progressive stages of development for nine years and includes multiple health areas, including HIV, tuberculosis, maternal and child health, laboratory information, pharmacy dispensing data, community health information, and hospital discharge information.

Use cases of these national health data warehouses include uses at all levels, including national, regional, healthcare facility, and individual clinician-patient level.

1. At the national and regional level, uses include: Data visualisations and dashboards for program management; routine report generation for program reporting; and customised data downloads for research purposes.
2. Healthcare facility level: At more advanced maturity models of a national data repository when healthcare facilities can access merged data from the NHDW, use cases include patient reports & line listings for health facility-based interventions (e.g., from linking clinic data with laboratory data) and automated patient flags.
3. At the individual clinician-patient level, at advanced maturity, clinicians can interact with a patient's historical clinical data from a web-based single patient viewer or from the site-based electronic medical record system, retrieving additional data from the NHDW. Use cases include a single patient viewer showing patient's records from multiple data sources, patient-level automated alerts, and machine-learning-based predictions for patient care.

Annexure Five. Data Source Assessment

Initial work on the data infrastructure assessment was carried out during a workshop in July 2024.

Source Systems

The following initial categories of source systems that could contribute data to the NHDW have been identified (see: DW Source systems):

- Disease Management Systems
- Regulatory Systems
- Malaria Disease Prevalence Systems
- Service Delivery Systems
- Human Resource Systems
- Laboratory Information Systems
- Healthcare Financing Systems
- Household Model Community Level Systems
- Commodity Management Systems

Mapping of data source system, data type, period of data sync, data target

The source systems that inform the above use cases were discussed aligned to the above subject areas. Below is an example of the data sources and the canvas.

Table 25: Data Sources

Disease Management					
Source System	eHA Category	System Type	Data Type	Period of data sync	Data Description
eCHIS	Point of Service	EMR: community level	Individual	Nearly real time	The platform is CommCare; it uses Postgresql DBMS. It is difficult to extract data from the database (postgresql) directly. Dimagi prepared a tool called Commcare Sync (CC) to extract meaningful data from the CommCare application. The eCHIS (electronic Community Health Information System) is a suite of mobile applications with a web-based monitoring portal. The eCHIS system includes a suite of CommCare applications for Health Extension Workers (HEWs), their supervisors, and health facility-level users that acts as an integrated primary health care service delivery and reporting tool for the health extension program (HEP). eCHIS digitalises the existing manual family folder and service workflows to record and report the households and members' health and related data.
IVR	Point of Service	Logistic / supply chain	Individual	Nearly real time	
DHIS2: HMIS	Institution based	Reporting	Aggregate	Monthly	The Ethiopian DHIS2 contains information for measuring primary care performance, using simple analytical methods, at national and regional levels and by facility type. Despite remaining data quality issues, DHIS2 is an important data source for generating health system performance assessment measures on a national scale. The Ethiopian DHIS2 contains regularly submitted data, typically monthly, by all 30 192 health facilities in the country. The system includes reports on health service activities and the conditions for which people seek care. It uses PostgreSQL.

Excel line list (PHEM)	Institution based	Tracker	Individual	Nearly real time	
DHIS2: PHEM	Institution based	Reporting	Aggregate	Weekly	
Excel aggregate (PHEM)	Institution based	Reporting	Aggregate	Weekly	
EMR Bahmni	Point of Service	Tracker	Individual	Nearly real time	
Smart care(HIV)	Point of Service	Tracker	Individual	Nearly real time	
ART(HIV)	Point of Service	Tracker	Individual	Nearly real time	
TB Tracker	Point of Service	Tracker	Individual	Nearly real time	
Simple app/ hypertension and DM	Point of Service	Tracker	Individual	Nearly real time	
Redcap	Population-based HIS & Data Sources	CBS	Individual	Nearly real time	
Dagu	Point of Service	Logistic / supply chain		Daily	An inventory and logistics management information system that manages store-based transactions (requisition and issue fulfilment) and allows facilities to generate commodity requests to hubs. It is fully functional in more than 750 facilities, representing about 75% of the value of health commodities distributed. Dagu enables submission of an electronic report and requisition form (eRRF). The eRRF removes the need to physically visit an EPSS hub to report on stock and order. It is interoperable with electronic auditable pharmaceutical transaction services (eAPTS), a dispensary unit application. The integration enables intra-facility visibility of stock status (e.g., the pharmacist can see what is being prescribed and its availability within the site's store).
SAP / Vitas - Center	Point of Service	Logistic / supply chain		Daily	

SAP / Vitas - Hub	Point of Service	Logistic / supply chain		Daily	
mBrana	Point of Service	Logistic / supply chain		Daily	mBrana is an application designed to manage vaccine and long-lasting insecticidal nets (LLINs) distribution and inventory at woreda levels. It is fully integrated with Vitas at central and regional/hub levels, providing visibility into stock status all the way down to woredas as well as into the quantities of vaccines issued to health facilities. It is fully functional in 137 sites nationally, served by 4 hubs.
Census	Population-based HIS & Data Sources	Population Data	Aggregate	10 years and above	
Excel/database	Institution based	Disease		TBD	
Regulatory					
Source System	eHA Category	System Type	Data Type	Period of data sync	Data Description
iHRIS-License/Facility Licencing/	Institution based	Registration and Licencing	Individual and Facility	Daily/Realtime	
MFR (service component)	Shared Service	Registration and Licencing	Individual and Facility		The MFR database captures vital information about each facility, including its name, location, services, infrastructure, and human resources. The MFR serves as a single source of truth for facility data within the national digital health system.
ERIS (Regulatory Information System)	External Systems	Drug and Medical Equipment Registration and Licencing	Drug/Medical Equipment	Daily	
EQA (External Quality Assurance System)	Institution based	External quality assurance System	Aggregate	Weekly	

Malaria: Disease Prevalence					
Source System	eHA Category	System Type	Data Type	Period of data sync	Data Description
eCHIS	Point of Service	EMR: community level	Individual	Nearly real time	The platform is CommCare; it uses Postgresql DBMS. It is difficult to extract data from the database (postgresql) directly. Dimagi prepared a tool called Commcare Sync (CC) to extract meaningful data from the CommCare application. The eCHIS (electronic Community Health Information System) is a suite of mobile applications with a web-based monitoring portal. The eCHIS system includes a suite of CommCare applications for Health Extension Workers (HEWs), their supervisors, and health facility-level users that acts as an integrated primary health care service delivery and reporting tool for the health extension program (HEP). eCHIS digitalises the existing manual family folder and service workflows to record and report the households and members' health and related data.
IVR	Point of Service	Logistic / supply chain	Aggregate	Nearly real time	
DHIS 2: HMIS	Institution based	Reporting	Aggregate	Monthly	The Ethiopian DHIS2 contains information for measuring primary care performance, using simple analytical methods, at national and regional levels and by facility type. Despite remaining data quality issues, DHIS2 is an important data source for generating health system performance assessment measures on a national scale. The Ethiopian DHIS2 contains regularly submitted data, typically monthly, by all 30 192 health facilities in the country. The system includes reports on health service activities and the conditions for which people seek care. It uses PostgreSQL.
Excel line list	Point of Service	Tracker	Individual	Nearly real time	
DHIS2: PHEM	Institution based	Reporting	Aggregate	Weekly	
Excel aggregate (PHEM)	Institution based	Reporting	Aggregate	Weekly	

Dagu	Point of Service	Logistic / supply chain		Daily	An inventory and logistics management information system that manages store-based transactions (requisition and issue fulfilment) and allows facilities to generate commodity requests to hubs. It is fully functional in more than 750 facilities, representing about 75% of the value of health commodities distributed. Dagu enables submission of an electronic report and requisition form (eRRF). The eRRF removes the need to physically visit an EPSS hub to report on stock and order. It is interoperable with electronic auditable pharmaceutical transaction services (eAPTS), a dispensary unit application. The integration enables intra-facility visibility of stock status (e.g., the pharmacist can see what is being prescribed and its availability within the site's store).
SAP / Vitas - Center	Point of Service	Logistic / supply chain		Daily	
SAP / Vitas - Hub	Point of Service	Logistic / supply chain		Daily	
mBrana	Point of Service	Logistic / supply chain		Daily	mBrana is an application designed to manage vaccine and long-lasting insecticidal nets (LLINs) distribution and inventory at woreda levels. It is fully integrated with Vitas at central and regional/hub levels, providing visibility into stock status all the way down to woredas as well as into the quantities of vaccines issued to health facilities. It is fully functional in 137 sites nationally, served by 4 hubs.
Service Delivery					
Source System	eHA Category	System Type		Period of data sync	Data Description
EMR	Point of Service	Individuals. Structured, Unstructured, Semi-Structured	Individual	Nearly real time	

ART Software	Point of Service	Structured	Individual	Nearly real time	
CBS Redcap	Population-based HIS & Data Sources	Structured	Individual	Nearly real time	The objective of the HIV CBS is to identify and describe geographic locations associated with newly diagnosed cases, including recent infections, and inform geographic prioritisation of HIV prevention interventions. The system also aims to monitor and describe epidemiological trends in newly diagnosed HIV cases by demographics, behaviour, mode of transmission, and recency of HIV infection. Additionally, the system aims to monitor and describe trends of clinical status (WHO Stage, initial CD4 and other opportunistic infections) at the point of diagnosis. At the health facility level, HIV case reports are generated at the point of service delivery, such as the hospital outpatient department or inpatient department, the Family and Child Health Unit (FCH), and the Voluntary Male Medical Circumcision Unit (VMMC). REDCap (Research Electronic Data Capture) is a browser-based, metadata-driven electronic data capture software and workflow methodology for designing clinical and translational research databases. At health facilities that use REDCap, all surveillance data elements are recorded on a single HIV case report form (CRF). The data from this case report form is then transferred into the REDCap system using an on-site computer device. Data from both e-first and e-last clinic-based REDCap databases are merged in the national shared health record component through web-based applications with secure internet connectivity. Deduplicated and anonymised individual health records are transmitted to the national server at the EPHI warehouse from the shared health record component.
Viral load and exposed infant test database	Population-based HIS & Data Sources	Structured	Individual	Nearly real time	
eCHIS	Point of Service	Structured	Individual	Nearly real time	The platform is CommCare; it uses Postgresql DBMS. It is difficult to extract data from the database (postgresql) directly. Dimagi prepared a tool called CommCare Sync (CC) to extract meaningful data from the CommCare application. The eCHIS (electronic Community Health Information System) is a suite of mobile applications with a web-based monitoring portal. The eCHIS system includes a suite of CommCare applications for Health Extension

					Workers (HEWs), their supervisors, and health facility-level users that acts as an integrated primary health care service delivery and reporting tool for the health extension program (HEP). eCHIS digitises the existing manual family folder and service workflows to record and report the households and members' health and related data.
Dagu/eAPTS	Point of Service	Structured	Individual	Nearly real time	An inventory and logistics management information system that manages store-based transactions (requisition and issue fulfilment) and allows facilities to generate commodity requests to hubs. It is fully functional in more than 750 facilities, representing about 75% of the value of health commodities distributed. Dagu enables submission of an electronic report and requisition form (eRRF). The eRRF removes the need to physically visit an EPSS hub to report on stock and order. It is interoperable with electronic auditable pharmaceutical transaction services (eAPTS), a dispensary unit application. The integration enables intra-facility visibility of stock status (e.g., the pharmacist can see what is being prescribed and its availability within the site's store).
LIS	Point of Service	Structured	Individual	Nearly real time	
RIS	External Systems	individuals, semi-structured and unstructured	Individual	Nearly real time	
DHIS2	Institution based	Structured	Aggregate	Monthly, Quarterly	
Other External data sources	External Systems	Structured	Individual	Annual, Daily	
Master data registers	Shared Service				
Human resource					
Source System	eHA Category	System Type	Data Type	Period of data sync	Data Description

HRIS	Institution based	Individuals. Structured, Unstructured, Semi-Structured	Individual	Nearly real time	
Student Information System	External Systems		Individual	Nearly real time	
Alumni Information Management System	External Systems		Individual	Nearly real time	
Excel-based HR data	Institution based		Individual	Nearly real time	
DHIS2 /HMIS	Institution based		Aggregate	Monthly/ 6-month, annual	
Other External source	External Systems		Individual		
National ID/Fayda	Shared Service		Individual	Nearly Real time	Fayda is the national ID system producing a 12-digit ID. Captures Demographic and Biometric data
Laboratory Information System					
Source System	eHA Category	System Type	Data Type	Period of data sync	Data Description
LDM LIS	Point of Service	Routine data	Individual	Nearly real time	LDM is a proprietary Laboratory Information System (LIS) that covers all functional areas of multi-speciality laboratories. LDM provides sample barcode labelling and tracking, test routing, interfaces with automated analysers and middleware, results reviewing and reporting, electronic delivery of results, and quality control monitoring and management. LDM integrates with hospital information, EMR systems, and information systems. LDM supports ICD10,

					LOINC, and CPT4 codes and HL7 integration with other systems. LDM has both desktop and web-based versions.
Polytech LIS	Point of Service	Routine data	Individual	Nearly real time	At least 25 laboratories in Ethiopia have installed the proprietary electronic Polytech Laboratory Information System (LIS), which is interoperable and networked with DHIS2. The Polytech LIS expedites patient records and automates data collection and reporting. Polytech sends health care providers timely information and simplifies reporting for lab technicians. Polytech is used in labs doing a wide variety of testing. Comp Pro Med has developed more than four hundred HL7-compliant direct interfaces for real-time communication within the lab environment and with clients. Polytech LIS is built on Edge Computing, which optimises hardware and software applications by bringing computing closer to the source of the data. This minimises the need for long-distance communications between client and server, or cloud applications, which minimises computing time and bandwidth usage.
Labxpert	Point of Service	Reporting	Aggregate	Nearly real time	LabXpert is a proprietary software solution from Med X International that reports the results of GeneXpert testing for tuberculosis diagnosis and to identify treatment resistance. Adoption of the LabXpert connectivity solution was selected by the Ethiopia National TB Program as a sustainable approach to connect the country's GeneXpert instruments to rapidly transmit testing results to clinicians to enable timely enrolment of patients onto treatment; automatically transmit data to real-time dashboards for use at site, regional, and national levels to monitor instrument functionality, maintenance status, and inventory management; and improve the programmatic data reporting system to strengthen overall lab programmatic management. LabXpert dashboards have been customised by MedX International according to the needs of the Ethiopia National TB Program, including results visualisation, graphs, aggregated data, rifampicin resistance reporting, module functionality status, supply status over time, treatment indicators, and maintenance status. The LabXpert connectivity project is being implemented in all 11 regions and two city administrations in Ethiopia. All data are hosted by the Ethiopian Public Health Institute
EIDVL	External Systems	Reporting	Individual	instantly	

EQA (External Quality Assurance System)	Institution based	Reporting	Aggregate	Weekly	
Health Care Financing					
Source System	eHA Category	System Type	Data Type	Period of data sync	Data Description
Paper, excel	External Systems	Data capture, Tracking & Reporting	Individual	Annually	
CBHI, Social Health Insurance	Institution based	Data capture, Tracking & Reporting	Individual	Annually	
CBHI, Social Health Insurance, EMR	Institution based	Tracking & Reporting	Individual, aggregate	quarterly, semiannual, or annual	
NHA, IBEX, EMR, APTS	Point of Service	Tracking & Reporting	aggregate	quarterly, semiannual, or annual	
eFMIS?, NHA, CBHA, Grant Management System	Institution based	Tracking & Reporting	aggregate and survey	Quarterly, semiannual, annual	
EMR	Point of Service				

APTS	Point of Service				
Household model community level					
Source System	eHA Category	System Type	Data Type	Period of data sync	Data Description
eCHIS	Point of Service	EMR: community level	Individual	Nearly real time	The platform is CommCare; it uses Postgresql DBMS. It is difficult to extract data from the database (postgresql) directly. Dimagi prepared a tool called Commcare Sync (CC) to extract meaningful data from the CommCare application. The eCHIS (electronic Community Health Information System) is a suite of mobile applications with a web-based monitoring portal. The eCHIS system includes a suite of CommCare applications for Health Extension Workers (HEWs), their supervisors, and health facility-level users that acts as an integrated primary health care service delivery and reporting tool for the health extension program (HEP). eCHIS digitises the existing manual family folder and service workflows to record and report the households and members' health and related data.
DHS/HDSS	External Systems	Reporting	Aggregate	Monthly	
DHIS 2: HMIS	Institution based	Reporting	Aggregate	Monthly	The Ethiopian DHIS2 contains information for measuring primary care performance, using simple analytical methods, at national and regional levels and by facility type. Despite remaining data quality issues, DHIS2 is an important data source for generating health system performance assessment measures on a national scale. The Ethiopian DHIS2 contains regularly submitted data, typically monthly, by all 30 192 health facilities in the country. The system includes reports on health service activities and the conditions for which people seek care. It uses PostgreSQL.
DHIS2 - PHEM	Institution based	Reporting	Aggregate	Weekly	
Excel aggregate (PHEM)	Institution based	Reporting	Aggregate	Weekly	

Dagu	Point of Service	Logistic / supply chain	Aggregate	Daily	An inventory and logistics management information system that manages store-based transactions (requisition and issue fulfilment) and allows facilities to generate commodity requests to hubs. It is fully functional in more than 750 facilities, representing about 75% of the value of health commodities distributed. Dagu enables submission of an electronic report and requisition form (eRRF). The eRRF removes the need to physically visit an EPSS hub to report on stock and order. It is interoperable with electronic auditable pharmaceutical transaction services (eAPTS), a dispensary unit application. The integration enables intra-facility visibility of stock status (e.g., the pharmacist can see what is being prescribed and its availability within the site's store).
Commodity Management					
Source System	eHA Category	System Type	Data Type	Period of data sync	Data Description
eCHIS	Point of Service	Community health IS	Individual	Nearly real time	The platform is CommCare; it uses Postgresql DBMS. It is difficult to extract data from the database (postgresql) directly. Dimagi prepared a tool called Commcare Sync (CC) to extract meaningful data from the CommCare application. The eCHIS (electronic Community Health Information System) is a suite of mobile applications with a web-based monitoring portal. The eCHIS system includes a suite of CommCare applications for Health Extension Workers (HEWs), their supervisors, and health facility-level users that acts as an integrated primary health care service delivery and reporting tool for the health extension program (HEP). eCHIS digitises the existing manual family folder and service workflows to record and report the households and members' health and related data.
IVR	Point of Service	Health Commodity	Aggregate	Daily	
eAPTS	Point of Service	Health Commodity	Individual	Daily	

Dagu	Point of Service	Health Commodity	Individual	Daily	An inventory and logistics management information system which manages store-based transactions (requisition and issue fulfilment) and allows facilities to generate commodity requests to hubs. It is fully functional in more than 750 facilities, representing about 75% of the value of health commodities distributed. Dagu enables submission of an electronic report and requisition form (eRRF). The eRRF removes the need to physically visit an EPSS hub to report on stock and order. It is interoperable with electronic auditable pharmaceutical transaction services (eAPTS), a dispensary unit application. The integration enables intra-facility visibility of stock status (e.g., the pharmacist can see what is being prescribed and its availability within the site's store).
MEMIS	Point of Service	Health Commodity	Individual	Daily	
Online Request	Point of Service	Health Commodity	Individual	Daily	
EMR (including trackers)	Point of Service	Medical Record	Individual	Never	
DHIS 2: HMIS	Institution based	HMIS	Aggregate	Monthly	The Ethiopian DHIS2 contains information for measuring primary care performance, using simple analytical methods, at national and regional levels and by facility type. Despite remaining data quality issues, DHIS2 is an important data source for generating health system performance assessment measures on a national scale. The Ethiopian DHIS2 contains regularly submitted data, typically monthly, by all 30 192 health facilities in the country. The system includes reports on health service activities and the conditions for which people seek care. It uses PostgreSQL.
Polytech	Point of Service	LIS	Individual	Never	
LDM	Point of Service	LIS	Individual	Nearly real time	
EIDVL Database / HIV	External Systems	LIS	Individual	Nearly real time	

Lab Expert / TB	Point of Service	LIS	Individual	Nearly real time	
mBrana	Point of Service	Health Commodity	Individual	Nearly real time	mBrana is an application designed to manage vaccine and long-lasting insecticidal nets (LLINs) distribution and inventory at woreda levels. It is fully integrated with Vitas at central and regional/hub levels, providing visibility into stock status all the way down to woredas as well as into the quantities of vaccines issued to health facilities. It is fully functional in 137 sites nationally, served by 4 hubs.
MFR	Shared Service	Facility Registry	Individual	Nearly real time	Postgresql and HAPI FHIR
DHIS 2: WBP	Institution based	Population registration	Aggregate	Annual	PostgreSql
Vitas / ERP	Institution based	Health Commodity	Individual	Nearly real time	
Vitas Disposal	Institution based	Disposal system	Individual	Nearly real time	
eRIS	External Systems	Regulatory	Individual	Nearly real time	
NPC	Shared Service	Product catalogue	Individual	Nearly real time	
NHDD	Shared Service	National disease dictionary	Individual	Nearly real time	
QAT	Institution based	Forecasting		Nearly real time	
Retailer systems	External Systems				
Private Pharmacy systems	Point of Service				

Electronic single window	Shared Service				
International product catalogue	Shared Service				
Procurement service agent systems	Institution based				
GS1 Database	Shared Service				
Vitas	Institution based				
Dagu / eAPTS	Point of Service				

Additional Data sources					
Source System	eHA Category	System Type	Data Type	Period of data sync	Data Description
Ethiopia socioeconomic survey	Population-base HIS & Data Sources				
Crop and Livestock Product Utilisation	External Systems				
Area and Production for Major Crops	External Systems				

HDSS	External Systems				
DHS	Population-base HIS & Data Sources				
Cancer registry	Population-base HIS & Data Sources				
Geospatial data	Population-base HIS & Data Sources				
Animal health surveys	Population-base HIS & Data Sources				
One health survey	Population-base HIS & Data Sources				
Research Repositories	Shared Service				
Environmental surveys	Population-base HIS & Data Sources				
MDR TB tracker	Population-base HIS & Data Sources				
Climate surveys	Population-base HIS & Data Sources				
PMA surveys	Population-base HIS & Data Sources				
NDMC (research data)	Shared Service				
AHRI research repository	Shared Service				

Step Survey	Population-base HIS & Data Sources				
WaSH Data	External Systems				
Malaria Indicator Survey	Population-base HIS & Data Sources				
GBD	Population-base HIS & Data Sources				
Agricultural researches/surveys	Population-base HIS & Data Sources				
The Malaria Research Network of Ethiopia	Shared Service				
LEAP (mobile application)	Point of Service				AMREF (Leap revolutionise education by utilising the power of mobile technology to reach learners across diverse demographics, regardless of their location and time)
Electronic Immunization Record/DHIS2 Tracker/Wellness Pass	Point of Service				

Important data source systems: further information

Smartcare

SmartCare is a portable, integrated EMR system that is among the largest EMR systems in use in Africa. The system was designed in Africa to be robust in environments with limited infrastructure [31]. The EMR was initially deployed in 5 hospitals in Ethiopia and has since been scaled up nationally. It is used to capture individual patient-level treatment information, including antiretroviral treatment drug use, patient interactions with the healthcare system, and laboratory results.

The SmartCare system has different components (modules) that can be used in the various units of healthcare facilities. The main modules of SmartCare include registration, outpatient department, inpatient (to admit, follow, and discharge patients in wards), tuberculosis, paediatrics, HIV/AIDS (to manage patients in antiretroviral therapy clinics), antenatal care, postpartum, pharmacy, drug stock control, laboratory (to store and send laboratory results to the requesting clinic), eHMIS (to generate monthly, quarterly, and annual reports), and finance. It produces a number of reports and visualisations for reporting at the facility level. Data is used to calculate important HIV indicators for reporting, including TX_NEW, TX_CURR, viral load suppression indicators, and patient treatment outcomes.

Addis Ababa Central Data Repository for HIV data

The Central Data Repository (CDR) installed at the Addis Ababa City Administration Health Bureau (AACAHB) for HIV data management centralises HIV data from > 65 HIV treatment facilities in the city. The data centralisation and integration platform uses open-source software, adhering to the Open Health Information Exchange (OpenHIE) architecture specification and employing the Fast Healthcare Interoperability Resources (FHIR™) standard by HL7. This platform was chosen for its compatibility with global health information exchange standards and the flexibility that the FHIR standard provides in terms of data interoperability. This CDR is among the largest operational data repositories in low- and middle-income countries to use FHIR standards for data interoperability and integration.

The system has been implemented at the AACAHB data centre with a trained team that manages the infrastructure, coordination, and troubleshooting support as required. The AACAHB team generates CDR reports, which are shared with relevant teams for evidence-based decision-making. The data management and HIV program teams are using the CDR report for HIV program monitoring, and relevant feedback is provided to health facilities.

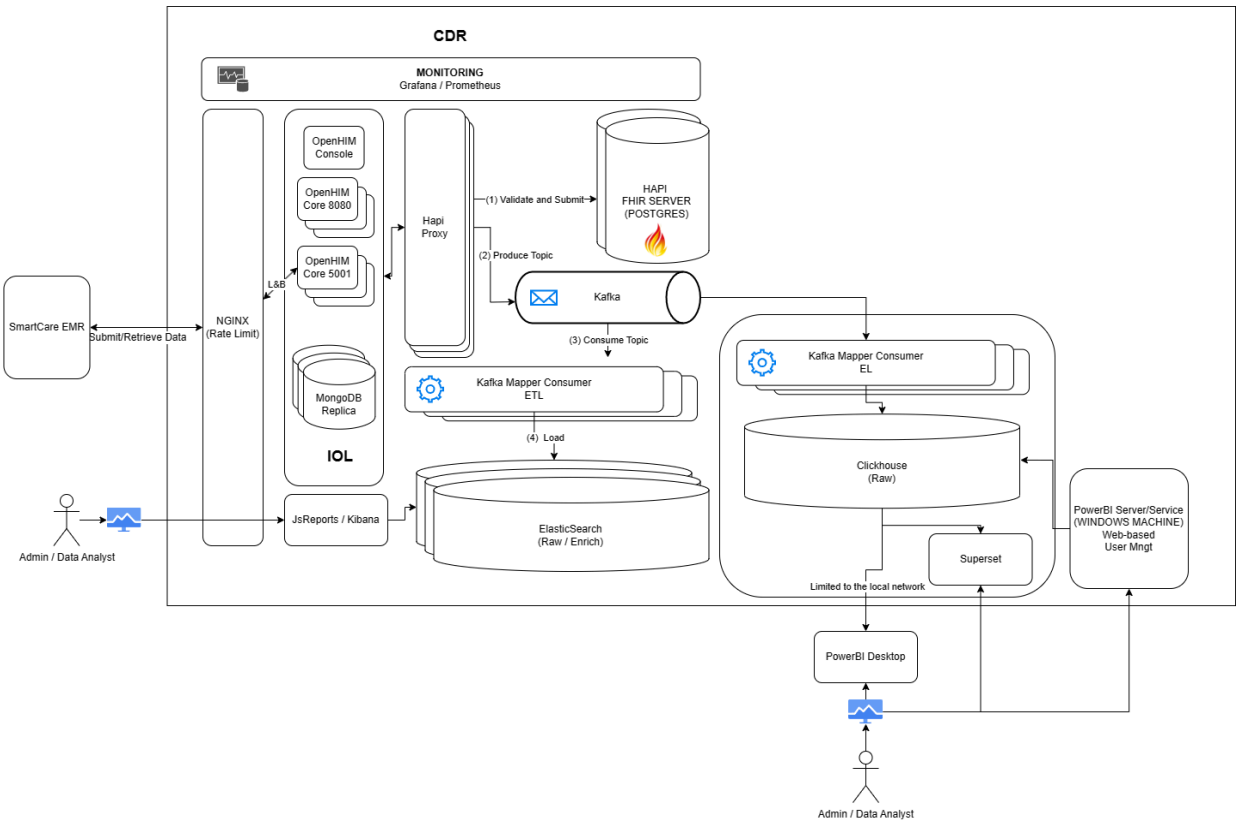


Figure 22: CDR Architectural Diagram

HIV case-based surveillance system using Research Electronic Data Capture (RedCAP).

The objective of the HIV CBS is to identify and describe geographic locations associated with newly diagnosed cases, including recent infections, and inform geographic prioritisation of HIV prevention interventions. The system also aims to monitor and describe epidemiological trends in newly diagnosed HIV cases by demographics, behaviour, mode of transmission, and recency of HIV infection. Additionally, the system aims to monitor and describe trends of clinical status (WHO Stage, initial CD4 and other opportunistic infections) at the point of diagnosis.

At the health facility level, HIV case reports are generated at the point of service delivery, such as the hospital outpatient department or inpatient department, the Family and Child Health Unit (FCH), and the Voluntary Male Medical Circumcision Unit (VMMC).

REDCap (Research Electronic Data Capture) is a browser-based, metadata-driven electronic data capture software and workflow methodology for designing clinical and translational research databases. At health facilities that use REDCap, all surveillance data elements are recorded on a single HIV case report form (CRF). The data from this case report form is then transferred into the REDCap system using an on-site computer device. Data from both e-first and e-last clinic-based REDCap databases are merged in the national shared health record component through web-based applications with secure internet connectivity. Deduplicated and anonymised individual health records are transmitted to the national server at the EPHI warehouse from the shared health record component.

eCHIS

The eCHIS (electronic Community Health Information System) is a suite of mobile applications with a web-based monitoring portal. The eCHIS system includes a suite of CommCare applications for

Health Extension Workers (HEWs), their supervisors, and health facility-level users that acts as an integrated primary health care service delivery and reporting tool for the health extension program (HEP). It captures electronic data on HEP and other community-level services, as well as utilising this data to improve HEP performance and community health outcomes and also serving as a job aid to HEWs [29]. eCHIS digitalises the existing manual family folder and service workflows to record and report the households and members' health and related data.

The eCHIS is primarily a mobile-based application that works in an offline environment. However, it needs connectivity for data synchronisation to the server and to facilitate digital referral linkages. It is also capable of creating a digital link between unique identifiers and health information about households and individuals and preventing the creation of duplicative household or individual records. It also enables the electronic sharing of household and individual information between HEWs and other staff. The application makes it easy for the health extension workers to easily monitor updates on patient status. One other important advantage of the application is that it facilitates a referral linkage between the health post and health centres (MOH website). The eCHIS digitises the CHIS content into a mobile platform for use by HEWs and health workers throughout the country.

CommCare is an open source mobile platform designed for data collection, client management, decision support, and behaviour change communication. CommCare consists of two main technology components: CommCare Mobile and CommCare HQ. The mobile application is used by client-facing community health workers in visits as a data collection and educational tool and includes optional audio, image, and video prompts. Users access the application-building platform through the website, which operates on a cloud-based server.

DHIS2

District Health Information System 2 (DHIS2) is an open-source, flexible offline and online software that collects, validates, analyses, and presents data tailored to manage and aggregate patient-based statistical data and integrated health information. Motivated by a need to improve data use for health system improvement, the Ethiopian health ministry adopted DHIS2 in 2016 as part of its digital health plan, achieving national implementation in 2019. The system captures aggregate-level indicators at health facilities. The Ethiopian DHIS2 contains information for measuring primary care performance, using simple analytical methods, at national and regional levels and by facility type. Despite remaining data quality issues, DHIS2 is an important data source for generating health system performance assessment measures on a national scale. The Ethiopian DHIS2 contains regularly submitted data, typically monthly, by all 30 192 health facilities in the country. The system includes reports on health service activities and the conditions for which people seek care. Concerns about data quality have hampered use of DHIS2. As a result, its rapid and extensive scaling up has not been matched by a corresponding increase in data use by policymakers and researchers. The Ethiopian health ministry conducts annual reviews of health system performance using DHIS2 data. However, the majority of indicators monitored focus on coverage estimates (e.g. antenatal care coverage or immunisation coverage) that do not reflect the quality of the services provided and rely on denominators that are sometimes inaccurate (e.g., estimated number of pregnant women, expected number of infants [37].

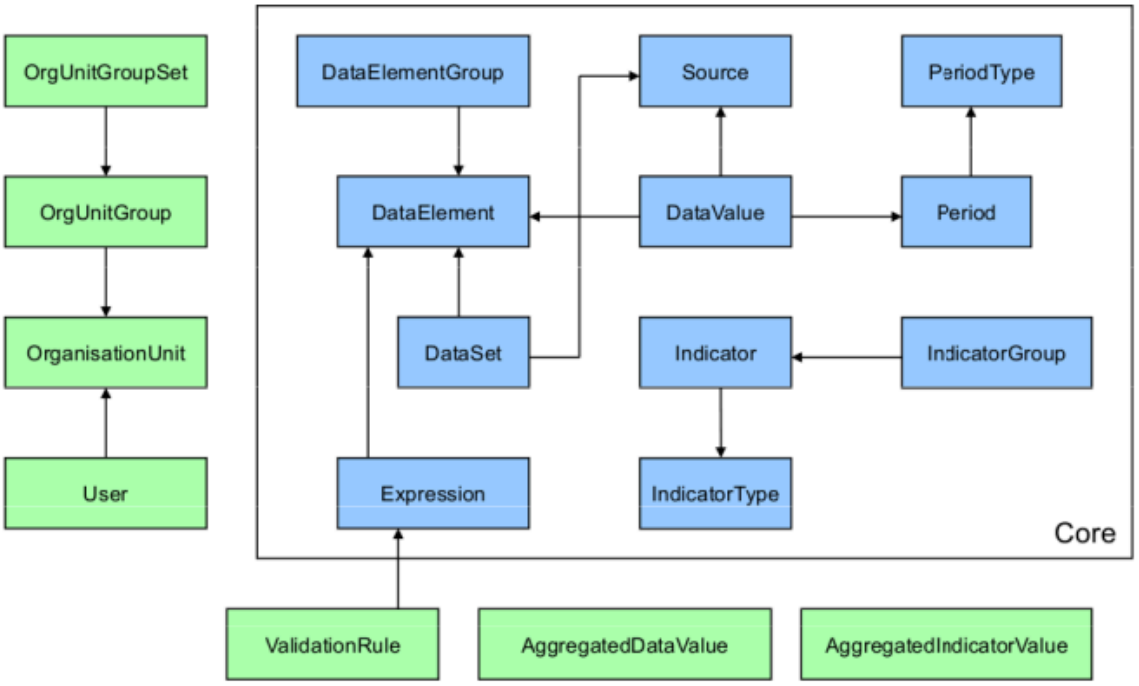


Figure 23: Simplified domain model of DHIS2

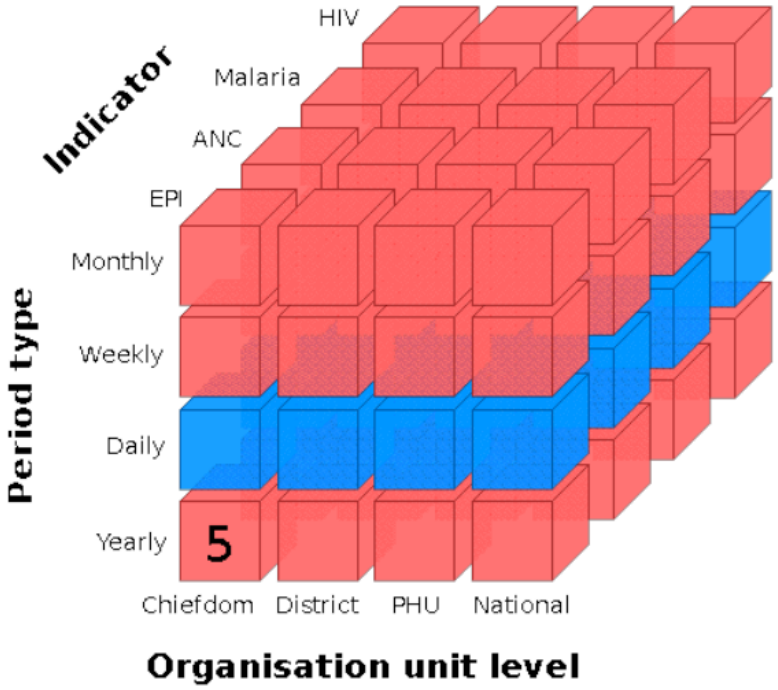


Figure 24: An illustration image of an OLAP-cube slice

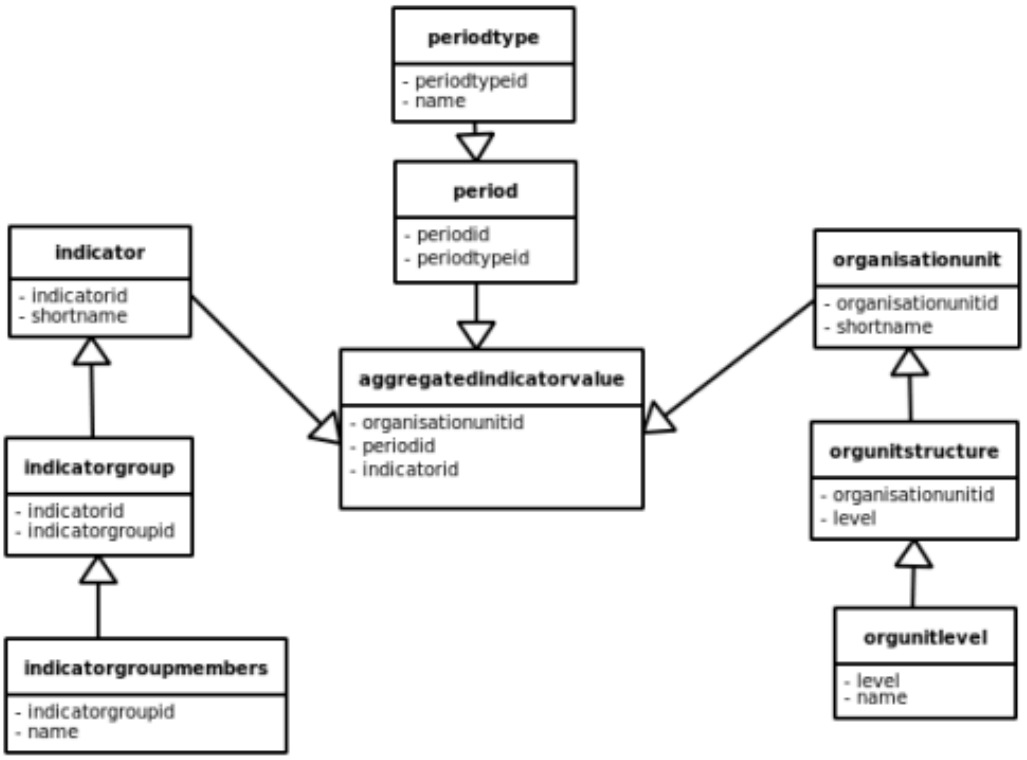


Figure 25: Schema of the basic OLAP cube for DHIS 2

Bahmni

Bahmni is an open-source Hospital Information System (HIS) and Electronic Medical Record (EMR) under development since November 2012 . Bahmni aims to meet the needs of hospitals in low-resource environments by leveraging a variety of existing open source products. The information that Bahmni makes accessible helps healthcare providers to improve efficiency and quality of patient care, reduce errors in clinical encounters, and advocate for issues related to public health.

Bahmni manages patient information across registration, point of care, investigations, and billing. It is simple to use at the point of care, with minimal training required. It is infrastructure-appropriate, hosted, and operated at the hospital site, requiring no dependence on the Internet. Bahmni can be used on a variety of devices, including tablets and laptops. The system is a global open EMR tool that meets the Ethiopian EHR standard and provides end-to-end automation of health facility services.

Bahmni allows clinicians, researchers, & hospital administrators to use SNOMED CT as a reference terminology. They can use SNOMED CT in their health facilities to record diagnosis & other clinical data, as well as for reporting and Clinical Decision Support (CDS) use-cases. Users can search and save diagnoses, receive alerts when a drug-diagnosis interaction is contraindicated, generate diagnosis reports, and create forms. With the integration of SNOMED CT, patient responses can be standardised and therefore more easily exchanged with other electronic health records. Bahmni has also integrated open source Snowstorm Terminology Server over FHIR. This allows for directly recording patient clinical data with SNOMED CT terms. Data can be exported in FHIR format with SNOMED CT-coded terms, including in Snolytical (SNOMED Analytics tool).

Bahmni architecture diagrams

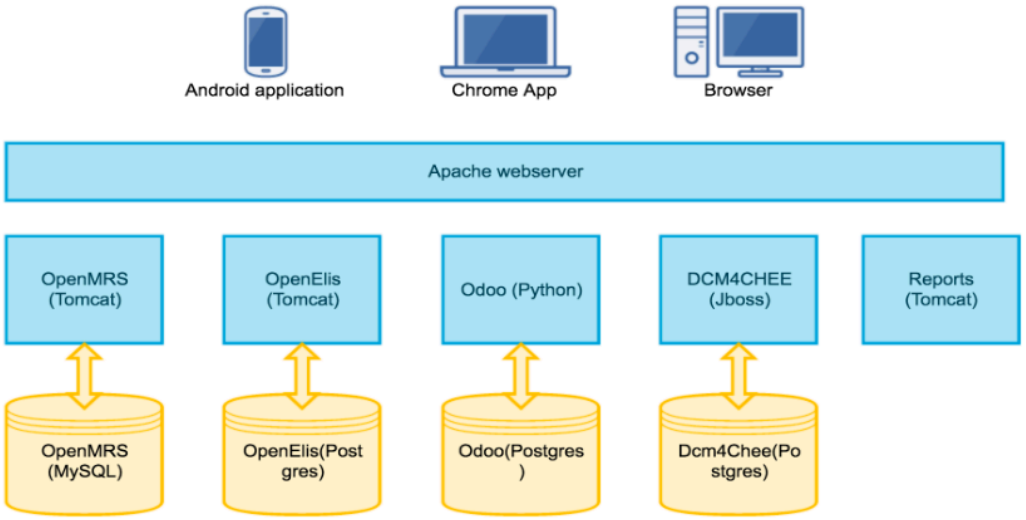


Figure 26: Bahmni Architecture > Component Architecture

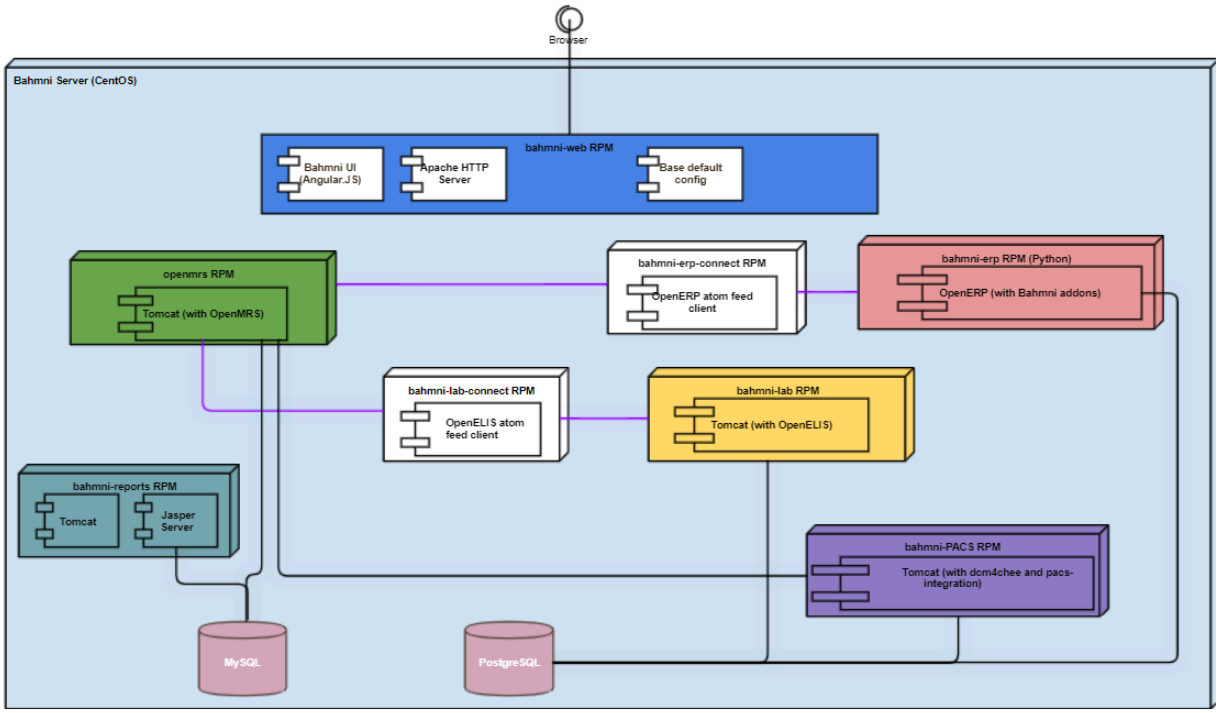


Figure 27: Bahmni Server (CentOS)

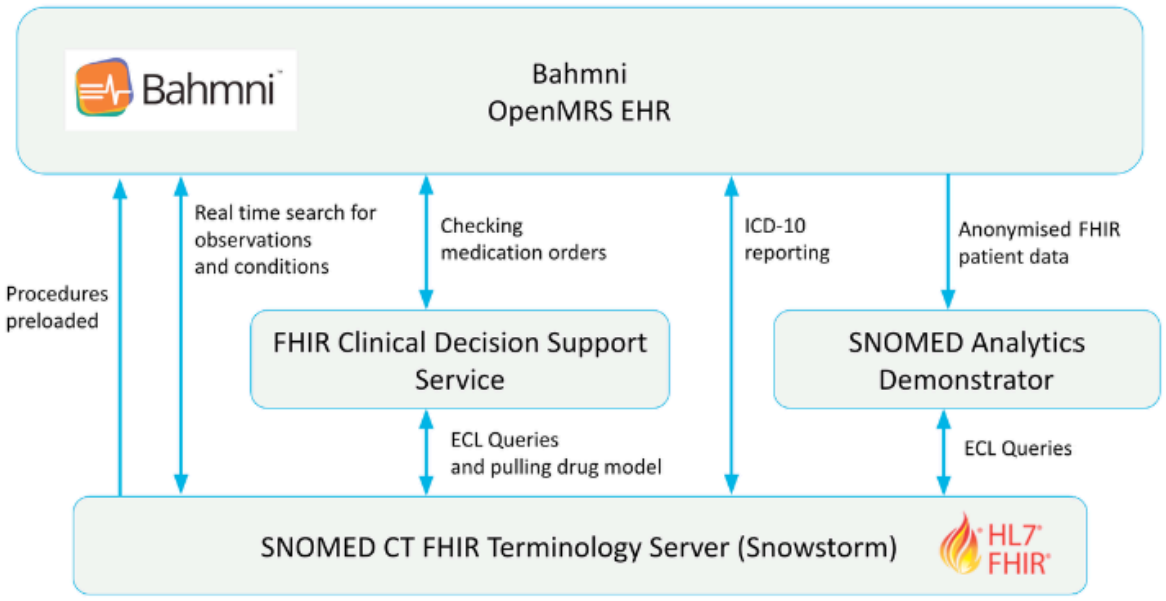


Figure 28: Bahmni SNOMED CT Integration Architecture

Supply chain management systems

A number of interrelated systems are used for supply chain management in Ethiopia, including [32].

1. Vitas

Vitas is an inventory and warehouse management tool used by EPSS (Ethiopian Pharmaceutical Supply Services) to ensure that each facility has the commodities required to provide high-quality services. It is a warehouse and logistics management information system that manages: demand and procurement planning, costing, financial management, requisition fulfilment, warehouse management, and inventory control. Vitas controls inventory of health commodities, from receipt to issue, and manages movement and storage at EPSS centres and hub warehouses. It allows effective demand planning, resource management, warehouse management, and inventory control of health commodities. Vitas is interoperable with downstream supply chain tools: Dagu, a warehouse management tool at facility level, and mBrana, a vaccine management tool at woreda (district) health office level.

2. Fanos

Fanos is a dashboard used at the EPSS centre and its hubs that allows supply chain data visibility in real time through visual presentation of selected supply chain indicators. It supports procurement, supply, and distribution decision-making for all commodities managed by EPSS. A web-based dashboard that displays Vitas data to guide supply chain performance monitoring. Sample metrics monitored include: stock on hand, months of stock, average monthly consumption, expiries, and pipeline. Fanos is also available as an Android application.

3. Dagu

An inventory and logistics management information system that manages store-based transactions (requisition and issue fulfilment) and allows facilities to generate commodity requests to hubs. It is

fully functional in more than 750 facilities, representing about 75% of the value of health commodities distributed.

Dagu enables submission of an electronic report and requisition form (eRRF). The eRRF removes the need to physically visit an EPSS hub to report on stock and order. It is interoperable with electronic auditable pharmaceutical transaction services (eAPTS), a dispensary unit application. The integration enables intra-facility visibility of stock status (e.g., the pharmacist can see what is being prescribed and its availability within the site's store).

4. mBrana

mBrana is an application designed to manage vaccine and long-lasting insecticidal nets (LLINs) distribution and inventory at woreda levels. It is fully integrated with Vitas at central and regional/hub levels, providing visibility into stock status all the way down to woredas as well as into the quantities of vaccines issued to health facilities. It is fully functional in 137 sites nationally, served by 4 hubs.

It is also used to manage the receipt and issue of vaccines and LLINs. mBrana generates electronic vaccine request forms (eVRF), enabling a rapid submission of supply requests electronically.

5. iImport

A software application developed for the Food, Medicine, and HealthCare Administration and Control Authority of Ethiopia that allows importers to apply for and receive permits to import medicines online and FMHACA staff to manage these applications online.

6. iRegister

A public domain, web-based management information system developed for the Food, Medicine, and Health Care Administration and Control Authority of Ethiopia that automates medicine, product registration, application submission, screening, review, and market authorisation processes.

7. eAPTS

A facility-level dispensary tool designed to facilitate the transaction during medication dispensing and management at the dispensary unit (pharmacy/medical supply site) of a health facility. Prior to the deployment of eAPTS, manual payment processing took longer, shortening patient counselling time. The implementation of eAPTS reduced patient waiting time, enabling longer counselling sessions.

Laboratory Information Systems

1. Polytech Laboratory Information System (COMP Promed Inc)

At least 25 laboratories in Ethiopia have installed the proprietary electronic Polytech Laboratory Information System (LIS), which is interoperable and networked with DHIS2 [33]. Polytech sends health care providers timely information and simplifies reporting for lab technicians. Comp Pro Med has developed more than four hundred HL7-compliant direct interfaces for real-time communication within the lab environment and with clients. Polytech LIS is built on Edge Computing, which optimises hardware and software applications by bringing computing closer to the source of the data. This minimises the need for long-distance communications between client and server, or cloud applications, which minimises computing time and bandwidth usage.

Polytech LIS interfaces to over 200 major clinical laboratory instruments and supports barcode and label printers. Polytech LIS is able to interface with billing and accounting systems, EMR systems, hospital information systems, and Windows programs. Polytech Interface Engine supports industry

standard HL7 and ASTM protocols, Windows TCP/IP, and others. Polytech is able to import and export data from other databases.

2. Laboratory Data Manager (LDM) (National Technology)

LDM is a proprietary laboratory information system that covers all functional areas of multi-speciality laboratories. LDM provides sample barcode labelling and tracking, test routing, interfaces with automated analysers and middleware, results reviewing and reporting, electronic delivery of results, and quality control monitoring and management. LDM integrates with hospital information systems, EMR systems, and financial systems. LDM supports ICD10, LOINC, and CPT4 codes and HL7 integration with other systems. LDM has both desktop and web-based versions.

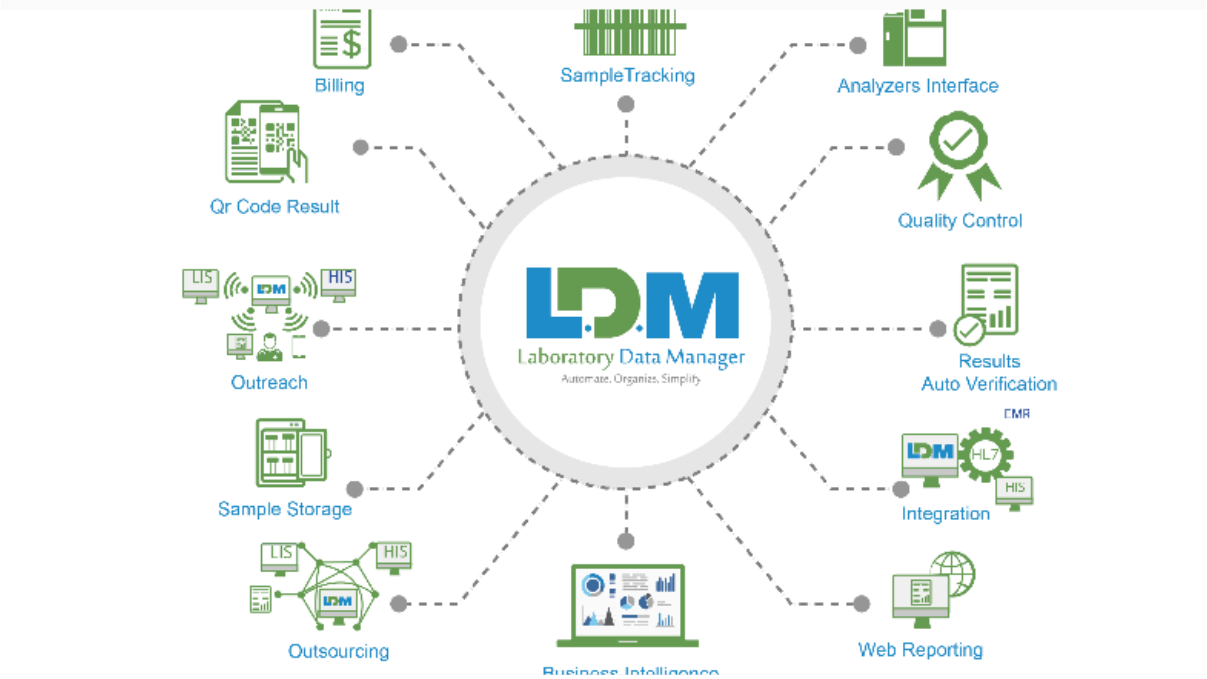


Figure 29: Laboratory Data Manager modules

3. LabXpert

LabXpert is a proprietary software solution from Med X International that reports the results of GeneXpert testing for tuberculosis diagnosis and to identify treatment resistance. Adoption of the LabXpert connectivity solution was selected by the Ethiopia National TB Program as a sustainable approach to connect the country's GeneXpert instruments to rapidly transmit testing results to clinicians to enable timely enrolment of patients onto treatment; automatically transmit data to real-time dashboards for use at site, regional, and national levels to monitor instrument functionality, maintenance status, and inventory management; and improve the programmatic data reporting system to strengthen overall lab programmatic management. LabXpert dashboards have been customised by MedX International according to the needs of the Ethiopia National TB Program, including results visualisation, graphs, aggregated data, rifampicin resistance reporting, module functionality status, supply status over time, treatment indicators, and maintenance status. The LabXpert connectivity project is being implemented in all 11 regions and two city administrations in Ethiopia. All data are hosted by the Ethiopian Public Health Institute (EPHI) [33]. The LabXpert system allows for monitoring of TB laboratory inventory of GeneXpert cartridges and sputum cups, as well as other

supplies, to ensure proper stock management. The LabXpert system can also visualise test performance using regional geolocations (figure below).

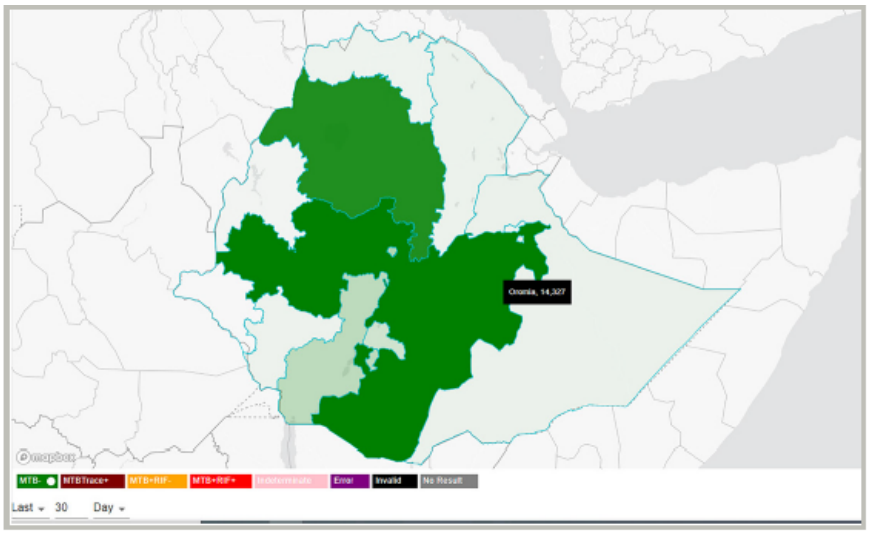


Figure 30: LabXpert map of TB test results

Annexure Six. Industry Standards and Best Practices for Data Warehousing

There are many ways to implement a data warehousing solution and many options an organisation needs to consider before designing a solution. In this section, we highlight some best practices for designing data warehousing and analytics solutions. These are derived from industry leaders in big data processing through their publications and architecture guides, as well as from our experience building Big Data projects. A list of references is provided below.

Table 26: List of Best Practices:

Standard	Priority	Best Practices
1. Verifying Data Source Systems and Analytics Platforms are Fully Functional	Required	Validate the data quality of source systems before transferring data for analytics.
	Required	Assuring Continuous Integration and Deployment of Analytics Solutions
2. Modernise deployment of the analytics jobs and applications	Recommended	Use version control for job and application changes.
	Recommended	Create test data and provision staging environments.
	Recommended	Test and validate analytics jobs and application deployments.
	Recommended	Build standard operating procedures for deployment, testing, rollback, and backfill tasks
3. Structuring Data Platforms to Uphold Governance and Compliance Standards	Required	Privacy by design.
	Required	Classify and protect data
	Required	Understand data classifications and their protection policies.
	Required	Identify the source data owners and have them set the data classifications.
	Required	Record data classifications into the Data Catalogue so that the analytics workload can be understood.
	Required	Implement encryption policies.
4. Establishing Data Access Controls	Recommended	Enforce downstream systems to honour the data classifications.
	Required	Allow data owners to determine which people or systems can access data in analytics and downstream workloads.
	Required	Build user identity solutions that uniquely identify people and systems.
	Required	Implement the required data authorisation models.
	Recommended	Establish an emergency access process to ensure

		that admin access is managed and used when required.
	Recommended	Track data and database changes.
5. Regulating Access to Work Infrastructure for Reliability	Required	Prevent unintended access to the infrastructure.
	Required	Implement least privilege policies for source and downstream systems.
	Required	Monitor the infrastructure changes and the user activities against the infrastructure.
	Required	Secure the audit logs that record every data or resource access in analytics infrastructure.
6. Creating Resilience in analytical Workloads	Required	Create an illustration of data flow dependencies.
	Required	Monitor analytics systems to detect analytics or extract, transform, and load (ETL) job failures.
	Required	Notify stakeholders about analytics or ETL job failures.
	Recommended	Automate the recovery of analytics and ETL job failures.
	Recommended	Build a disaster recovery (DR) plan for the analytics infrastructure and the data.
7. Govern data and metadata change	Required	Build a central data catalogue to store, share, and track metadata changes.
	Required	Monitor for data quality anomalies.
	Required	Trace data lineage.
8. Identifying the Most Efficient Computing Solution	Recommended	Identify analytics solutions that best suit your technical challenges.
	Recommended	Provision compute resources to the location of the data storage.
	Recommended	Define and measure the computing performance metrics.
	Recommended	Continuously identify underperforming components and fine-tune the infrastructure or application logic.
9. Selecting Optimal Storage Solutions	Highly recommended	Identify critical performance criteria for your storage workload.
	Highly recommended	Identify and evaluate the available storage options for your compute solution.
	Recommended	Choose the optimal storage based on access patterns, data growth, and the performance requirements.
10. Choose the best-performing file format and partitioning	Recommended	Select format based on data write frequency and patterns for append-only compared to in-place update.
	Recommended	Choose data formatting based on your data access

		pattern
	Recommended	Utilise compression techniques to both decrease storage requirements and enhance I/O efficiency.
	Recommended	Partition your data to enable efficient data pruning and reduce unnecessary file reads.
11. Choosing Economical Compute and Storage Solutions Synchronised with Workload Usage Patterns	Recommended	Decouple storage from compute.
	Recommended	Plan and provision capacity for predictable workload usage.
	Recommended	Use On-Demand Instance capacity for unpredictable workload usage.
	Recommended	Use auto-scaling where appropriate.
12. Formulating Financial Responsibility Models for Data and Workload Utilisation	Recommended	Measure data storage and processing costs per user of the workload.
	Recommended	Balancing agility and skill sets—when to build locally compared to centralised data analytics platforms.
	Recommended	Build a common, shared processing system and measure the cost per analytics job.
	Recommended	Restrict and record resource allocation permissions using AWS Identity and Access Management (IAM).
13. Manage cost over time	Recommended	Remove unused data and infrastructure.
	Recommended	Reduce overprovisioning infrastructure.
	Recommended	Evaluate and adopt new cost-effective solutions.
14. Adopting Best-suited Pricing Models Based on Infrastructure Utilisation Patterns	Recommended	Evaluate the infrastructure usage patterns, then choose payment options accordingly
	Recommended	Consult with your finance team and determine optimal payment models.
15. Guide on Implementing Sustainable Practices.	Recommended	Define your organisation's current environmental impact
	Recommended	Encourage sustainable thinking
	Recommended	Encourage a culture of data minimization
	Recommended	Implement data retention processes to remove unnecessary data from your analytics environment
	Recommended	Optimise your data modelling and data storage for efficient data retrieval
	Recommended	Prevent unnecessary data movement between systems and applications.
	Recommended	Efficiently manage your analytics infrastructure to reduce underutilised resources

Annexure Seven: TOGAF resources to facilitate the NHDW architecture

Note: TOGAF resources are not publicly available and requires an Open Group account to access the resources

- **TOGAF® Series Guide: Business Intelligence and Analytics:**
  - **Directly relevant:** This guide focuses on how to use TOGAF to support BI and analytics initiatives, which are closely tied to data warehousing.
  - **Key topics:** It covers topics like defining business information needs, designing data architectures for analytics, and governing data for BI and analytics.
  - **Helpful for:** Understanding how your data warehouse fits into the broader information architecture and supports decision-making processes within the organisation.
- **TOGAF® Series Guide: Information Architecture—Metadata Management:**
  - **Foundational guide:** This provides comprehensive guidance on applying TOGAF principles to data architecture, a core component of your data warehousing project.
  - **Key topics:** It covers data modelling, data integration, data governance, and data security, all essential for a successful data warehouse implementation.
  - **Helpful for:** defining a robust data architecture that aligns with your organisation's overall enterprise architecture and supports the specific needs of your data warehouse.

- **TOGAF® Series Guide: Value Streams:**
  - **Business alignment:** This guide helps you map your data warehouse project to specific value streams within your organisation.
  - **Key topics:** It covers identifying value streams, analysing their information needs, and designing architectures that support those needs.
  - **Helpful for:** ensuring your data warehouse delivers tangible value to the organisation by supporting critical business processes and outcomes.
- **TOGAF® Series Guide: Information Mapping:**
  - **Focus:** This guide provides a structured approach to understanding and representing the information that drives your organisation's operations.
  - **Key Topics:** It covers the identification of key information assets, the creation of visual maps to illustrate information flows, and the alignment of information architecture with business needs and strategic goals.
  - **Helpful for:** Applying this guide to your data warehouse project can help you define the scope of your data warehouse, understand how data flows within your organisation, and effectively communicate the role of the data warehouse to stakeholders.